

3DSS-DX and 3DSS-iDX Sonar Manual

VERSION 2.3
APRIL 18TH 2024





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PATENTS

US Patent #6130641, "Imaging methods and apparatus using model-based array signal processing."

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DOCUMENTATION FEEDBACK

Ping DSP is committed to the quality and continuous improvement of our products and services. We urge you to provide Ping DSP with any feedback regarding this guide by writing to the following email address: support@pingdsp.com

DOCUMENT REVISIONS

Date	Version	Document Changes
2014-06-12	1.0	Initial Draft
2015-01-12	1.1	All sections revised.
2015-03-22	1.2	Minor corrections and format changes
2015-03-27	1.3	3DSS-DX Sonar Control UI and time synchronization updates
2015-10-22	1.4	Added bathymetry, and sound velocity updates.
2016-06-08	1.5	Added XTF converter and target logger instructions. Added care and maintenance.
2016-10-27	1.6	Added HYPACK clarification.
2016-11-25	1.7	Minor changes throughout. Added Chart Display. And Troubleshooting section. Added SonarWiz 6 setup.
2017-06-26	1.8	Added multibeam bathymetry processing to 3DSS-DX Control Improvements to 3DSS Bathymetry Profile Display Improvements to 3DSS Chart Display Application
2018-03-02	1.9	Added CARIS import instructions. Added fuses specification, and extended DC power specs.
2018-05-14	1.9B	Added QINSy instructions. Added preliminary PDS 2000 instructions.
2019-01-10	2.0	Added 3DSS-iDX specific information.
2020-11-05	2.1	Added HYPACK configuration for 3DSS-iDX. Added 3DSS-iDX-PRO specific information.
2023-09-21	2.2	Changed Sound Velocity max value from 1600 to 2500. Updated list of allowable range values.
2024-04-18	2.3	Updated to include Simple GUI explanation

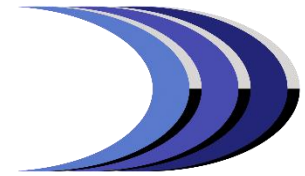


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WHAT'S INCLUDED

The **3DSS-DX-450** or **3DSS-iDX-450** Sonar System hardware contents include the following:

- 1 x **3DSS-DX / DX+** Sonar head with internal MRU (VN-100).
or
1 x **3DSS-iDX-BASE / FULL** Sonar head with SBG Ellipse-2 and AML Micro-X.
or
1 x **3DSS-iDX-PRO** Sonar head with SBG Navsight Ekinox and AML Micro-X.
- 1 x **3DSS-DTB** 3DSS™-DX Topside Box
or
1 x **3DSS-SIU** 3DSS™-iDX Sonar Interface Unit
- 1 x **3DSS-DSC-10** 10m 3DSS™-DX Sonar Cable
- 1 x **3DSS-DPC** Power Cable
- 1 x **3DSS-PPS** PPS Cable
- 1 x **3DSS-DZA** Zinc Optional Anode
- 2 x **3DSS-DF2A** Spare Fuses - 2A 507-1210-ND
- 2 x **3DSS-OR016** Spare 2-016 O-rings 1176N25
- 2 x **3DSS-OR017** Spare 2-017 O-rings 1884T24
- 1 x **3DSS-DPC** Pelican Case

Figure 1 shows the **3DSS-DX-450** Sonar, Topside Box and Sonar Cable in its Pelican case.

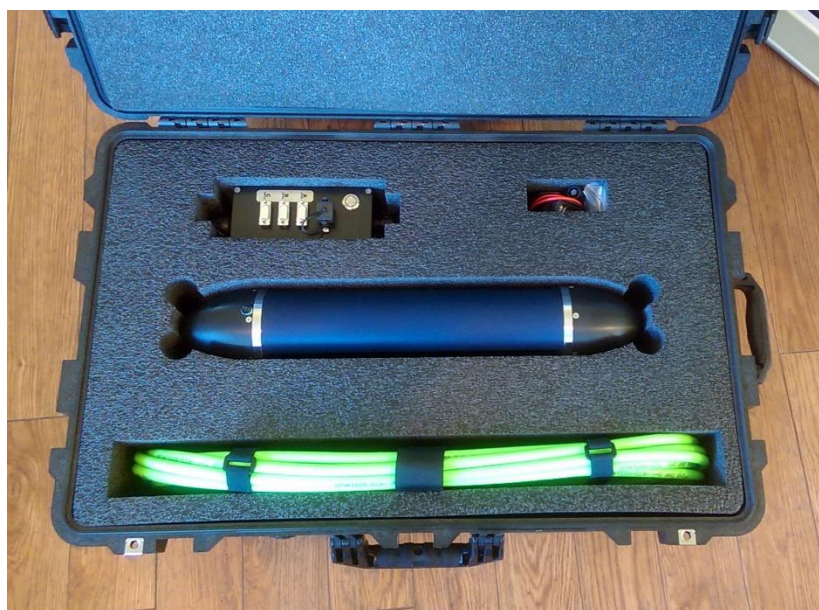


FIGURE 1. 3DSS-DX-450 SONAR SYSTEM HARDWARE.



The software and documentation for the **3DSS-DX-450** and **3DSS-iDX-450** is delivered over the internet and includes

- **3DSS-DCS** *3DSS™-DX Sonar Control Software*
- **3DSS-DDS** *3DSS™-DX Sidescan-3D Display Software*
- **3DSS-DPS** *3DSS™ Profile Display Software*
- **3DSS-DMS** *3DSS™ Chart Display Software*
- **3DSS-DUS** *3DSS™ Utility Software*
- **3DSS-DUG** *3DSS™-DX User Guide*

INSTALLATION REQUIREMENTS

Installation and operation of the **3DSS-DX/iDX** sonar requires the following:

1. Sonar mounting fixture (e.g. 1.5" ID pipe).
2. 24V DC power supply capable of 1A average, 3A for 2 seconds, and 7A peak current.
3. Computer or laptop with equal or better than an Intel Quad Core 2.4GHz CPU, 8GB of memory and discrete GPU for the 3DSS 3D Display application.
4. Microsoft Windows 7 or 8 - 64 bit edition operating system.
5. Temporary internet connection to download operating software and obtain a license.
6. Suitably long Cat5e Ethernet cable to connect the computer to the Topside Box.
7. Microsoft .NET 4.0. and Microsoft Visual Studio Redistributable 2010.

The following equipment may be optionally installed but is not required:

1. GPS or Dual Antenna GPS with an RS232 serial output.
2. External Motion Reference Unit (MRU) with an RS232 serial output and that can be configured to output TSS1 or NMEA protocol sentences.
3. Ethernet switch and additional client computers.

HARDWARE INSTALLATION INSTRUCTIONS

Hardware installation consists of the following steps:

1. Mounting the sonar.
2. Locating or mounting the Topside Box.
3. Connecting the sonar cable between the sonar and the Topside Box.
4. Connecting power to the Topside Box.
5. Connecting the Computer or Laptop or an Ethernet switch for multiple computer clients to the Topside Box via Ethernet
6. Connecting an optional GPS to the Topside Box.
7. Connecting an optional external MRU to the Topside Box.
8. Connecting an optional PPS signal to the Topside Box.
9. Connecting an optional Trigger In or Trigger Out to the Topside Box.

Figure 2 provides an overview of possible sonar configurations.

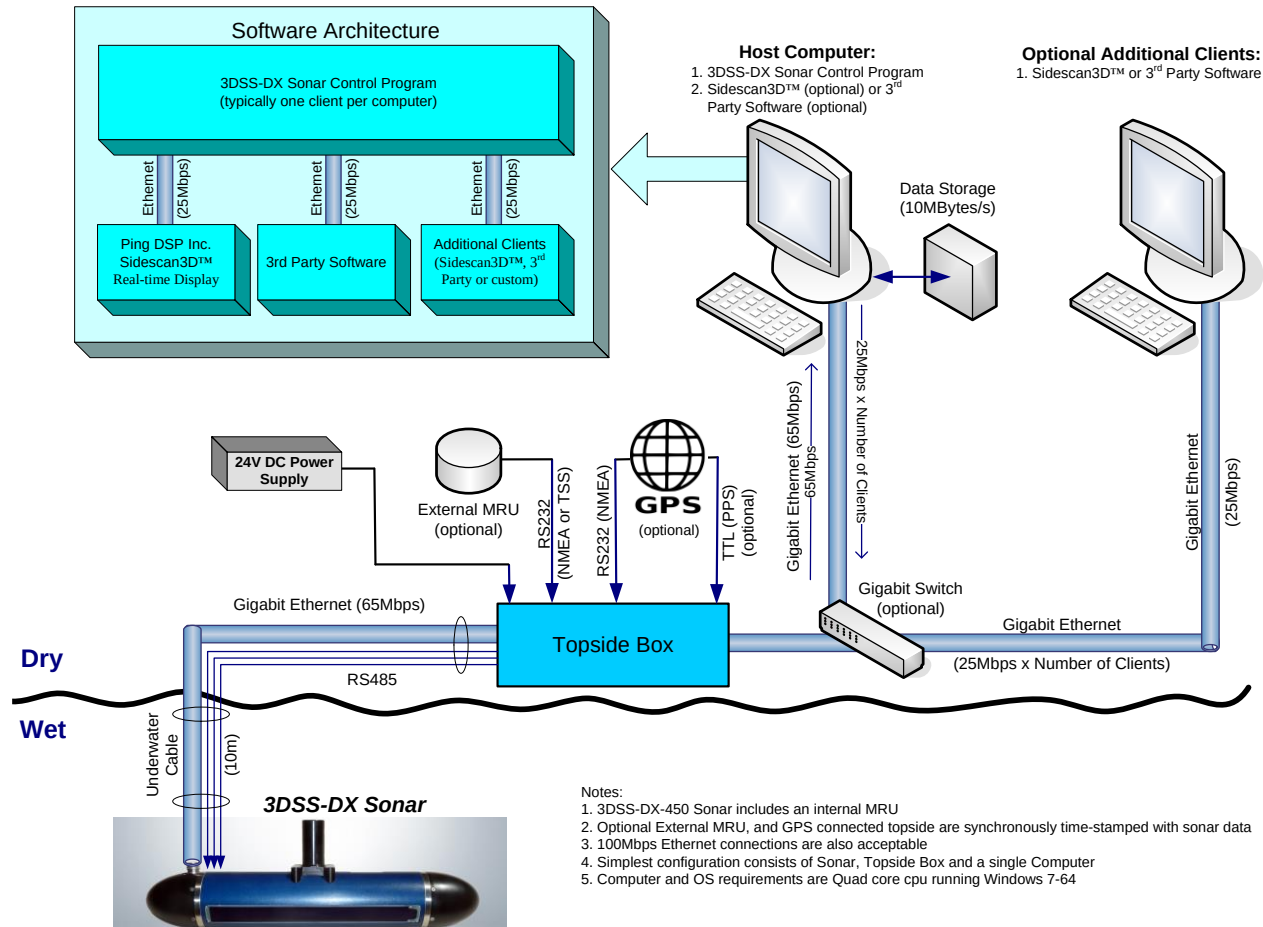


FIGURE 2.3DSS-DX-450 SYSTEM CONFIGURATION REQUIREMENTS AND OPTIONS.

MOUNTING THE SONAR

Figure 3 shows the correct 3DSS-DX mounting orientation.

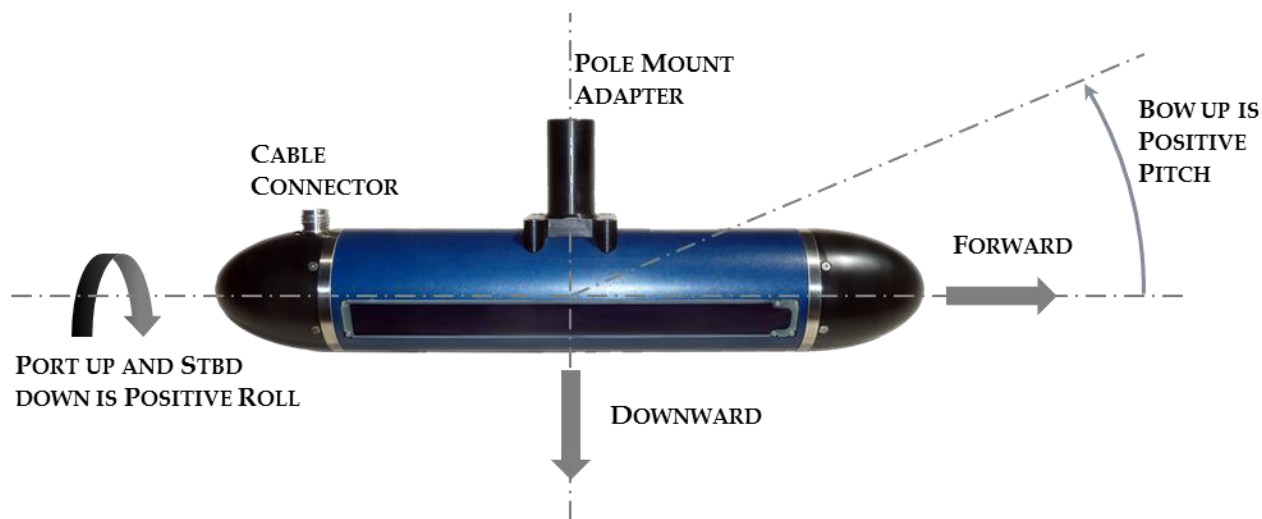


FIGURE 3. 3DSS-DX-450 MOUNTING ORIENTATION.

Ideally the sonar is mounted approximately 1m below the surface and/or boat hull in order to avoid a secondary transmit source arising due to close proximity of strong reflector above the sonar. Mounting the sonar within 1m of the surface or boat hull is possible, however adjustments to the transmit beampattern and/or beamsteering direction may be necessary in order to minimize the possibility and effect of an interfering secondary transmit pulse. This surface bounce multipath may arise on receive as well as transmit, however the receive multipath is separated using CAATI processing while the transmit multipath may produce a seafloor interference pattern characterized by periodic fluctuations in signal amplitude that in turn produce fluctuations in bathymetric variance and thereby limit the achievable swath width.

Thus, for sonar installations with less than 1m clearance between the sonar and the surface or boat hull, care should be taken to review data results and adjust the transmit beamwidth and/or beamsteering direction if necessary in order to minimize the possibility of a secondary transmit pulse (please see Chapter 3).

ENVIRONMENTAL CONSIDERATIONS

The **3DSS-DX-450** is designed to withstand harsh environmental conditions; however, the following limits should be adhered to when operating and storing the sonar:

- Sonar operating temperature: 0°C to +40°C (32°F to +104°F) (water temperature)
- Topside Box operating temperature: -20°C to +60°C (-4°F to +140°F)
- Sonar maximum depth: 5m

- Topside Box humidity: 95% non-condensing
- Sonar and Topside Box storage temperature: -20°C to +60°C (-4°F to +140°F)

The **3DSS-DX/iDX** transducers are protected from accidental damage by recessing them as much as possible into the protective housing, however a direct impact, or dropping or jarring the sonar can result in serious damage to the transducers and therefore the sonar should be handled with care at all times.

The sonar housing has been hard anodized to protect against corrosion and while this coating is durable, care should be taken to avoid marring or damaging the anodization layer. As further protection in applications requiring extended periods immersed in seawater, a sacrificial anode can be mounted on the **3DSS-DX/iDX** by replacing one of the nose cone screws with the anode and its mounting screw. The anode is included with the sonar and is pictured, along with the mounting location in Figure 4.



FIGURE 4. OPTIONAL ANODE INSTALLATION ON 3DSS-DX-450 NOSE CONE.

MOUNTING WITH THE POLE ADAPTER

There are two ways of attaching the **3DSS-DX/iDX** to a mounting fixture. The first makes use of the included pole adapter Figure 5 shows a drawing of the **3DSS-DX/iDX** anodized aluminum pole adapter. The adapter is designed to fit a standard 1.5" ID pipe and will also fit 1.75" or 2" Telespar square tubing. The pole adapter can also be used to attach the sonar to mounting fixtures other than a pole.



The technical drawing consists of two views of the SCS-007 sensor unit:

- Top View (Left):** Shows the overall dimensions of the unit. The total length is 22.1 units, and the diameter at the rounded ends is $\varnothing 3.88$. A central rectangular feature is shown.
- Side View (Right):** Shows the profile of the unit. It includes a label with the following information:
 - SCS-007
 - Sensor - Housing
 - 0000-0000
 - THE ABC DATA
 Dimensions include a vertical distance of 2.200 between mounting holes and a horizontal distance of 2.200 from the centerline to one hole. Four mounting holes are specified as:
 - 4 x $\varnothing 0.20 \nabla 0.80$
 - 1/4-20 UNC $\nabla 0.65$
 - $\varnothing 0.30 \times 90^\circ$, Near Side

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MOUNTING THE TOPSIDE BOX

The Topside Box is used for the 3DSS-DX and its connectors are shown in Figure 7. For information on the 3DSS Sonar Interface Unit (SIU) used with the 3DSS-iDX and optionally used with the 3DSS-DX please refer to the iDX Chapter in this manual.

The Topside Box connectors include:

1. **3DSS-DX-450** Sonar cable connector
2. 24V DC power connector
3. Computer Ethernet RJ45 connector
4. Three DB9 connectors (the inputs are compatible with both TTL and RS232 signal levels).



Top Face Ethernet and RS485 connectors.



Power connector and fuse.



3DSS-DX-450 Cable connector.

FIGURE 7. TOPSIDE BOX AND ITS CONNECTORS.

Subsequent sections describe the use of each of the connectors.

The Topside Box is splash proof with all connectors sealed to prevent moisture from finding its way to the interior of the Box. However, the Topside Box should be installed in a dry area. For permanent or semi-permanent installations, four mounting holes are located on the bottom flange of the Topside Box as shown in Figure 8.

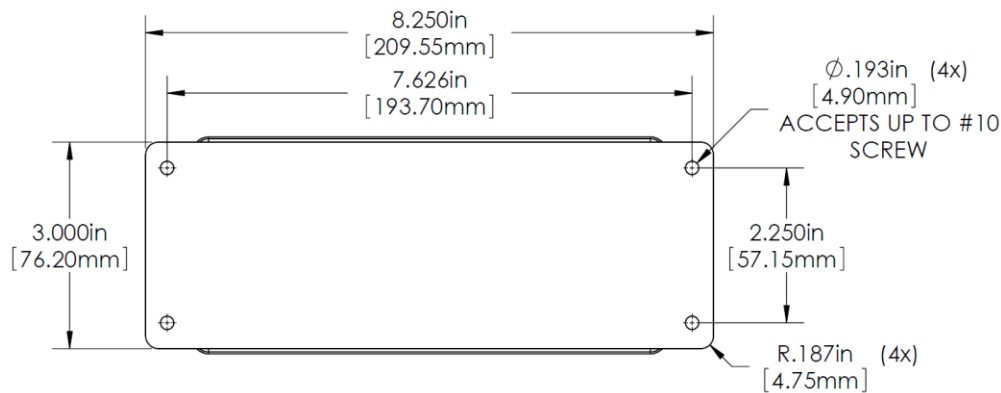


FIGURE 8. TOPSIDE BOX DIMENSIONS AND MOUNTING HOLES.

CONNECTING THE SONAR CABLE

The included sonar cable has a Teledyne Impulse MHDG-CCP-16 underwater connector that mates with the MHDG-FCR-16 on the sonar. When making this connection it is important that the 2-017 O-ring (green¹) in the CCP and the 2-016 O-ring (blue²) in the FCR have been installed and have not been damaged. Both connectors should be free from dirt and debris and pins on the CPP connector should be clean. Figure 9 shows excerpts from the MHDG connector manual with the location for proper installation of both O-rings. The **3DSS-DX-450** sonar system is supplied with both O-rings pre-installed but the **O-rings should be visually inspected each time the cable is connected** and should be replaced at the first sign of damage or once a year at a minimum.

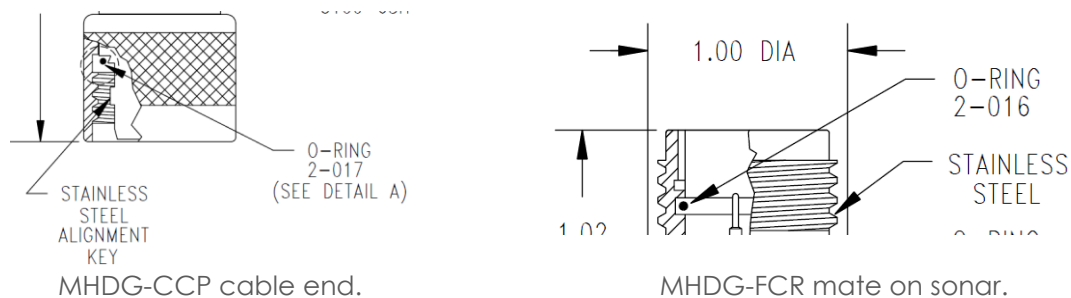


FIGURE 9. SONAR CABLE CONNECTOR O-RING LOCATIONS.

Care should be taken to ensure the cable is never bent in such a way as to cause a kink.

¹ McMaster-Carr Part Number: **1884T24**

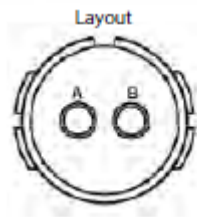
² McMaster-Carr Part Number: **1176N25**

CONNECTING POWER

The Topside Box power connector is a Souriau UTS78E2P. The mating connector is a Souriau UTS6JC8E2S and is included with the sonar system attached to red and black power leads. The red lead is connected to pin A and 24VDC and the black lead is connected to pin B which is ground. Figure 10 shows the connectors and the pinout for the Topside Box power connector.



Topside Box UTS78E2P Power Connector. Pin A is +24V and pin B is ground.



UTS6JC8E2S Power Connector Mate.



Sonar Power cable.

FIGURE 10. TOPSIDE BOX POWER CONNECTOR, MATE, AND INCLUDED POWER CABLE.

A fuse holder is located next to the input power connector, and contains a 20mm by 5mm – 2A rated fuse (P/N: 507-1210-ND).

Power requirements for the sonar is a DC supply or battery with a voltage between 10.5V and 28.5V DC. Typically, the sonar is run from 24V with the following current requirements:

- Average Current 1A @ 24V
- Surge Current (2 seconds surge) 1.5A @ 24V
- Peak Current 3.5A @ 24V (50ms duration)

For operation from a 12V supply, the currents can be roughly doubled and the externally accessible fuse should be replaced by a 4A rated one (recommended P/N: F2692-ND).

Care should be taken to ensure the input power connection is free from spurious electromagnetic noise since this can significantly degrade sonar performance.

CONNECTING COMPUTERS

One or more computers may be connected to the **3DSS-DX-450** sonar via the Ethernet connector on the Topside Box. One of these computers will be denoted as the “Host” computer and will run the **3DSS-DX Sonar Control** software as described in the Chapter 3. If the Host computer is the only computer to be connected, it can be directly connected to the Topside Box Ethernet



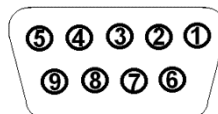
connector. If connections to additional computers are desired (e.g. to run multiple display and acquisition programs), the Topside Box Ethernet port should be connected to an Ethernet switch as shown in Figure 2.

The sonar IP address is 192.168.228.1 and computers connecting to the sonar should include this address within their subnet mask. For example, a connected computer with an IP address of 192.168.228.113 and a subnet mask of 255.255.255.0 will be able to communicate with the sonar.

CONNECTING AN OPTIONAL GPS

A GPS serial port may be connected to the Topside Box through the Port B DB9 connector shown in Figure 7. Port B is an RS232 port configured for 38.4kbaud, 8 bits, 1 start bit and no stop bits (i.e. 8N1). If the GPS device cannot be configured for 38.4Kbaud or a higher baud rate is required please contact Ping DSP Inc.

The pinout for all three DB9 connectors (ports A, B, and C) on the Topside Box is the same and is shown in Figure 11.



Pin 1: NC
Pin 2: NC
Pin 3: **RX**
Pin 4: NC
Pin 5: **GND**

Pin 6: NC
Pin 7: NC
Pin 8: NC
Pin 9: NC

FIGURE 11. DB9 PINOUT FOR TOPSIDE BOX PORTS A, B, AND C.

During operation the GPS should be configured to provide at least these output sentences and rates:

1. \$GPRMC at 10Hz (used for time, position, velocity, and true course)
2. \$GPHDT at 10Hz (20Hz recommended) only if available (used for heading)
3. \$GPGGA at 10Hz (reports position and *altitude* and is useful for RTK tide corrections)
4. \$GPGST at 5Hz or 10Hz (position error statistics used in GPS aiding of Ellipse2 MRU)
5. \$GPZDA at 1Hz

All sentences received by the 3DSS-DX Sonar are forwarded to third party software which may require additional sentences for proper operation. For example, the following sentences are often useful:

1. \$GPGGK at 10Hz (reports position and *altitude* and is useful for RTK tide corrections)



2. \$GPHEV at 10Hz (20Hz recommended) only if available (used for heave)

The table below specifies what GPS NMEA sentences are required for each 3DSS software package. Requirements for third party software need also be met.

TABLE 1 - NMEA REQUIREMENTS PER SOFTWARE PACKAGE

Data	Sidescan3D Display	Chart Display	XTF Converter	3 rd Party Software
Position	RMC or GGA	RMC, GGA, GSK	RMC, GGA, GSK	Make sure data requirements for any 3rd party software are met.
Velocity	RMC or VTG	-	RMC or VTG	
Heading	HDT, RMC*, VTG*	HDT, RMC*	HDT, (RMC, VTG)*	
TSS	TSS**	-	TSS or Internal MRU	
Altitude	-	-	GGA, GSK	

* Course made good only, HDT is preferred for true heading

** Displayed but not required (Uses sonar internal MRU only)

CONNECTING AN OPTIONAL EXTERNAL MRU

An external Motion Reference Unit (MRU) may be connected to the Topside Box through the Port C DB9 connector shown in Figure 7. Port C is an RS232 port configured for 115.2Kbaud, 8 bits, 1 start bit and no stop bits (i.e. 8N1). If the MRU device cannot be configured for 115.2Kbaud or a different baud rate is required please contact Ping DSP Inc.

The pinout for all three DB9 connectors (ports A, B, and C) on the Topside Box is the same and is shown in Figure 11.

The MRU should be configured to output TSS1 protocol sentences at a minimum rate of 20Hz (50Hz recommended).

CONNECTING AN OPTIONAL PPS SIGNAL

In the case where close time synchronization is desired between the **3DSS-DX-450** sonar and another system or computer, a positive logic (active high) PPS signal can be used to align the sonar clock to the nearest GPS second. When the sonar clock is locked to a valid PPS signal, the accuracy of the lock is indicated in the Sonar Status display of the **3DSS-DX Sonar Control** program (please see Chapter 2). Please note however that the indicated lock is only to the second and that the sonar clock must first be updated to within 0.5 seconds of GPS time to ensure that it locks to the correct second. For example, if a GPS is connected to the Topside Box as described above, the sonar clock is aligned to the correct GPS second using the UPDATE TIME button in the **3DSS-DX Sonar Control** program (please see the UPDATE TIME section of Chapter 2).

In order to check that the sonar clock is locked to the correct second, the sonar timestamp and the GPS time can be compared in the HEAD's UP DISPLAY section of the **Sidescan3D™** real-time data display (please see Chapter 4).

The 3DSS-DX-450 comes included with a BNC to RS232 adapter cable for connecting a variety of PPS sources. The Topside Box PPS input level is both TTL and RS232 level compatible. The pinout for all three DB9 connectors (ports A, B, and C) on the Topside Box is the same and is shown in Figure 11.



FIGURE 12 - PPS CABLE

CONNECTING AN OPTIONAL TRIGGER IN

The **TRIGGER IN** interface is shared with the other external device interface options on any of the pre-assigned DB9 ports (i.e. GPS, MRU and PPS). By default, the **TRIGGER IN** option is disabled in favour of the other port options. To enable the external trigger option in exchange for any of the pre-assigned interfaces, please contact Ping DSP Inc.

When enabled, the **TRIGGER IN** signal is TTL or RS232 positive logic.



CARE AND MAINTENANCE

IMPORTANT INFORMATION

The 3DSS-DX is ruggedly designed and built, and therefore requires little maintenance. However, to ensure long-lasting, reliable service, some periodic maintenance is recommended.

The 3DSS-DX / IDX is not designed for long-term deployments, and it is not recommended to keep the system in the water for more than two weeks at a time. If the system is to be deployed for 2+ days of continuous submersion – the provided zinc anode should be installed. After each deployment, the system should be rinsed thoroughly with freshwater, and its port and starboard transducer arrays inspected for any organism growth or damage. To remove any buildup of debris, the transducer faces should be washed using a soft cloth and a mild non-abrasive hand soap with water.

IMPORTANT: DO NOT USE HARSH CHEMICAL DETERGENTS OR SOLVENTS ON THE TRANSDUCER FACES.

Any time the sonar cable is disconnected from the sonar head the O-rings and connector pins in both the cable end and connector end should be inspected for any grime buildup. The connector on the sonar cable should be clean and inspected for the presence of a single green 2-017 O-ring. The connector on the sonar housing should be clean and inspected for the presence of a blue 2-016 O-ring. If grime buildup does occur, the O-rings should be removed the connectors cleaned with isopropyl and a low lint Q-tip, and new greased O-rings must be installed. To extend the life and increase the reliability of the O-ring seal, apply a thin film of an O-ring lubricant before installation of the O-ring. The underwater connector on the sonar unit, as well as, the mating end on the underwater cable should always be kept clean and protected when the cable is not connected.

UV EXPOSURE WARNING: Do not allow the 3DSS sonar head to sit in direct sunlight when not in use. Prolonged exposure to UV will damage the polyurethane face of transducers. In order to protect the hardware components of the sonar the following should be observed:

- DO NOT connect or disconnect the main sonar cable with the power on.
- DO NOT attempt to open and service the 3DSS sonar head.
- DO NOT attempt to open and service the Sonar Interface Unit unless specifically instructed to by Ping DSP.
- DO NOT place the 3DSS sonar head where the transducers may be scratched or damaged.
- DO NOT exceed the bend radius the main sonar cable (100mm).
- DO NOT clean the sonar head using a pressure washer – the high-velocity spray will cut through the polyurethane transducer face.



SOFTWARE INSTALLATION INSTRUCTIONS

Software installation consists of the following steps:

1. Downloading the **3DSS-DX Sonar Control** program to the Host computer or laptop.
2. Installing the **3DSS-DX Sonar Control** program.
3. Obtaining a license for the **3DSS-DX Sonar Control** program.
4. Downloading the **Sidescan3D™** real-time display program to the Host or other computer or laptop.
5. Installing the **Sidescan3D™** real-time display program.
6. Obtaining a license for the **Sidescan3D™** real-time display program.
7. Download and Install the **3DSS™-DX Profile Display** application.
8. Download and install the **3DSS™ Chart Display** OpenCPN plugin (optional).

After software installation a sample data file can be optionally downloaded to test the installation or to explore the software before using it with the **3DSS-DX-450**.

3DSS™-DX SONAR CONTROL PROGRAM

DOWNLOADING THE 3DSS-DX SONAR CONTROL PROGRAM

Please email support@pingdsp.com to receive a download link to the most up to date **3DSS-DX Sonar Control** program. The program is contained within a single executable install file that can be quickly downloaded over the internet.

Notifications and links to program updates are also be sent via email.

INSTALLING THE 3DSS-DX SONAR CONTROL PROGRAM

Once downloaded, simply double click on the install file and follow the instructions. The program requires both Microsoft .NET 4.0 and the Microsoft Visual Studio 2010 Redistributable Library. The installer will therefore check whether or not these libraries are already present on the Host computer and will install them as necessary.

Program updates are installed in exactly the same way as the original program and will overwrite a previous installation.

OBTAINING A SONAR CONTROL PROGRAM LICENSE

The first time the **3DSS-DX Sonar Control** program is executed a dialog box will appear to request a license key. A license key is obtained via email by clicking on the "Request New License" button in the License Key dialog box. An email will be sent to Ping DSP Inc. and a license code will then be provided by return email.

On receiving the license code, simply copy it into the License Code entry box and click continue. The program is now ready for use.



A license key is required for each computer that executes the **3DSS-DX Sonar Control** program. Program updates do not affect the license key.

3DSS™-DX SIDESCAN3D DISPLAY SOFTWARE

DOWNLOADING THE SIDESCAN3D™ REAL-TIME DISPLAY PROGRAM

Please email support@pingdsp.com to receive a download link to the most up to date **Sidescan3D™** program. The program is contained in a single executable install file that can be quickly downloaded over the internet.

Notifications and links to program updates are also sent via email.

INSTALLING THE SIDESCAN3D™ REAL-TIME DISPLAY PROGRAM

Once downloaded, simply double click on the install file and follow the instructions. The program requires Microsoft .Net4.0 and the installer will install this library if necessary.

Program updates are installed in exactly the same way as the original program and will overwrite a previous installation.

OBTAINING A SIDESCAN3D™ REAL TIME DISPLAY PROGRAM LICENSE

The first time the **Sidescan3D™** program is executed a dialog box will appear to request a license key. A license key is obtained via email by clicking on the "Request License" button in the License Key dialog box. An email will be sent to Ping DSP Inc. and a license code will then be provided by return email.

On receiving the license code, simply copy it into the License Code entry box and click continue. The program is now ready for use.

A license key is required for each computer that executes the **Sidescan3D™** program. Program updates do not affect the license key.

3DSS™-DX PROFILE DISPLAY

Please email support@pingdsp.com to receive a download link to the most up to date **3DSS™-DX Profile Display** program. The program is contained in a single executable install file that can be quickly downloaded over the internet. Once downloaded, simply double click on the install file and follow the instructions.

DOWNLOADING A 3DSS-DX-450 SAMPLE DATA FILE

Please email support@pingdsp.com to receive a download link to the most up to date list of **3DSS-DX-450** sample data files. The data file named **3DSS-DX Demo Data (Sidney_9).3dss-dx** is referred to in Chapter 3 as part of the **SOFTWARE QUICK START** instructions and therefore should be downloaded to help become familiar with the software before actual sonar operation.



3DSS™-DX CHART DISPLAY SOFTWARE (OPTIONAL)

DOWNLOADING THE OPENCPN 3DSS™DX CHART DISPLAY SOFTWARE

Please email support@pingdsp.com to receive a download link to the most up to date **3DSS™ DX Chart Display** program. The program is contained in a single executable install file that bundles OpenCPN and the 3DSS™-DX Chart Display plugin.

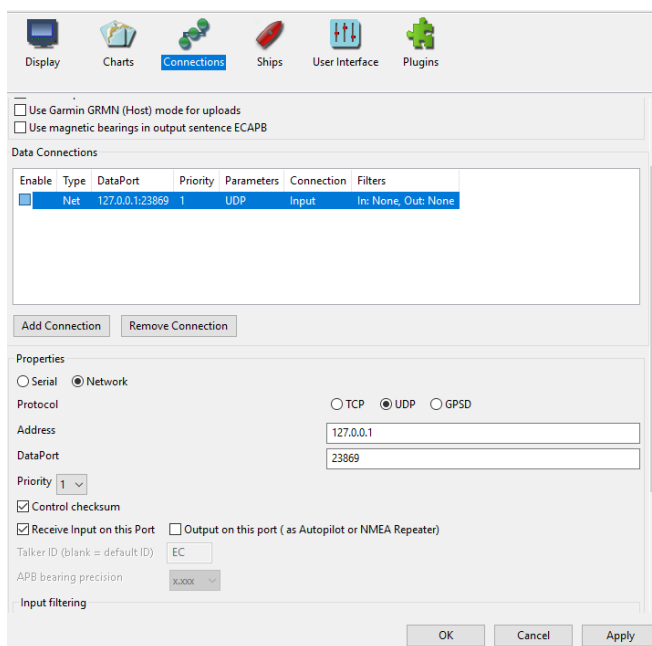
CONFIGURING THE OPENCPN 3DSS™DX CHART DISPLAY SOFTWARE

1. Run the provided executable and follow the steps for installing OpenCPN. This will automatically install the 3DSS™DX Chart Display plugin as well.
2. Once installed, run OpenCPN and navigate to the "Options" menu. In the Options window select the "Connections" tab. Add a new UDP connection by selecting "Add Connection" and entering the following under "properties" as shown in the figure below.

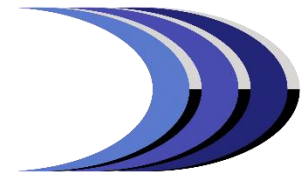
Address: **127.0.0.1**

Data Port: **23869**

Protocol: **UDP**



3. Navigate to the "Plugins" tab and enable the 3DSS Chart Display plugin by selecting the "Enable" button next to the plugin name.
4. To load navigational charts, select the "Charts" tab and then select "Add Directory". A window will prompt you to select a folder. Choose the folder containing your S57 or S63 vector charts.



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WHAT'S INCLUDED

The operation of the 3DSS-iDX is nearly identical to the operation of the 3DSS-DX. The main features of the iDX are the inclusion of an integrated SBG GNSS aided inertial navigation system and an AML Micro-X sound velocity sensor; both are integrated into the sonar head. The 3DSS-iDX also uses a new and improved Sonar Interface Unit (SIU) on the top side. This new SIU features

- full isolation on all input ports,
- isolated output power on 2 of the input ports,
- an Ethernet switch for easier deployment, and
- an optional integrated dual antenna GNSS receiver (Septentrio AsteRx4).

3DSS-iDX SONAR HEAD

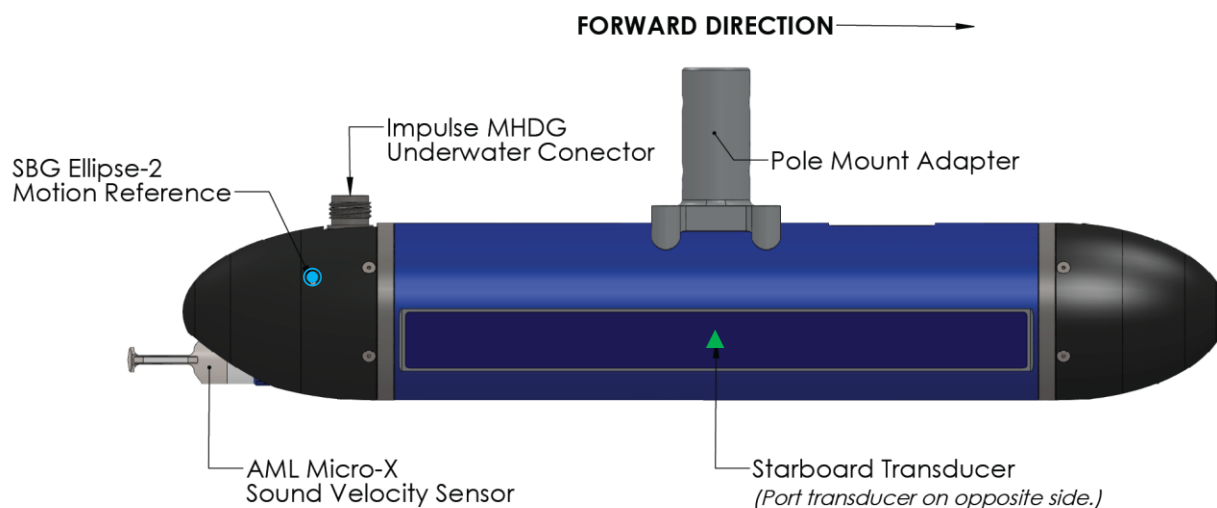


FIGURE 13. SIDE VIEW OF 3DSS-iDX SONAR

AML MICRO-X SOUND VELOCITY PROBE

The iDX integrates the OEM variant of the Micro•X - SV•Xchange™ sound velocity probe from AML Oceanographic. The sensor can be seen at the rear of the sonar head in Figure 13. The sensor is used to measure the sound velocity at the transducer face, which is subsequently used in the angle-of-arrival calculation.

To use the sound velocity values, the SOURCE of the sound velocity in 3DSS Control **must** be set to PROBE. The SVP indicator light will light up green when properly operating.

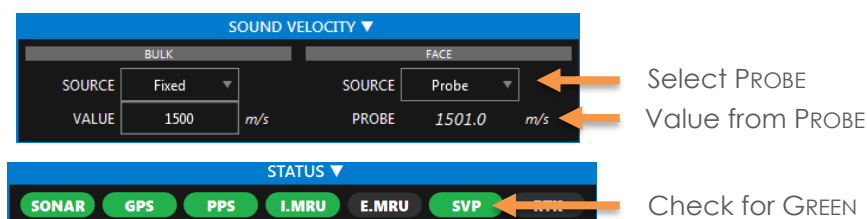


FIGURE 14. 3DSS CONTROL - SOUND VELOCITY SETTING

SBG ELLIPSE-2 INS

The iDX-BASE and iDX-FULL both integrate the OEM variant of the Ellipse2-E INS from SBG. The Ellipse2-E is a miniature high-performance MEMS based inertial navigation system that can be aided by an external GNSS receiver to improve orientation accuracy. The Ellipse-2 is located in the rear section of the 3DSS-iDX sonar head and is **not** user accessible.



FIGURE 15. ELLIPSE2-E OEM VARIANT

ELLIPSE2-E PERFORMANCE SPECIFICATIONS

The performance of the Ellipse2 is sufficient for shallow water applications:

Positioning Mode	Roll / Pitch	Heading	Heave
No Aiding	0.1°	0.8° yaw	5cm or 5%
GPS (non-RTK)	0.1°	0.8°	5cm or 5%
Dual Antenna GPS	0.1°	0.2°	5cm or 5%
RTK	0.05°	0.2°	4cm
PPK (SBG QInertia software)	0.05°	0.1°	2cm

SBG NAVSIGHT EKINOX INS

The iDX-PRO integrates the OEM variant of the Navsight Ekinox INS from SBG. The Ekinox is a miniature high-performance MEMS based inertial navigation system that is aided by a GNSS receiver to improve orientation accuracy. The Ekinox's IMU is located in the rear section of the 3DSS-iDX sonar head, while its processing module is integrated into the SIU top-side; neither is user accessible.



FIGURE 16. NAVSIGHT EKINOX OEM VARIANT; IMU AND PROCESSING MODULE

EKINOX PERFORMANCE SPECIFICATIONS

The performance of the Ekinox is listed below:

Positioning Mode	Roll / Pitch	Heading	Heave
GPS (non-RTK)- Dual Antenna	0.03°	0.08°	2.5cm or 2.5%
RTK	0.02°	0.08°	2.5cm or 2.5%
PPK (SBG QInertia software)	0.015°	0.03°	2.5cm or 2.5%
GNSS Outage 30s RTK	0.05°	0.13°	2.5cm or 2.5%
GNSS Outage 30s PPK	0.04°	0.05°	2.5cm or 2.5%

GPS/GNSS AIDING - NMEA-0183

It is highly recommended to provide GPS aiding to the Ellipse2-E / Ekinox by connecting a GPS NMEA-0183 source to the iDX Sonar Interface Unit (or using the internal GPS option). The GPS should be configured to output

- \$GPRMC @ 5 Hz (or higher)
- \$GPGGA @ 5 Hz (or higher)
- \$GPGST @ 5 Hz (or higher)
- \$GPHDT @ 5 Hz (or higher) (if applicable, on dual antenna systems), and
- \$GPZDA @ 1 Hz.

In addition, the GPS's PPS output signal **must** be connected to the Sonar Interface Unit (SIU) on either the SMA connector labeled 'PPS', or on pin 9 of the serial connector (DB-9) labeled 'GPS'. The SIU's PPS light will illuminate green when a proper PPS signal is detected.

To verify that GPS aiding is being used by the SBG INS please look at the I.MRU indicator in 3DSS Control as shown below:

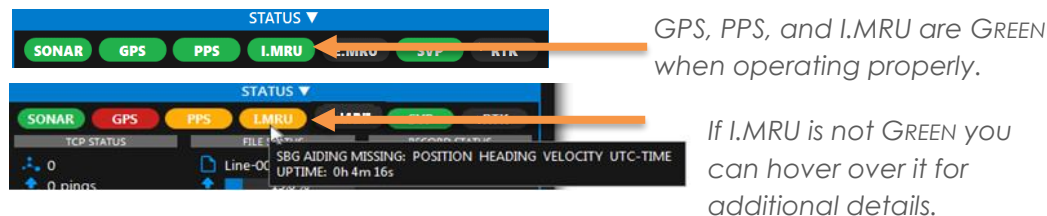
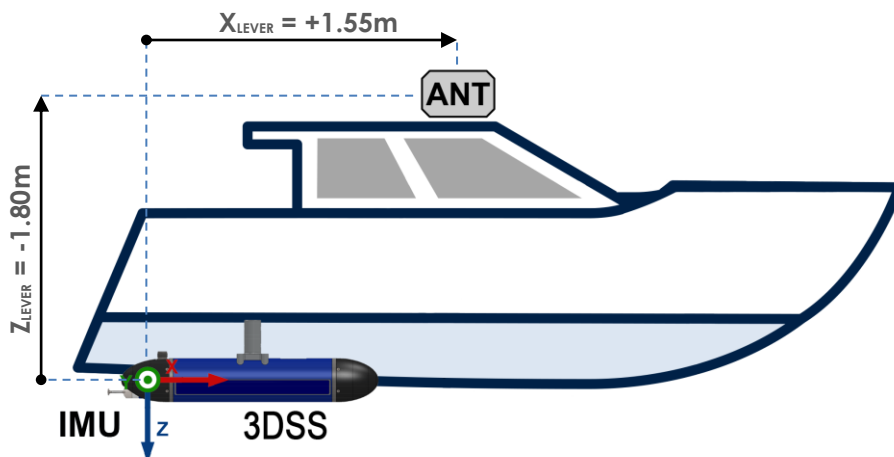


FIGURE 17. 3DSS CONTROL - I.MRU STATUS DISPLAY

GPS/GNSS AIDING - LEVER ARM - SINGLE ANTENNA GPS/GNSS

The GNSS lever arm is the signed distance, expressed in the vehicle coordinate frame, **FROM** the internal IMU (Ellipse-2 / Ekinox) **TO** the primary GPS antenna, and it **must** be programmed into the INS for proper operation. The lever arm should not exceed 10 m for best performance.



While operating with only a single antenna GNSS receiver as an aiding reference is possible, it is discouraged in favour of dual antenna GNSS configurations. When a single antenna GNSS is used, magnetic calibration is highly recommended. Please contact Ping DSP for instructions on how to perform magnetic calibration if this configuration is of interest.

GPS/GNSS AIDING - LEVER ARM - DUAL ANTENNA GNSS

The installation of a dual antenna GPS system will require special care in order to obtain optimal performance:

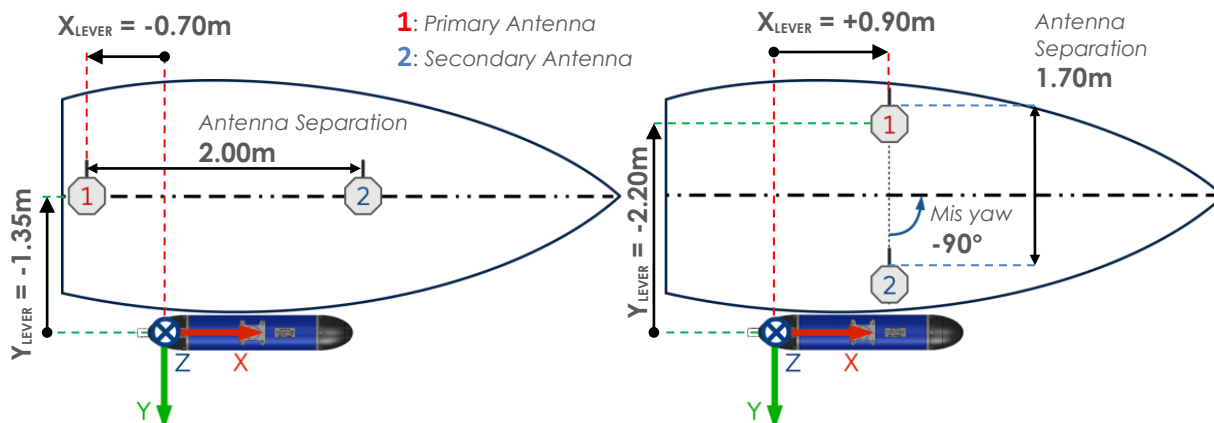
- Maximum separation between the two antennas should not exceed 3 m.
- The same antenna type and same cables with identical lengths must be used for both antennas.
- Both antennas must be mounted in the same orientation with respect to the vessel.
- Both antennas must have the same view of the sky when mounted on the vessel.

Once installed, the GNSS lever arm must be measured **from** the IMU to the **primary** GPS antenna. It must be measured to an accuracy of 5 cm or better.

Then, the absolute distance between the primary and secondary antennas must be measured.

Finally, the alignment angle between the antennas and the vessel coordinate frame must be accurately measured within 0.5°. Or specified as a nominal 0°, +90° or -90°, and further refined in the heading portion of a patch test.

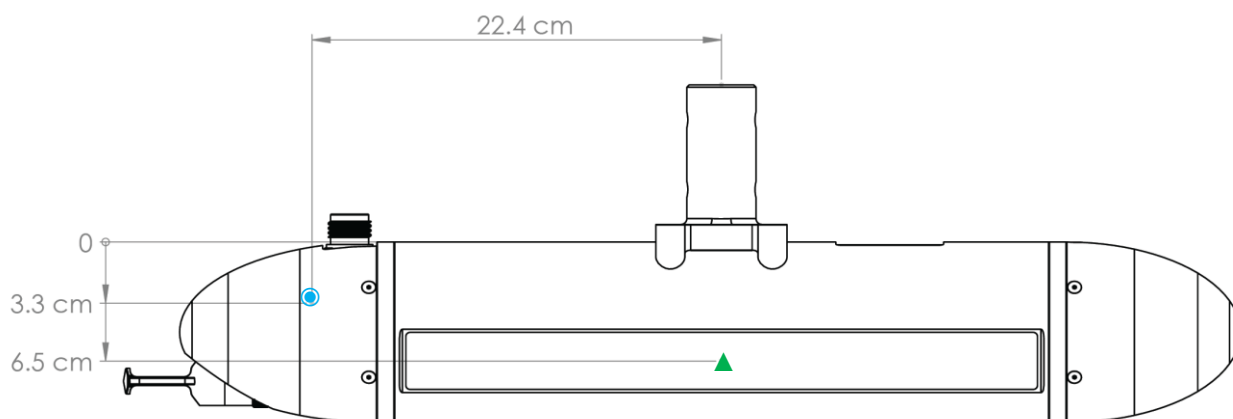
The following diagrams show two typical installations for the 3DSS-iDX over the starboard side of a vessel. The first one with the GPS antennas installed with a zero angle offset (*the preferred configuration*), and the second one, with a -90° installation:



NOTE: The yaw misalignment value should only be -90° or $+90^\circ$ when the HDT heading output by the GNSS is misaligned – does not output the vehicle heading. If the GNSS corrects and outputs the heading in the vehicle coordinate frame, the yaw misalignment value should be set to 0° .

EXACT POSITION OF SBG IMU INSIDE THE 3DSS-iDX SONAR HEAD

The exact position of the Ellipse-2 and Ekinox inside the 3DSS sonar head is shown below:



Where ● marks the position of the center of measurement for both the Ellipse-2 and Ekinox, and ▲ marks the acoustic center of the starboard transducer.



PROGRAMMING THE LEVER ARM INTO THE ELLIPSE-2

Once the lever arm is measured it must be programmed into the Ellipse-2 through the use of the 3DSS Control Console interface. The lever arm is specified in **meters** and **degrees**.

The lever arm value is stored inside the Ellipse-2 and needs to be updated when the installed orientation of the sonar head or GPS antennas changes thereby changing the lever arm.

To program the lever arm into the Ellipse-2 inside the iDX sonar head follow these steps:

1. Turn on the sonar head and connect to it using 3DSS Control.
2. Click the chevron in the top-right of the 3DSS Control application to open up the command console. See Figure 18 marker 2.
3. You will now be able to enter command in the console to query the Ellipse-2 and also check and set the lever arm. See marker 3.
4. Run **sbg --get-offsets** to check the offsets programmed into the Ellipse-2.
5. Run **sbg --set-offsets=x:y:z:pitch:yaw:antenna** to update the lever arm values stored in the Ellipse-2. Where **x** is the XLEVER offset, **y** is YLEVER offset, **z** is the ZLEVER offset, all specified in meters, and pitch is the **pitch** misalignment in degrees (typically 0), **yaw** is the yaw misalignment in degrees, and **antenna** is the antenna separation distance specified in meters.

For example, to set the lever arm of XLEVER =+0.90m, YLEVER =-2.20m, and ZLEVER =-1.80m in a single antenna system run:

```
sbg --set-offsets=0.90:-2.20:-1.80
```

For a dual-antenna system you will have the additional parameters of yaw misalignment and antenna separation distance. For a yaw misalignment of -90° and an antenna separation of 1.70m, run the following command:

```
sbg --set-offsets=0.90:-2.20:-1.80:0:-90:1.70
```

6. Finally, you **must save the settings** in the Ellipse-2 by running the following command:
sbg --save
7. After a couple seconds – you can run the **sbg --get-offsets** command again to confirm that the correct offsets were written to the Ellipse-2.

Figure 18 on the next page shows the 3DSS Control application after all of the commands in steps 4 – 7 have been executed.

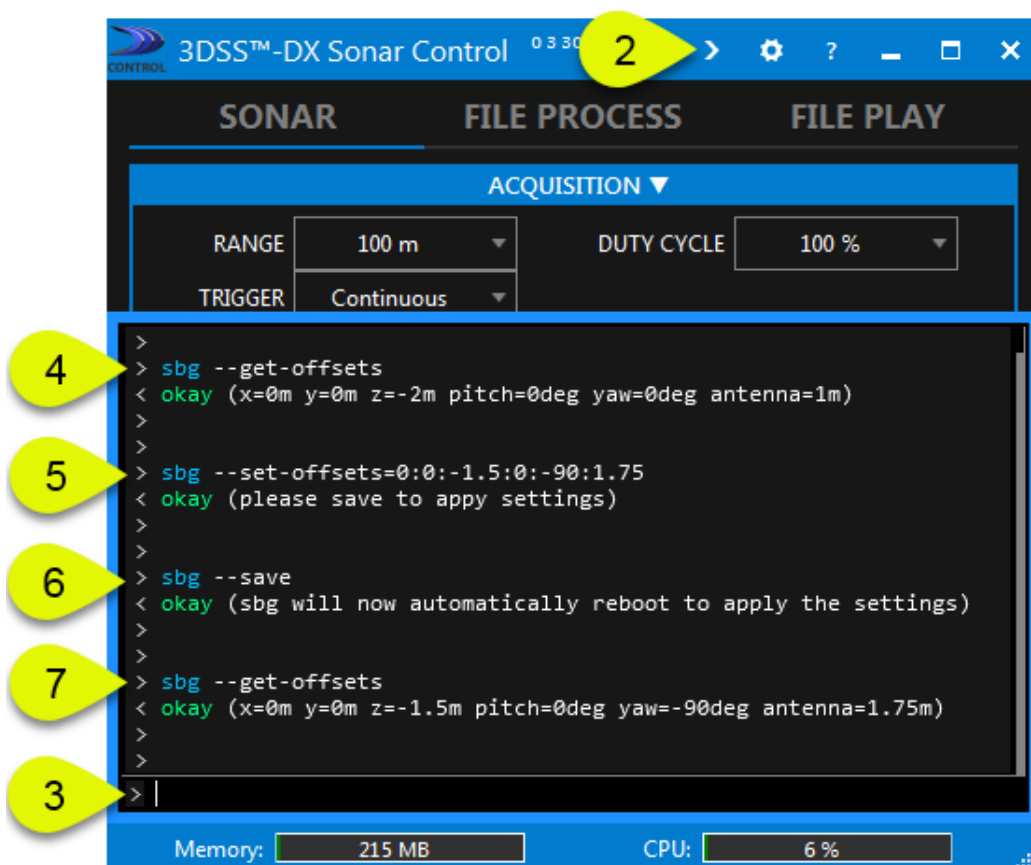


FIGURE 18. PROGRAMMING ELLIPSE-2 LEVER ARM USING COMMAND CONSOLE

PROGRAMMING THE LEVER ARM INTO THE NAVSIGHT EKINOX

Once the lever arm is measured it must be programmed into the Navsight Ekinox through the use of its web interface. The lever arm is specified in **meters**.

The lever arm value is stored onboard the Navsight Processing Module inside the SIU and needs to be updated when the installed orientation of the sonar head or GPS antennas changes thereby changing the lever arm.

To program the lever arm into the Navsight Processing Module inside the SIU follow these steps:

1. Turn on the SIU, and ensure that your local Ethernet adapter is configured for operations with the 3DSS-iDX (static IP), and the network cable is plugged into the SIU.
2. Open a web browser and browse to <http://192.168.228.92>. This accesses the internal web interface for configuring the SBG Navsight unit.



3DSS-DX and 3DSS-iDX Sonar Manual

VERSION 2.3
APRIL 18TH, 2024

SBG SYSTEMS NAVSIGHT-E-01 : 053000213 Marine View Configure

General Status Calibration Information Raw Values

Ship Navigation LATITUDE 0° 0' 0" (± 1000.00 m) LONGITUDE 0° 0' 1" (± 1000.00 m) TRACK 0.0° SPEED 1.2kt	Ship Attitude ROLL -4.08° (± 0.099 °) PITCH -0.36° (± 0.099 °) HEADING 355.2° (± 13.820 °)	UTC Time DATE 03/21/2013 TIME 00:01:53
Ship Motion Sensor location motion HEAVE 0.1m MAX 0.3m PERIOD 0.0s SURGE 0.0m SWAY -0.0m Deported location HEAVE 1 0.1m HEAVE 2 0.1m HEAVE 3 0.1m		Status Summary SYSTEM ✓ ALIGNMENT NOT ALIGNED CLOCK No UTC FREE RUNNING GNSS MODE INSUFFICIENT OBS. DUAL ANTENNA INSUFFICIENT OBS. DELAYED HEAVE ✗

- Click on Configure in the top-right corner, and select Aiding Settings:

Device Settings

Installation Overview IMU Setup Sensor Aiding Assignment **Aiding Setting** Inputs/Outputs Data Output Advanced Administration

GPS 1

GNSS Setup
Select the receiver model and if you plan to use single or dual antenna mode.
Dual antenna heading is useful for low dynamics applications and to initialize the INS in static conditions.

Receiver Model Septentrio Select Antenna Model JAV_GRANT-G3T NONE
GNSS Heading Mode Dual antenna (known lever arm)

GNSS Lever Arms
Please enter the primary and secondary lever arms FROM the INS, TO the GNSS antenna ARP with an accuracy better than 1 cm.
If you plan to use the calibration mode, please at least provide lever arms with an accuracy better than 20 cms

Primary Antenna (X,Y,Z) -0.270 0.020 -1.080 m Secondary Antenna (X,Y,Z) 0.710 0.020 -1.051 m

Aiding Use and Rejection
You can change the rejection filter to define how the INS should use the measurements coming from this GNSS.
To use in the INS solution both Position/Velocity and True Heading measurements from this GNSS receiver, please select Auto Rejection.
Auto rejection is the preferred mode as the INS will detect and ignore inconsistent measurements automatically.

Position/Velocity Auto rejection True Heading Auto rejection

Save Cancel

- For the forward-aft dual antenna configuration where the aft antenna is Primary. Enter, the primary and secondary antenna lever arms in meters.



5. For the port-starboard dual antenna configuration where the GNSS is outputting true heading, i.e. is adjusted such that ϕ_{HDT} heading points in the forward direction of travel. Enter the measured lever arm for the primary antenna. However, for the secondary antenna, enter the same Y and Z values as for the primary antenna, and add the antenna separation distance to the X value of the primary antenna and enter the aggregate for the X value of the secondary antenna.
6. For the port-starboard dual antenna configuration where the port antenna is Primary and the GNSS is outputting a heading that is offset by 90 degrees. Enter the appropriate measured lever arms for both the primary and secondary antennas.
7. Click Save to apply the configuration, and reboot the SIU.

RECORDING RAW MEASUREMENTS FROM ELLIPSE-2 / EKINOX FOR PPK

The raw binary sensor measurements from the Ellipse-2 and Ekinox are broadcast over Ethernet UDP and can be recorded by a computer on the top side (typically along with raw GPS measurements) for post processing in Qinetica PPK software. You can use the 3DSS iDX Binary Logger application to log these measurements to a file for post-processing. Additionally, the Navsight processing module logs raw data to its onboard memory and can be accessed using the Navsight web interface at <http://192.168.228.92>.

ADVANCED USER: SBCENTER FOR ELLIPSE-2E

The Ellipse-2 inside the iDX sonar head can connect to SBG Center over Ethernet UDP with the additional software. Ping DSP uses Eltima's *Serial to Ethernet Connector* software in order to create a virtual COM port on the PC that connects to over Ethernet UDP port 24333 to the sonar head. This virtual serial port then has direct access to the Ellipse-2 IMU, and can be used in SBCENTER to configure any and all settings of the Ellipse-2. The baud-rate for the Ellipse-2 ports should not be changed and Port A should be set to 921,600 bps. Access to the Ellipse-2 through SBG Center is only needed in special circumstances such as diagnostics or firmware upgrades.

3DSS-iDX SONAR INTERFACE UNIT

The 3DSS-iDX sonar uses a new and enhanced set of top-side electronics – the Sonar Interface Unit (SIU). The SIU features

- versatile power input internally isolated,
- two isolated RS-232 input ports (GPS, MRU) and an isolated serial RS-232 in-out port(AUX),
- isolated 12V 1A output power on GPS DB-9 port,
- isolated 12V 0.5A output power on AUX DB-9 port,
- isolated PPS input on SMA connector,
- isolated 5V logic trigger output on SMA connector,
- an integrated Ethernet switch with two external RJ45 connections,
- 3DSS sonar connector,
- indication LEDs for all ports, and
- an *optional* integrated GPS with dual TNC antenna connectors.

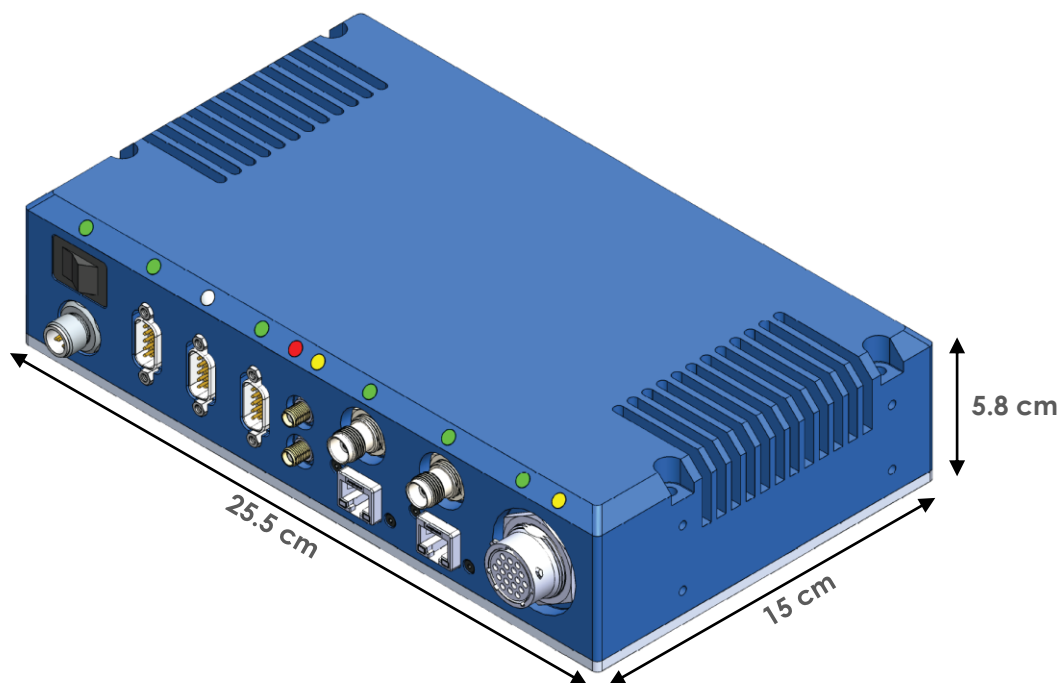


FIGURE 19. 3DSS-iDX SONAR INTERFACE UNIT

MECHANICAL DRAWING

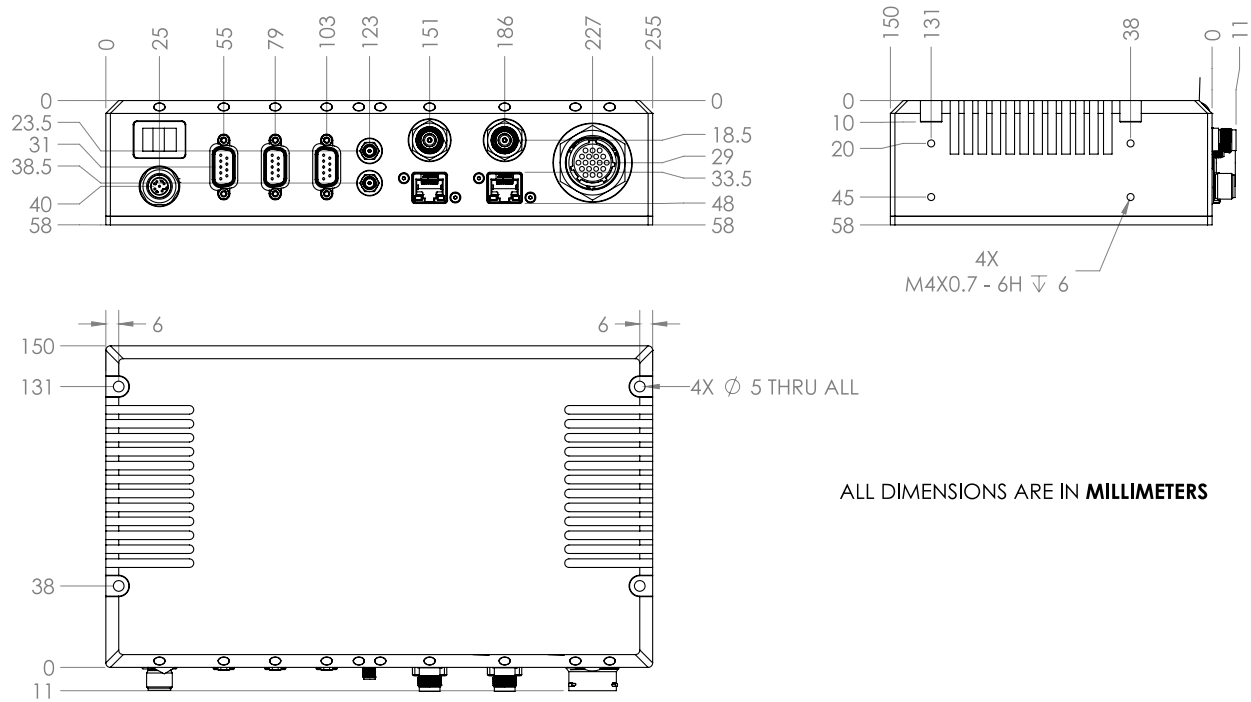


FIGURE 20. SIU MECHANICAL DRAWING



POWER REQUIREMENTS AND PIN-OUT

The SIU can be powered from a variety of DC power sources provided that the voltage is kept between +11.4V and +28V, and a sufficient current is available from the power source. This voltage range covers 12 volt and 24 volt batteries. If the input voltage ever drops below 10.8V – the SIU will automatically turn-off and will require a power-cycle to turn back on. The SIU has a soft-start circuit that limits in-rush current when the system is turned-on.

Electrical Characteristics	MIN	TYPICAL	MAX	UNITS
Input Voltage V_{IN}	11.4	12 or 24	28.0	V
Reverse Supply Protection	-31		31	V
Current I_{IN} @ 12V, Idle		1.7		A
Current I_{IN} @ 12V, Active		2.4		A
Current I_{IN} @ 12V, Burst (30ms)		3.5	TBD	A
Current I_{IN} @ 24V, Idle		0.9		A
Current I_{IN} @ 24V, Active		1.2		A
Current I_{IN} @ 24V, Burst (30ms)		1.7	TBD	A
Minimum Start-Up Voltage		11.4		V
Automatic Turn-Off Voltage		10.8		V
In-rush Current (100ms – 200ms)		1.0	1.2	A
External GPS Port Supply Voltage		12		V
External GPS Port Supply Current	0		1.25	A
External AUX Port Supply Voltage		12		V
External AUX Port Supply Current	0		0.5	A

*CONDITIONS: No external loads on GPS and AUX port, and no optional integrated GPS.

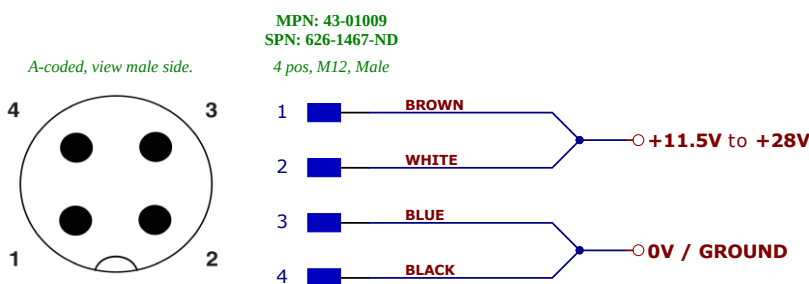


FIGURE 21. INPUT CONNECTOR PIN-OUT, M12 – 4 POSITION MALE

The recommended mating connector for the SIU power is one of the following:

- Phoenix Contact – MPN: 1681114 (277-4196-ND)
- Phoenix Contact – MPN: 1442719 (277-4824-ND)
- Any M12 – 4 position female connector with A coding.

At 19V or above, it is permissible to use only two of the contacts to power the SIU and system, however below 19V all four contacts should be used to provide power. Special care must be

taken at 12V where four 22 AWG conductors or two 18 AWG conductors with a length of less than 2 meters are recommended; longer runs can be accommodated using a thicker gauge wire.

POWER CABLE - 3DSS-SIU-PC150

The 3DSS-iDX is supplied with a power cable (3DSS-SIU-PC150) useful for connecting to DC sources such as 12V or 24V batteries, and regulated power supplies.

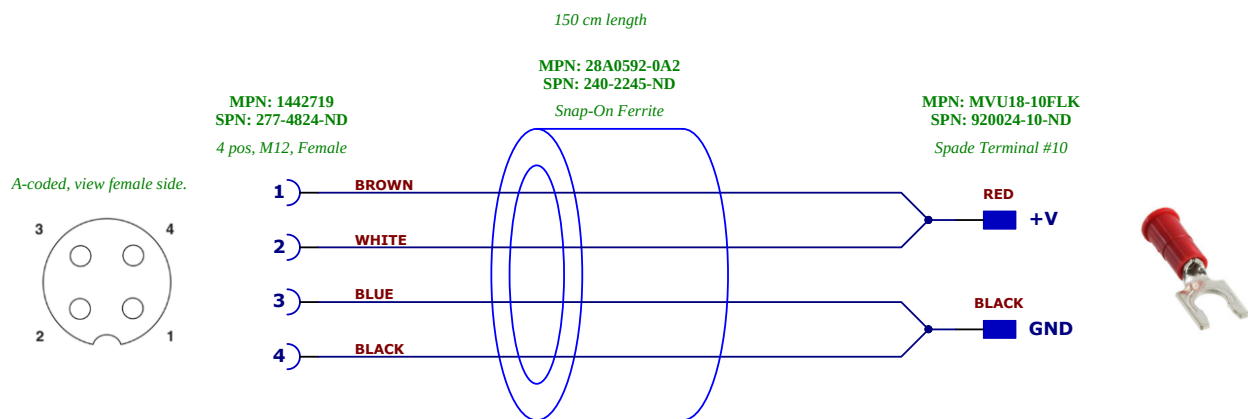


FIGURE 22. 3DSS-SIU-PC150 CABLE ASSEMBLY SCHEMATIC

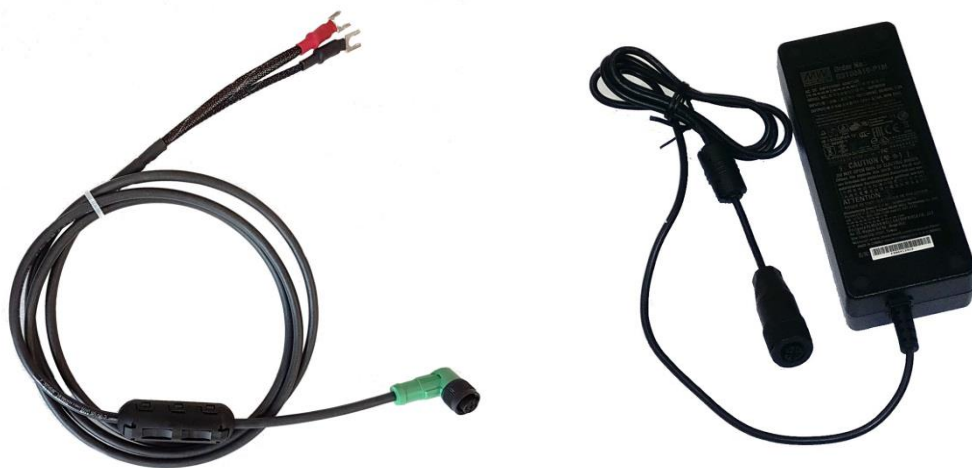


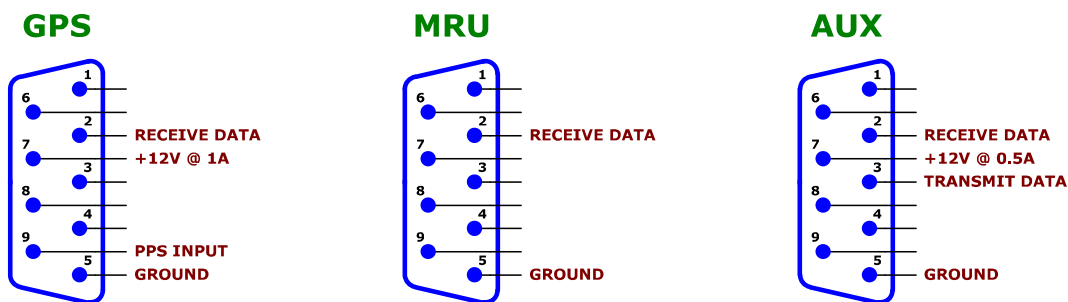
FIGURE 23. PICTURES OF 3DSS-SIU-PC150 (LEFT) AND 3DSS-SIU-PA90 (RIGHT)

POWER ADAPTER - 3DSS-SIU-PA90

The 3DSS-iDX can also be powered from AC power using the provided AC/DC adapter (3DSS-SIU-PA90). The adapter is universal AC input / full range unit, and an input range of 90V – 264V AC at 50 or 60 Hz. The adapter uses a standard IEC320-C13 power cable (not supplied).

SERIAL PORTS

The SIU has three serial ports on its front labelled GPS, MRU, and AUX from left to right. Each port is fully isolated from all other ports and connections on the SIU – in other words, the grounds of each port are independent.



GPS PORT

The GPS port is a receive only interface that is used to provide external GPS positioning information to the SIU and sonar head. The GPS port expects to receive a set of NMEA-0183 sentences at the default baud-rate of 38,400 bps. The required set of GPS sentences can be found in the Ellipse-2 GPS/GNSS aiding section on page 30, and a further recommended set of sentences can be found in the “Connecting an Optional GPS” section on page 18.

The GPS port also dedicates pin 9 as an optional PPS input – PPS signalling allows the SIU and the 3DSS sonar to maintain a lock to GPS time. The PPS signal can be supplied either through pin 9 of the GPS connector, or the dedicated SMA connector discussed later – PPS should **not** be provided on both. The PPS input is 3.3V and 5V logic compatible and a positive logic (rising edge) signal. It is recommended to set the duration of the active high of the PPS pulse to 1ms if possible.

The GPS port also has a +12V output that can be used to power a GPS. This power output has a maximum current output of 1.0A which is sufficient for many GPS units.

The SIU can have an optional internal GPS – connecting an external GPS will override the internal GPS.

MRU PORT

The MRU port is a receive only interface that is used provide an external motion information to the SIU and sonar head. The MRU port expects to receive either TSS1 sentences or NMEA-0183 sentences such as \$PASHR at the default baud-rate of 115,200 bps.

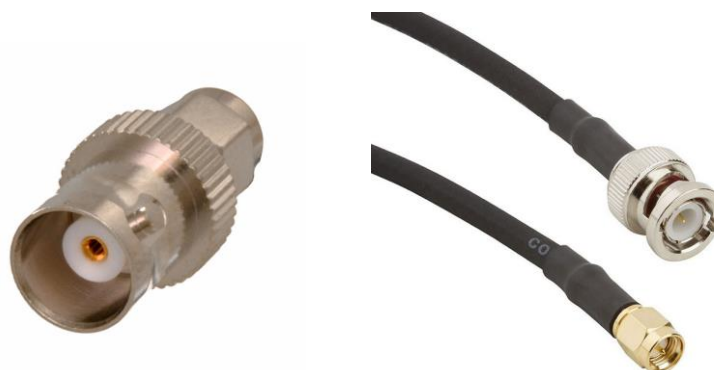
AUX PORT

The AUX port is a bi-directional port that is primarily used for firmware upgrades of the SIU. The AUX port also has a +12V 0.5A power output available on pin 7. If the SIU has the internal Septentrio GNSS option – the AUX port is connected to COM2 on the Septentrio and the GNSS can be

configured to accept RTCM or CMR corrections using its web user interface at <http://192.168.228.91/>.

LOGIC PORTS

The SIU has two logic level ports labelled PPS and TRIG. Each port is fully isolated from all other ports on the SIU and is compatible with 3.3V and 5.0V logic levels. The logic ports use an RF SMA connector which can easily be converted to BNC using an adapter or cable:



**SMA PLUG TO BNC JACK
ADAPTER
SPN: ACX1322-ND**

**SMA PLUG TO BNC PLUG
CABLE
SPN: J10118-ND**

The logic ports are TTL compatible and have the following electrical characteristics:

Electrical Characteristics	MIN	MAX	UNITS
High-level Output Current		-8.0	mA
Low-level Output Current		8.0	mA
High-level Output Voltage (-8mA load)	3.5		V
Low-level Output Current (8mA load)		0.5	V
High-level Output Voltage (-1mA load)	4.6		V
Low-level Output Current (1mA load)		0.2	V
High-level Input Voltage	2.1		V
Low-level Input Voltage		0.8	V
Input Hysteresis	0.2		V

PPS INPUT PORT

The 3DSS-iDX **requires** a Pulse Per Second (PPS) signal and the ZDA NMEA time sentence to synchronize the sonar head to the GPS time and accurately timestamp all the data. The PPS input timing must be stable to within $\pm 100\text{ppm}$ – an unstable PPS signal will cause locking issues. BNC-T adapters can be used to split the PPS signal to multiple devices. The PPS signal pulse length should be set to between $10\mu\text{s}$ and $1000\mu\text{s}$.

Timing Characteristics	MIN	MAX	UNITS
------------------------	-----	-----	-------



PPS Pulse Length	0.001	500	ms
Recommended PPS Pulse Length	10	1000	μs

PPS has an associated LED indicator that shows the status of the PPS signal:

LED Color	Meaning
Off	No PPS detected.
Red	PPS invalid or unstable.
Yellow	PPS stable and unlocked.
Green	PPS stable and locked.

Note that there are 3 possible sources for PPS: The SMA PPS Port, the PPS pin on the external GPS port, and the internal GPS. The LED indicator refers to all of these.

TRIGGER OUTPUT PORT

The second output port is labelled TRIG and is used to send trigger synchronization pulses to other devices. The TRIG line is positive logic with a 5V signal level. The output goes high (rising edge) when the 3DSS-iDX transmits a pulse, and remains high until the sonar has finished receiving the echo for the ping. Transmit occurs approximately 0.3ms after the trigger has gone high. The TRIG output can be inverted by the SIU if necessary – in this case, a falling edge indicates the start of transmit and the trigger remains low for the duration of the receive cycle.

The TRIG output has an associated LED indicator that flashes green every time the sonar transmits – at high ping rates it may appear solid.

RF ANTENNA PORTS

The SIU has two RF antenna ports that connect to the internal Septentrio AsteRx-4 GNSS. These ports are only fitted if the internal GNSS option is installed. The primary antenna must be connected to the ANT 1 port, and the secondary antenna must be connected to the ANT 2 port.

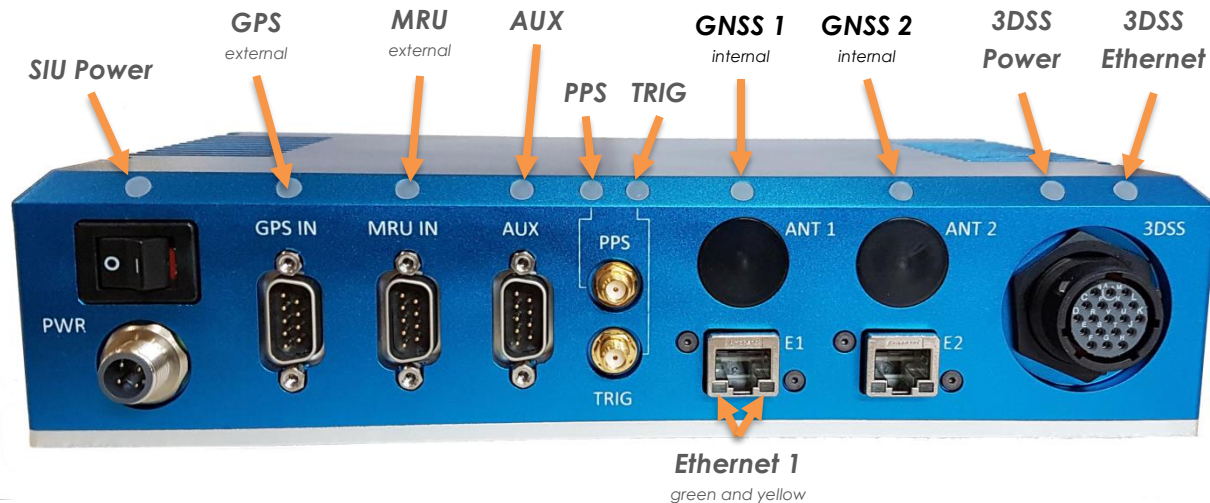
The two antennas must be connected with the same coaxial cable of the same length. The connectors on the coaxial cable should be IP-65 rated or better. The antenna connectors should be fitted with the provided IP-65 caps when the antennas are not connected.

ETHERNET PORTS

The SIU has two RJ-45 Ethernet ports that are connected to an internal Ethernet switch. Both ports have the same functionality and either can be used to operate the sonar. In order to maintain the IP-65 rating, a connector with a sealing boot must be used (PN: MUSBR-AHD2-241SK). Any Ethernet ports that are not in use should be fitted with the provided rubber caps.

LED INDICATORS

Below is a summary of the meaning of all LED indicators on the SIU:



LED	STATE	MEANING
SIU Power	Off Yellow Green	SIU off – no power. SIU firmware update. SIU on.
GPS (external) or GNSS 1 (internal)	Off Red Red (blinking) Yellow Green	No external NMEA-0183 detected. Invalid input data. GPS no fix. Incomplete GPS data. GPS fix and is outputting all required data.
MRU (external)	Off Red Green	No external MRU detected. Invalid input data. MRU TSS1 or NMEA detected.
AUX	Off	Not currently implemented.
PPS	Off Red Yellow Green	No PPS detected. Unstable or bad PPS. Stable PPS – sonar unlocked. Stable PPS – sonar locked.
TRIG	Off	Not currently implemented.
Ethernet 1 & 2	Off Green Yellow	No Ethernet link. Ethernet link up. Ethernet data active.
3DSS Power	Off Red Green	Sonar off. No sonar – power issue. Sonar on.
3DSS Ethernet	Off Green Green (flashing)	No Ethernet link. Ethernet link up. Ethernet data active.



3DSS-IDX INTEGRATED SEPTENTRIO ASTERX-4 GNSS (OPTIONAL)

The 3DSS-IDX SIU can come equipped with an integrated Septentrio Asterx-4 GNSS. The GNSS is connected to the internal Ethernet switch at can be accessed using a web browser.

Septentrio IP Address: 192.168.228.91
Septentrio Web Interface: <http://192.168.228.91> or <http://asterx4-serial/>
where <serial> is the serial number of the Asterx-4 board
and is available in the quick start guide

The Septentrio is preconfigured to output the necessary NMEA-0183 sentences to iDX the sonar head, as well as a set of binary messages over Ethernet UDP/IP. The binary SBF messages can be recorded to file using the 3DSS-IDX Binary Logger, and are required for the post-processing of position using SBG's Qinertia PPK software package.

ASTERX-4 SPECIFICATIONS

GNSS Technology

544 hardware channels for simultaneous tracking of all visible satellite signals

Supported signals:

- ▶ GPS: L1, L2, L5
- ▶ GLONASS: L1, L2, L3
- ▶ Galileo: E1, E5ab, AltBoc, E6¹
- ▶ BeiDou: B1, B2, B3¹
- ▶ SBAS: EGNOS, WAAS, GAGAN, MSAS, SDCM (L1, L5)
- ▶ IRNSS: L5¹
- ▶ QZSS: L1, L2, L5, L6

Septentrio's patented GNSS+ technologies:

- ▶ **AIM+** interference mitigation unit against narrow system against narrow and wideband interference with spectrum analyser
- ▶ **IONO+** advanced scintillation mitigation
- ▶ **APME+** a posteriori multipath estimator for code and phase multipath mitigation.
- ▶ **RAIM** (Receiver Autonomous Integrity Monitoring)

RTK (base and rover)¹

Integrated dual-channel L-band receiver
PPP (SECORX and VERIPOS services)^{1,2}

Moving base^{1,3}

Heading GNSS attitude¹

Position Accuracy^{4,5}

	Horizontal	Vertical
Standalone	1.2 m	1.9 m
SBAS	0.6 m	0.8 m
DGNSS	0.4 m	0.7 m
SECORX-D ⁶	6 cm	9 cm
SECORX-C ⁶	4 cm	6 cm
SECORX-60	4 cm	6 cm

RTK Performance^{4,5}

Horizontal accuracy	0.6 cm + 0.5 ppm
Vertical accuracy	1 cm + 1 ppm
Initialisation	7 s

GNSS attitude accuracy^{4,5,7,8}

	Heading	Pitch/Roll
Antenna separation		
1 m	0.15°	0.25°
5 m	0.03°	0.05°

Velocity accuracy^{4,5}

	0.03 m/s
--	----------

Maximum Update Rate

Position	100 Hz
----------	--------

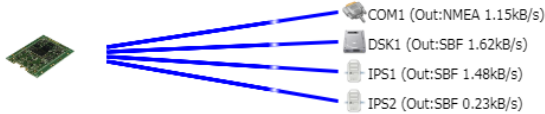
PROVIDING CORRECTIONS

RTCM or CMR corrections can be supplied to the Septentrio using the AUX port on the SIU. In order to do this, the Septentrio's COM2 input port must be configured using its web interface. Under CORRECTIONS > CORRECTIONS INPUT, COM2 must be set to RTCMv2, RTCMv3 or CMRv2.

Overview GNSS TERRASTAR Communication **Corrections** NMEA/SBF Out Logging Admin

Corrections > Corrections Input

Data Streams



COM1 (Out:NMEA 1.15kB/s)
DSK1 (Out:SBF 1.62kB/s)
IPS1 (Out:SBF 1.48kB/s)
IPS2 (Out:SBF 0.23kB/s)

Input Streams

Input	
COM1	CMD
COM2	auto
COM3	none
COM4	CMD
USB1	RTCMv3
USB2	CMRv2
IP10	DC1
IP11	DC2
IP12	RTCMV
	auto

And under COMMUNICATION > SERIAL PORT, the proper baud-rate must be selected for COM2.

Overview GNSS TERRASTAR **Communication** Corrections NMEA/SBF Out Logging Admin

Communication > Serial Port

COM Port Settings

	COM1	COM2	COM3	COM4
Baud rate	38400 baud	115200 baud	115200 baud	115200 baud
Data bits	8 bits	1200 baud	8 bits	8 bits
Parity	No	2400 baud	No	No
Stop bits	1 bit	4800 baud	1 bit	1 bit
Flow control	none	9600 baud	none	none
		19200 baud		
		38400 baud		
		57600 baud		
		115200 baud		
		230400 baud		
		460800 baud		

Default Ok

The settings must be saved under ADMIN > CONFIGURATIONS, by copying the CURRENT configuration to the BOOT configuration.

Overview GNSS TERRASTAR Communication Corrections NMEA/SBF Out Logging **Admin**

Admin > Configurations

1 Copy Configuration File

2 Source: Current Target: Boot

Default Ok 3 Press "OK" to apply the changes.

Receiver Configurations

Current	Different from factory default	✓	⬆
Boot	Different from factory default	✓	⬆
User1	Different from factory default	✓	⬆
User2	Equal to factory default		⬆



TERRASTAR / SECORX (PPP)

The Septentrio Asterx-4 receiver is fully compatible with TERRASTAR services. SECORX is a subscription based service providing over-the-air corrections for Precise Point Positioning (PPP) without the need for a local base station.

For further information please visit:

<https://www.septentrio.com/products/correction-services/secorx>

Configuring your receiver for SECORX positioning involves the following steps:

1. Contact your local Septentrio sales office or representative, and sign-up for the SECORX-60, SECORX-C, or SECORX-D subscription service. You will need the Product Activation Code and Augmentation User ID for your receiver – this information is shown in the receiver's web user interface under TERRASTAR.
2. Using the Septentrio receiver's web user interface (<http://192.168.228.91/>). Make sure that the PPP positioning mode is enabled on your receiver. Under **GNSS > POSITION, POSITION MODE** – the **PPP** box must be checked. It is checked by default.
3. Special care must be taken if PPP and RTK are used at the same time. The benefit of this configuration is that PPP can be used during RTK outages, so that the receiver continues to output high accuracy positions.

RTK positions are typically expressed in a regional datum which depends on your local RTK provider. Instead, SECORX-D PPP positions relate to the global ITRF2008 reference frame. To avoid coordinate jumps each time the PVT engine switches between RTK and PPP modes, and to ensure accurate seeding of the PPP engine from RTK, the regional datum used by your RTK provider must be provided to the receiver. For example, in Europe, the RTK datum will often be ETRS89, and you will need to change the receiver's datum to match under **GNSS > POSITION, GEODETIC DATUM**.

4. Enable the automatic seeding of the PPP engine with RTK positions. Check the **RTKFIXED** box under **GNSS > POSITIONS, PPP SEEDING MODE**.



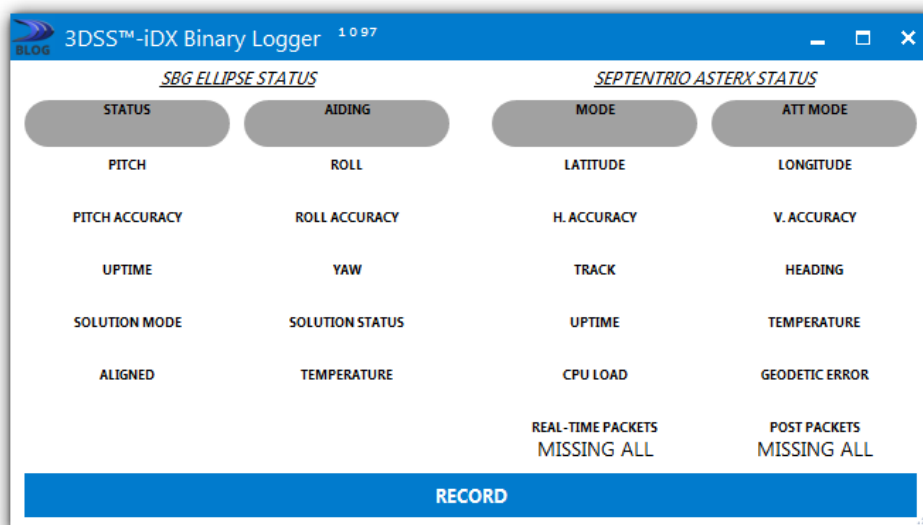
RECORDING BINARY ELLIPSE INS DATA AND BINARY GNSS DATA

The raw binary INS readings from the Ellipse-2E or Ekinox unit can be recorded using the 3DSS-iDX Binary Logger application. This data is required for post-processing in Qineria, and can be very useful for debugging any issues with the Ellipse-2E or Ekinox INS (such as incorrect lever arms). It is highly recommended that 3DSS-iDX users always record this binary data.

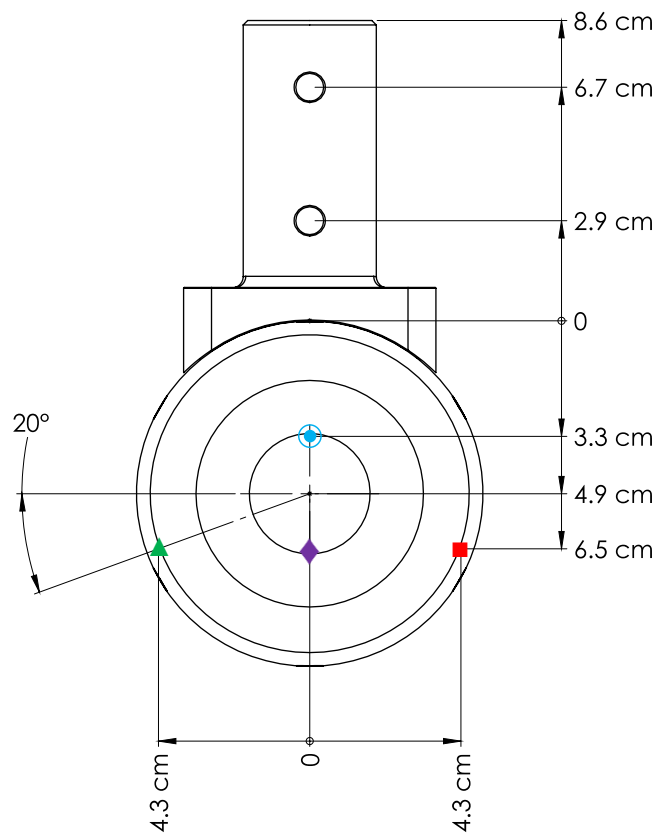
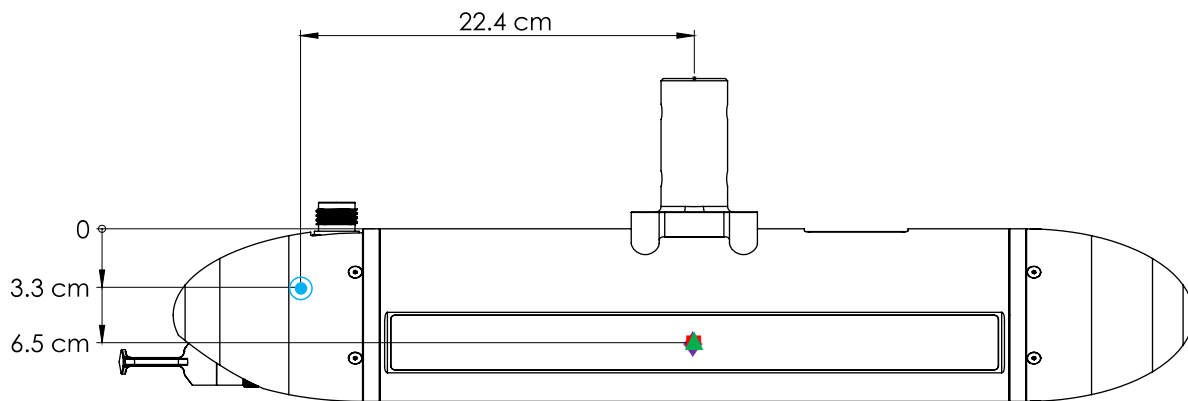
The 3DSS-iDX Binary Logger also records raw GNSS observations from the internal Septentrio GNSS receiver (*optional*). The GNSS raw observations are also needed by Qineria to perform tightly-coupled PPK computations, and output a post-processed SBET file for improved position and attitude. Loosely coupled solutions do not require raw observations; however, the result will not approach the accuracy of RTK / PPK solutions.

If an external GPS is being used with the 3DSS-iDX, the user must record the raw observables external to the 3DSS-iDX software suite in a format that is compatible with Qineria.

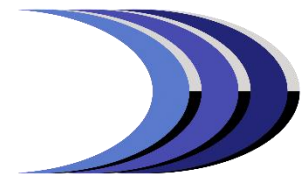
To record binary Ellipse-2E or Ekinox INS data and internal Septentrio GNSS raw observables; install and start the 3DSS-iDX Binary Logger application, click Record, and select a folder in which to record the files.



The files will be named using the date and time with the extension .bin for Ellipse-2E or Ekinox data, and extension .sbf for the Septentrio data.



- The position of the center of measurement for the Ellipse-2 or Ekinox.
- The acoustic center of the port transducer.
- ▲ The acoustic center of the starboard transducer.
- ◆ The reference for INS EKF processed positions output by the Ellipse-2 or Ekinox INS



CHAPTER 3: SONAR OPERATION



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SOFTWARE QUICK START

After installation of the **3DSS-DX Sonar Control** program and the **Sidescan3D™** display program, operation of the sonar can be simulated using a previously recorded data file. The following is intended to help get acquainted with the **3DSS-DX-450** software without requiring full sonar operation. This guide will also provide steps to run the optional 3DSS™-DX Chart Display and 3DSS™ Profile Display.



FIGURE 24. SOFTWARE QUICK START 3DSS-DX SONAR CONTROL PROGRAM SETTINGS.



CONFIGURING THE 3DSS™-DX SONAR CONTROL PROGRAM

1. Start the **3DSS-DX Sonar Control** program and maximize the program window.
2. Click on the words **FILE PROCESS** near the top of the GUI.
3. Under the **GAIN** heading set **LOG=30**, **LINEAR=0.2**, **CONSTANT =0** for both port and starboard by clicking on each setting and entering the value. Set the other parameters on the GUI to the settings shown in Figure 24.
4. Open the sample data file entitled **3DSS-DX Demo Data (Sidney_9).3dss-dx** (please see Chapter 1 for downloading instructions) by clicking on the **OPEN** button near the bottom left corner of the display and selecting the file.
5. Commit the settings displayed in the GUI by clicking on the **COMMIT** button beside the **OPEN** button.
6. Click on the up/down arrows beside the **COMMIT** button to set the processing rate to 1x (i.e. same as real-time).

CONFIGURING DISPLAY APPLICATIONS

1. Start the **Sidescan3D™** display program and size the window as desired.
2. On the sidebar menu under the heading **RENDER OPTIONS** click on **FOLLOW SONAR** and **ANTI-ALIASING** to turn both switches to the on position. Set the other parameters on the sidebar menu as shown in Figure 25. Note that slider settings are changed by clicking and dragging the slider and it is only necessary to set the parameters close to the settings in Figure 25.

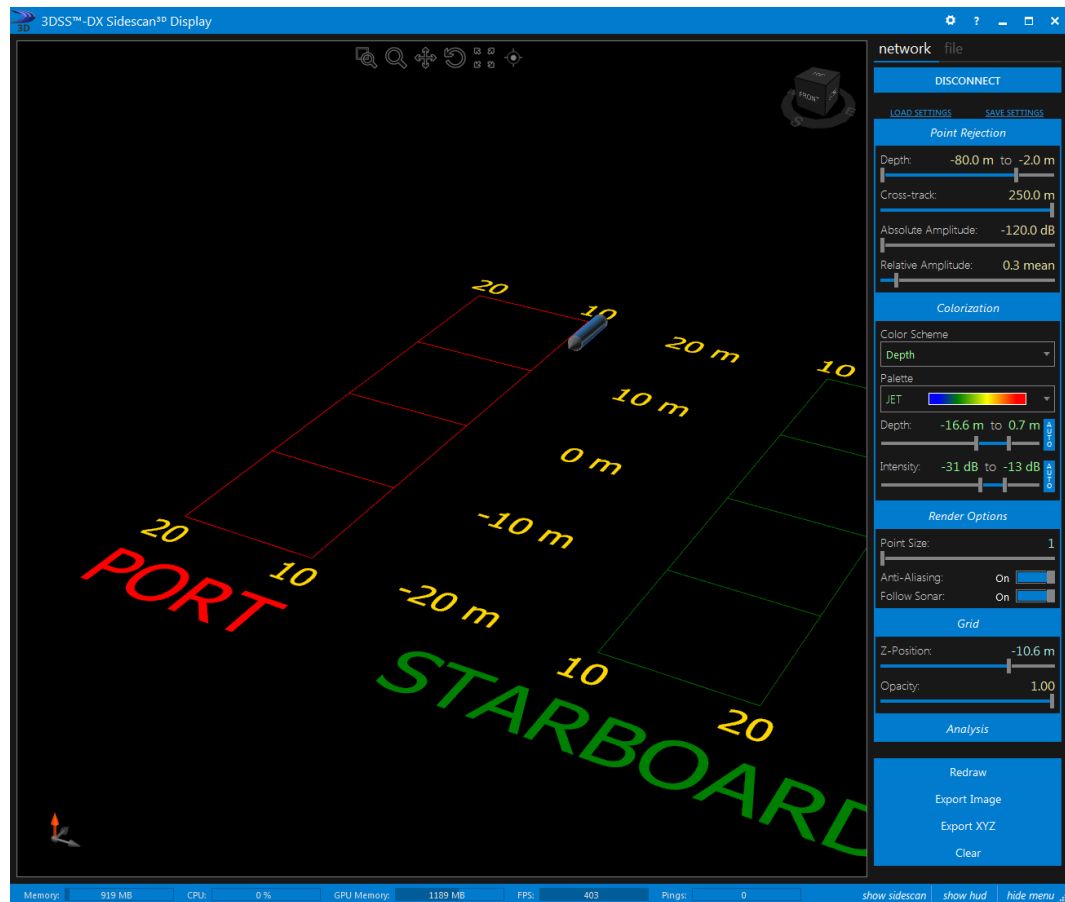


FIGURE 25. SOFTWARE QUICK START SIDESCAN3D DISPLAY SETTINGS.

3. At the top of the **Sidescan3D™** program sidebar menu click on **NETWORK** to make sure it is selected (i.e. underlined in blue).
4. Below **NETWORK** click on **CONNECT** and see it change its label to **DISCONNECT**. The **Sidescan3D™** display program is now connected to **the 3DSS-DX Sonar Control** program.
5. (Optional) Start the 3DSS™ Chart Display Software by first running OpenCPN. If the 3DSS™ Chart Display installation instructions were followed an icon should appear in the OpenCPN toolbar displaying the PingDSP logo. Clicking on this icon brings up a user interface for the 3DSS™ Chart Display. Press "Start".

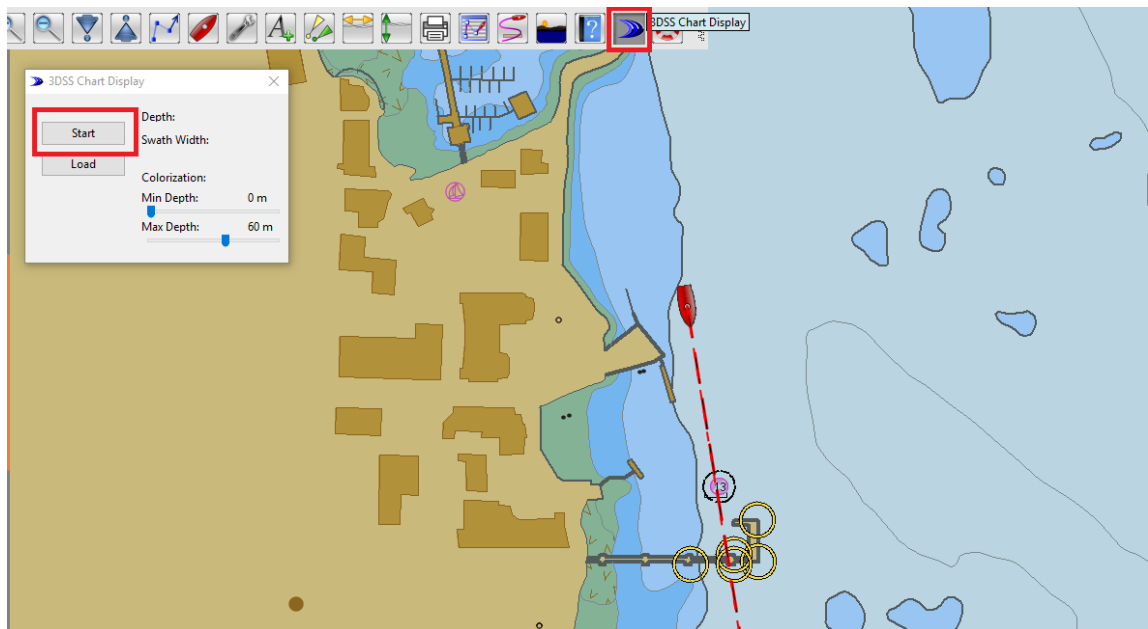


FIGURE 26- 3DSS™ CHART DISPLAY

6. (Optional) Start the 3DSS™-DX Profile Display. Press on the “Connect” button on the top right hand side of the 3DSS™-DX Profile Display GUI. The button should turn green indicating that it is connected to the 3DSS™ Control Software.

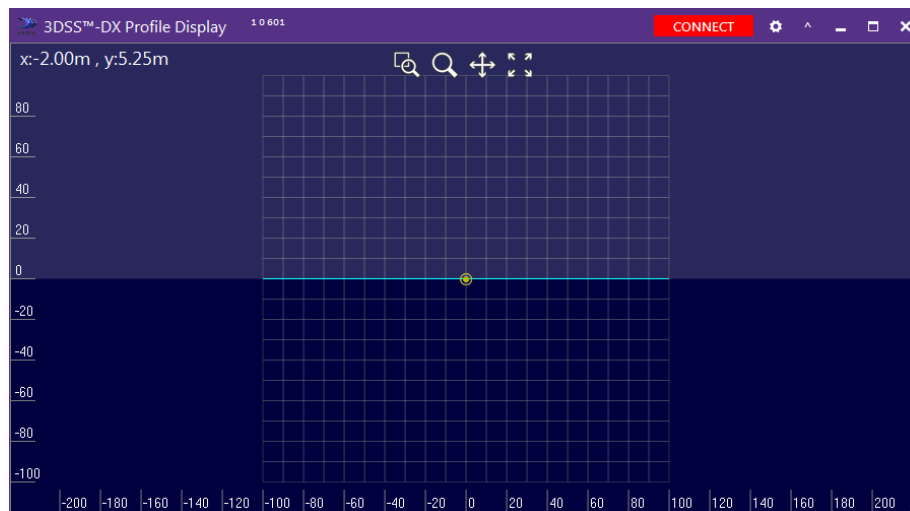


FIGURE 27 - 3DSS™-DX PROFILE DISPLAY

PLAYING BACK DATA

1. In the **3DSS-DX Sonar Control** program window click on the **PLAY** button to begin the simulation. The status display at the bottom of the window will flash status updates with each ping. In the **Sidescan3D™** window the sonar icon will begin moving forward and cross-sections of the seafloor and water-column will appear along the sonar track. The data is shown as a 3D point cloud colored according to depth.
2. Click on the **Sidescan3D™** display window near the top to select the pan tool  and then use the left mouse button to click and drag the display, the mouse wheel to zoom in or out, and clicking and dragging with the right mouse button to rotate the display.
3. Under the **COLORIZATION** heading change the **COLOR SCHEME** to **INTENSITY** and the **PALETTE** to **KAMELO**. The display is recolored to show the 3D point cloud colored by acoustic intensity (i.e. true 3D sidescan).
4. Try adjusting the **INTENSITY** to change the brightness and contrast of the point cloud.
5. Switch back to the **DEPTH** color scheme and play with the **DEPTH** sliders to vary the range of colours that span the bathymetry.
6. Experiment with all other settings in the sidebar and see Chapter 4 for detailed descriptions of the **Sidescan3D™** display controls and settings.
7. (Optional) If open, click on the **3DSS™-DX Profile Display** window. This shows the backscatter profile on a ping by ping basis. Like the **Sidescan3D™** display, it is possible to pan and zoom. Each is represented in meters with the y axis being height relative to the sonar, and the x axis being cross track distance from the sonar. Return from the starboard is colored in green and port colored in red. In Legacy Mode, the binned bottom track filter is made visible by the two white lines encompassing the bottom, while in MBES Mode the equidistant and equiangular sectors are shown.
8. (Optional) If open, click on the **OpenCPN** window. While the 3DSS data has been playing back in the 3DSS™-DX Control Software, the 3DSS™ Chart Display Software has been overlaying the bathymetry data onto a chart in OpenCPN. This overlay is colored by depth and can be adjusted with the “Min Depth” and “Max Depth” sliders.



SONAR SYSTEM QUICK START

The sonar system is ready for use after the **3DSS-DX-450** sonar and Topside Box have been mounted and connected as per Chapter 1 and both the **3DSS-DX Sonar Control** program and the **Sidescan3D™** display program are running on the Host computer.

The following is intended to help get quickly acquainted with the **3DSS-DX-450** sonar.

1. Power up the sonar by pressing the power on switch on the Topside Box. A blue LED ring will illuminate around the switch. Wait for 1 minute to allow the sonar to power up completely.
2. Start the **3DSS-DX Sonar Control** program and click on the word SONAR near the top of the GUI.
3. Click the CONNECT button in the bottom left corner of the GUI. The button will change to read DISCONNECT once a connection to the sonar is established (a few seconds).
4. If you are using the advanced GUI, under the GAIN heading set LOG=30, LINEAR=0.2, CONSTANT=0 for both port and starboard by clicking on each setting and entering the value. Set the other parameters on the GUI to the settings shown in Figure 28. If you are using the normal GUI, these parameters are set for you. Simply set the acquisition parameters to your preference.
5. Commit the sonar settings displayed in the GUI by clicking on the COMMIT button at the bottom of the GUI. Also click on the UPDATE TIME button to set the sonar clock to match the Host computer clock or GPS time if an optional GPS is connected as described in Chapter 1.
6. Start the **Sidescan3D™** display program and size the window as desired.
7. On the sidebar menu under the heading RENDER OPTIONS click on FOLLOW SONAR and ANTI-ALIASING to turn both switches to the on position. Set the other parameters on the menu as shown in Figure 29. Note that slider settings are changed by clicking and dragging the slider and it is only necessary to set the parameters close to the settings in Figure 29.
8. At the top of the **Sidescan3D™** program sidebar menu, click on NETWORK to make sure it is selected (i.e. underlined in blue).
9. Below NETWORK click on CONNECT and see it change its label to DISCONNECT.



3DSS-DX and 3DSS-iDX Sonar Manual

VERSION 2.3
APRIL 18TH, 2024



Default GUI

Advanced GUI

FIGURE 28. SONAR QUICK START CONTROL PROGRAM SETTINGS.

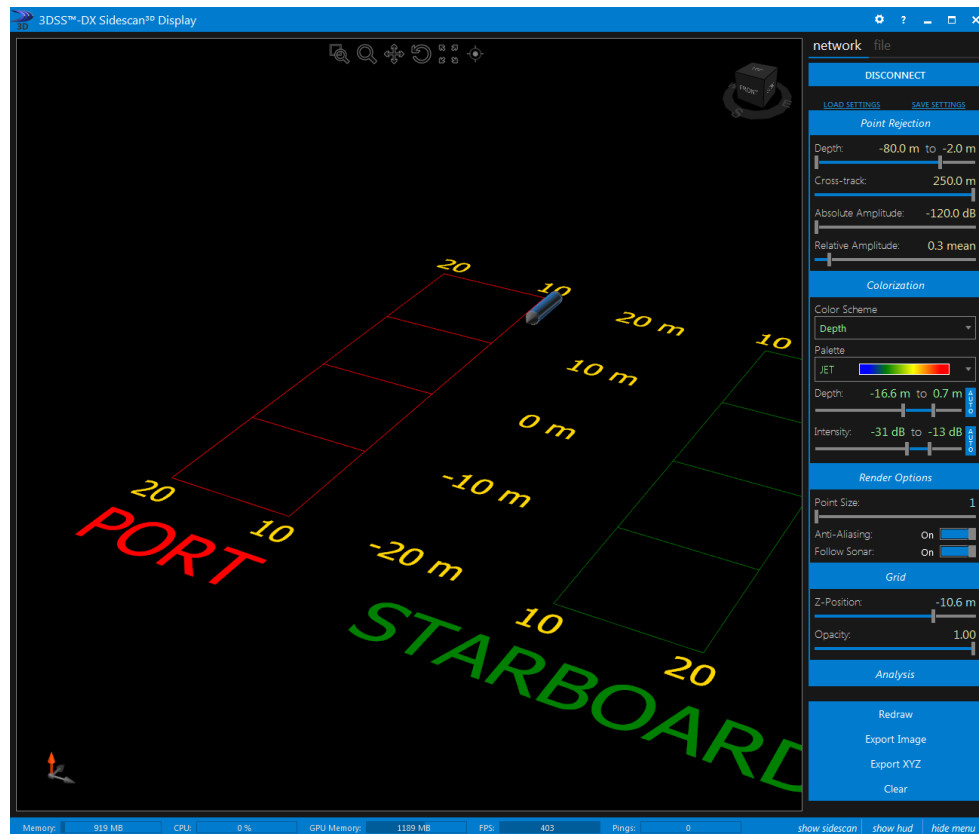


FIGURE 29. SONAR SYSTEM QUICK START SIDESCAN3D DISPLAY SETTINGS.

10. Back in the **3DSS-DX Sonar Control** program window click on the **PLAY** button to begin pinging. The status display at the bottom of the window will flash updates with each ping. In the **Sidescan3D™** window the sonar icon will begin moving forward and cross-sections of the seafloor and water-column will appear along the sonar track and the 3D point cloud will begin evolving. With the sonar moving forward, the ping-to-ping cross-sections will change with the bathymetry and water column objects to either side of the sonar. For these display settings the 3D point cloud is colorized according to depth.
11. Halfway up the **Sidescan3D™** sidebar click the **AUTO** button beside the depth slider. The slider will be automatically adjusted to span the depth range for the currently displayed data.
12. Adjust both ends of the **DEPTH** slider to fine tune the point cloud coloring by depth as desired.

13. Click on the **Sidescan3D™** display window near the top to select the pan tool  and then use the left mouse button to click and drag the display, the mouse wheel to zoom in or out, and clicking and dragging with the right mouse button to rotate the display.
14. Click on the SHOW SIDESCAN label near the bottom right hand corner of the window to enable the 2D sidescan display.
15. On the 2D sidescan display sidebar menu, click on the follow sonar switch to turn it on.
- Also click on the zoom-to-fit tool  at the top of the display.
16. Adjust both ends of the INTENSITY slider to achieve good image brightness and contrast on the 2D sidescan display.
17. Back on the 3D sidescan display, under the COLORIZATION heading, change the COLOR SCHEME to INTENSITY and the PALETTE to BBY. Also adjust the INTENSITY sliders to approximately match the slider values from the 2D sidescan display and then click REDRAW near the bottom of the sidebar. The 3D point cloud display is recolored to show acoustic intensity (i.e. true 3D sidescan).
18. Try adjusting the INTENSITY sliders followed by a REDRAW to change the brightness and contrast of the point cloud.
19. Change the color scheme to DEPTH HDR and press REDRAW again. The 3D point cloud is now colorized by a combination of both depth and intensity, with stronger more intense backscatter contributing more to the display than weaker returns.
20. Adjust the depth of the display grid using the Z-POSITION slider on the **Sidescan3D™** program sidebar. The grid can be depth aligned with a feature of interest on the seafloor in order to make accurate position and size estimates.
21. Please see below for more information regarding the sonar settings in the **3DSS-DX Sonar Control** program and Chapter 3 for more information regarding the display settings in the **Sidescan3D™** real-time display program.

3DSS-DX SONAR CONTROL PROGRAM

The **3DSS-Sonar Control** program provides a GUI for setting all sonar operating parameters as well as a data server to provide data to display programs such as the **Sidescan3D™** point cloud display or third-party acquisition software.

GUI LAYOUT

The **3DSS-DX Sonar Control** program advanced GUI consists of 7 general sections as shown in Figure 30. The default GUI will show a reduced set of parameters with sections consisting of Acquisition, Sound Velocity, Bathymetry, and Status.

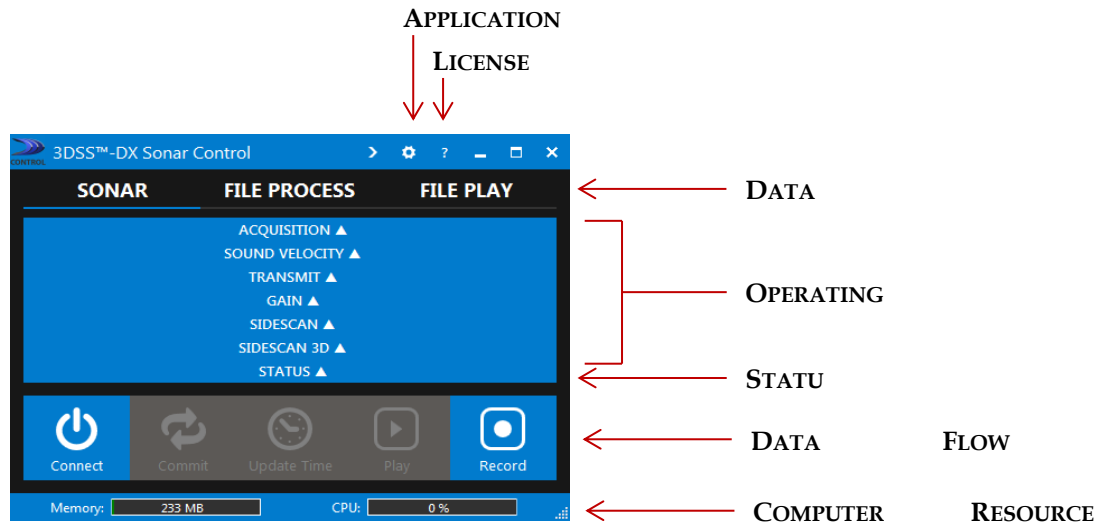


FIGURE 30. 3DSS-DX CONTROL APPLICATION GUI PANEL WITH ALL OPERATING PARAMETER SECTIONS COLLAPSED.

In addition, each of the headings in the OPERATING PARAMETERS section may be expanded by clicking to reveal a set of parameters specific to each heading. The small triangle beside each heading label indicates whether the heading is currently expanded or collapsed. Similarly, the STATUS section may also be expanded or collapsed.

For the three possible selections of DATA MODE, the current selection is indicated with a blue underline. Clicking on an alternative selection will change the selection.

For the buttons in the DATA FLOW section, buttons that are not currently available are grayed out. Clicking on an available button will invoke a data flow action. For example, if the DATA MODE is set to SONAR, clicking on the PLAY button will invoke sonar pinging and data collection.



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APRIL 18TH, 2024

Clicking on symbols on the title bar will open dialog boxes for APPLICATION CONFIGURATION and LICENSE INFORMATION. The size of the GUI panel in can be adjusted in the vertical direction by clicking and dragging the lower right corner of the panel. The fully expanded GUI panels are shown in Figure 31.



FIGURE 31. FULLY EXPANDED 3DSS-DX SONAR CONTROL PROGRAM GUI PANEL WITH EXAMPLE OPERATING AND STATUS PARAMETERS. DEFAULT GUI IS SHOWN ON THE LEFT. ADVANCED IS SHOWN ON THE RIGHT.

LICENSE INFORMATION AND APPLICATION CONFIGURATION

Clicking on the question mark in the title bar displays the LICENSE INFORMATION dialog box showing the version of the software as well as the license status and terms. If a license has not yet been obtained the dialog box provides a button for requesting a license as described in Chapter 1.

Clicking on the  symbol in the title bar opens the APPLICATION CONFIGURATION dialog box as shown in Figure 32.

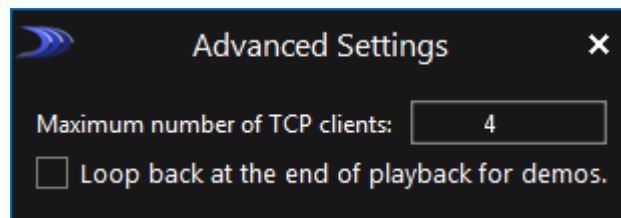


FIGURE 32. 3DSS-DX CONTROL APPLICATION CONFIGURATION DIALOG BOX.

MAXIMUM NUMBER OF TCP CLIENTS

Represents the maximum number of TCP clients that can be simultaneously connected to the **3DSS-DX Sonar Control** program and receive a copy of the data stream. For example, one client may be the Ping DSP **Sidescan3D™** point cloud display software running on the same computer. Another client may be third party acquisition software running on a second computer. As many as 8 simultaneous clients can be connected.

LOOPBACK AT END OF PLAYBACK FOR DEMOS

This checkbox is only used when the DATA MODE is set to FILE PROCESS or FILE PLAY. When enabled, the loop back feature automatically restarts data output from the beginning of file when the end of file is reached.



ENABLE ADVANCED SETTINGS IN THE GUI

This checkbox is used to toggle between the simple GUI and advanced GUI in Sonar Mode.

RECORDING SETTINGS

These radio buttons determine how **3DSS-DX Sonar Control** records data. The default option records both reprocessing and playback data to the same file. The second option will record playback data only. This will greatly reduce file size, but you will not be able to use file process mode on these playback-data only files. The third option will record both reprocessing and playback data to separate files. Unless specified otherwise by the user, playback data only files will have "NoRaw" prepended to them.

STARTUP PROGRAMS

Checking these boxes will ensure that these programs startup alongside **3DSS-DX Sonar Control**. Please note that these programs need to be in their default installation location for this feature to be functional.

DATA MODE

DATA MODE allows selection of the data source (sonar or file) and, in the case of file, whether the data is replayed or reprocessed. A selection is made by clicking on the desired DATA MODE and the current selection is indicated with a blue underline.

Note that the DATA MODE cannot be changed without either first releasing the connected sonar (i.e. pressing the DISCONNECT button) or closing an open data file. DATA MODE selections are grayed out when the selection cannot be changed.

SONAR

Selects the data source as real-time acquisition from the **3DSS-DX-450** sonar.

FILE PROCESS

Selects previously recorded data as the source and specifies that new OPERATING PARAMETER settings will be used for reprocessing the data. Computer resource requirements are greater than for FILE PLAY.

FILE PLAY

Selects previously recorded data as the source and specifies that the OPERATING PARAMETER settings will not be changed from the settings used at the time of recording. Computer resource requirements are less than for FILE PROCESS.

DATA FLOW CONTROL

The DATA FLOW CONTROL section provides a group of buttons for updating sonar and processing parameters that control data flow from the sonar or a file, and for recording data to a file. The function and availability of the buttons changes with the DATA MODE.

SONAR DATA FLOW CONTROL BUTTONS

Figure 33 shows the collapsed **3DSS-DX Sonar Control** program GUI and the DATA FLOW CONTROL buttons that are available when the DATA MODE is set to SONAR. Grayed out buttons require other operations before becoming active. For example, the PLAY button is grayed out until a connection is established with the sonar.

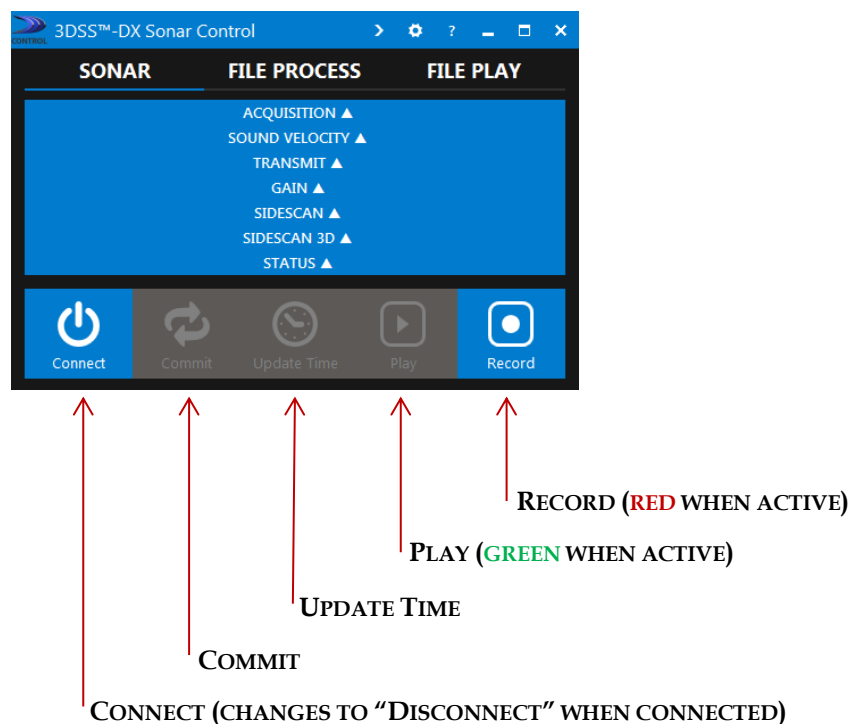


FIGURE 33. DATA FLOW CONTROL BUTTONS FOR SONAR DATA MODE.

CONNECT

Establishes a connection between the **3DSS-DX Sonar Control** program and the **3DSS-DX-450** Sonar. After the connection is established the button label changes to DISCONNECT to indicate that the connection was successful.



Pressing the DISCONNECT button releases the sonar from the connection. The sonar must be released before the DATA MODE setting can be changed. The sonar should also be released before it is powered off.

COMMIT

Updates the sonar with the current parameter settings from the GUI. If an operating parameter is changed in the GUI, please note that parameter changes will not affect sonar operation until the COMMIT button is pressed.

Also note that parameter settings can be changed on the fly during sonar operation by modifying the settings in the program GUI and pressing COMMIT. For some parameter changes, sonar pinging will pause briefly during the parameter update.

UPDATE TIME

Updates the **3DSS-DX-450** sonar with the current Host computer time or to GPS time (if a GPS is connected, please see Chapter 1) to within ~100ms. The sonar time must be updated each time the sonar is connected. After the initial sonar time update, it is only necessary to update again if it is important to maintain sonar and computer clock times to within a specified offset. For example, under typical operating conditions the sonar clock will drift from the Host computer clock by as much as 360ms per hour.

In applications with external sensors (e.g. DGPS or MRU) connected to the sonar through the Topside Box, actual sonar clock accuracy does not affect data synchronization since all data (i.e. sonar data and external sensor data) are time stamped by the same clock.

In applications where sonar data is to be time aligned with data from sensors collected outside of the sonar, the Topside Box can also accept an external PPS (Pulse Per Second) input from a GPS instrument. Please see Chapter 1 for further information regarding the PPS connection.

PLAY

Starts the sonar pinging (i.e. transmitting and receiving) with parameter settings from the most recent COMMIT. The PLAY button changes from blue to green to indicate that pinging is underway.

Note that parameter settings can be changed on the fly during sonar operation by modifying the settings in the program GUI and pressing COMMIT. For some parameter changes, sonar pinging will pause briefly during the parameter update.

RECORD

Opens a data file dialog box to allow specification of the file for saving the recorded data and then enables data recording. The RECORD button turns red to indicate that recording is enabled. Pressing the RECORD button when enabled, disables recording and closes the recording file.

Note that recording can be enabled before the PLAY button is pressed and data will not be written to the output file until the sonar begins pinging. Also note that if data flow is halted and then restarted (e.g. by disabling and then re-enabling PLAY), data will be appended to the same recording file as long as recording is still enabled.

FILE PROCESS DATA FLOW CONTROL BUTTONS

Figure 34 shows the collapsed **3DSS-DX Sonar Control** program GUI and the DATA FLOW CONTROL buttons that are available when the DATA MODE is set to FILE PROCESS. Grayed out buttons require other operations before becoming active. For example, the PLAY button is grayed out until a data file has been opened.

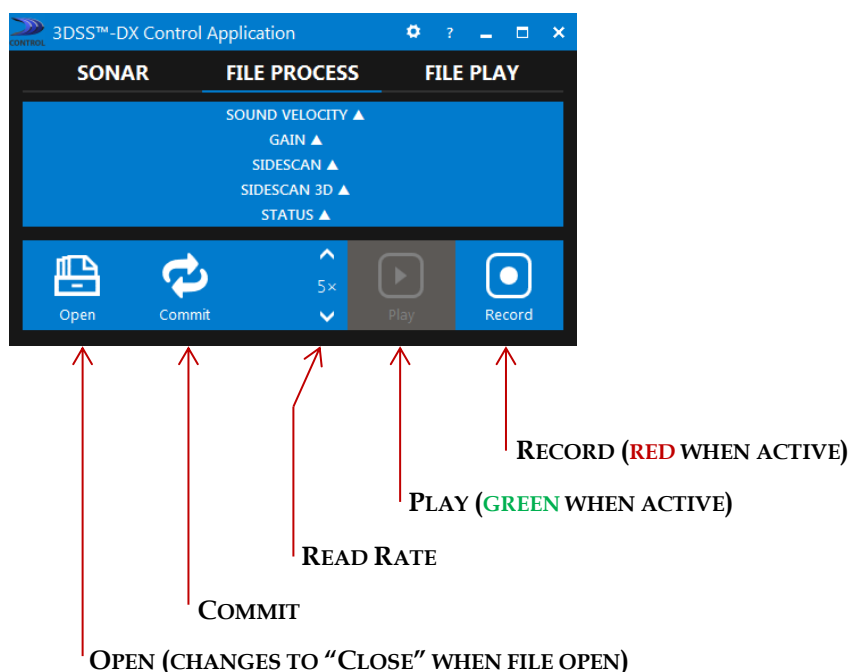


FIGURE 34. DATA FLOW CONTROL BUTTONS FOR FILE PROCESS DATA MODE.

OPEN

Opens a data file dialog box to select the file for reading. After the file has been opened the button label changes to CLOSE to indicate that the open operation was successful.



Pressing the CLOSE button closes the data file. The file must be closed before the DATA MODE setting can be changed.

COMMIT

Updates the sonar processor with the current parameter settings from the program GUI. If an operating parameter is changed in the GUI, please note that parameter changes will not affect sonar processing until the COMMIT button is pressed.

Also note that parameter settings can be changed on the fly during data read-back by modifying the settings in the program GUI and then pressing COMMIT.

READ RATE

This button includes both an UP and a DOWN arrow and allows the data reading rate from file to be increased or decreased relative to the ping rate at which it was recorded. For example, a READ RATE setting of 1x indicates that the reading rate from file is equal to the real time ping rate at the time the data was recorded. Values higher than 1x indicate faster than real time reading rates and values less than 1x indicate reading rates slower than real time.

PLAY

Starts reading data from file and reprocessing the data using the sonar parameter settings from the most recent COMMIT. The PLAY button changes from blue to green to indicate that data read-back is underway.

Note that parameter settings can be changed on the fly during data read-back by modifying the settings in the program GUI and then pressing COMMIT.

RECORD

Opens a data file dialog box to allow specification of the file for saving the recorded data and then enables data recording. The RECORD button turns red to indicate that recording is enabled. Pressing the RECORD button when enabled, disables recording and closes the recording file.

The file dialog box also has an option for starting a new file every x GB in a dropdown menu next to the Save / Cancel buttons which close the dialog box.

Note that recording can be enabled before the PLAY button is pressed and data is not written to the output file until reading from the input file begins. Also note that if data flow is halted and then

restarted (e.g. by disabling and then re-enabling play), data will be appended to the same recording file as long as recording is still enabled.

FILE PLAY DATA FLOW CONTROL BUTTONS

Figure 35 shows the fully collapsed **3DSS-DX Sonar Control** GUI with the DATA FLOW CONTROL buttons for the FILE PROCESS data mode. Grayed out buttons require other operations before becoming active. For example, the PLAY button is grayed out until a data file has been opened.

OPEN

Opens a data file dialog box to select the file for reading. After the file has been opened the button label changes to CLOSE to indicate that the open operation was successful.

Pressing the CLOSE button closes the data file. The file must be closed before the DATA MODE setting can be changed.

READ RATE

This button includes both an UP and a DOWN arrow and allows the data reading rate from file to be increased or decreased relative to the ping rate at which it was recorded. For example, a READ RATE setting of 1x indicates that the reading rate from file is equal to the real time ping rate at the time of recording. Values higher than 1x indicate faster than real time reading rates and values less than 1x indicate reading rates slower than real time.

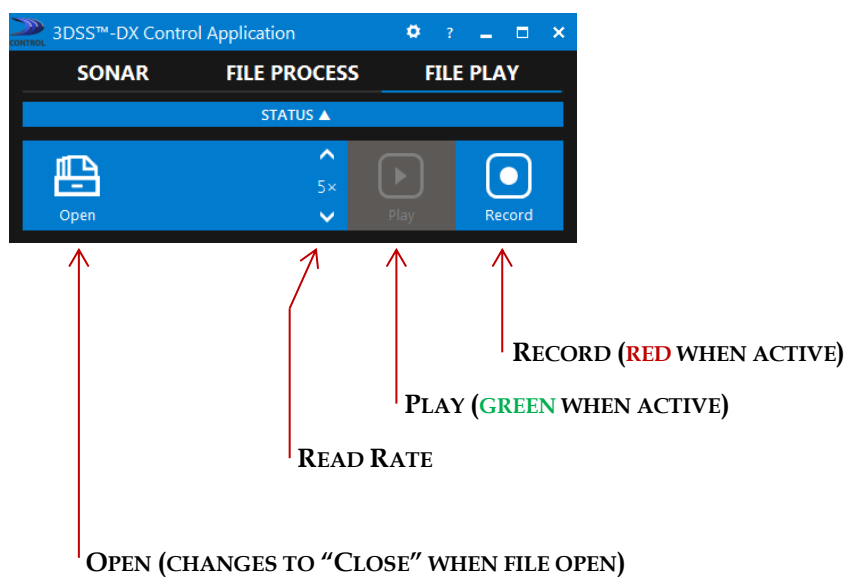


FIGURE 35. DATA FLOW CONTROL BUTTONS FOR FILE PLAY DATA MODE.



PLAY

Starts reading and replaying data from file without changing the processing parameters in effect at the time of recording. Processing settings cannot be changed in this data mode and are therefore not displayed. The PLAY button changes from blue to green to indicate that data read-back is underway.

RECORD

Opens a data file dialog box to allow specification of the file for saving the recorded data and then enables data recording. The RECORD button turns red to indicate that recording is enabled. Pressing the RECORD button when enabled, disables recording and closes the recording file.

Note that recording can be enabled before the PLAY button is pressed and data will not be written to the output file until reading from the input file begins. Also note that if data flow is halted and then restarted (e.g. by disabling and then re-enabling play), data will be appended to the same recording file as long as recording is still enabled.

OPERATING PARAMETERS

The OPERATING PARAMETERS section of the program GUI provides 4 or 6 expandable headings for specifying the **3DSS-DX-450** sonar and processing settings. Each heading can be expanded or collapsed by clicking within the blue heading bar.

Two of the headings, ACQUISITION and SOUND VELOCITY, include parameters that are used identically for both port and starboard sonar operation. The remaining 4 headings, TRANSMIT, GAIN, SIDESCAN, and SIDESCAN 3D, have independent settings for PORT and for STARBOARD.

Only relevant OPERATING PARAMETER headings are displayed for each DATA MODE. When DATA MODE is set to SONAR, any of the parameters can be set and all are displayed. When DATA MODE is set to FILE PROCESS, the ACQUISITION and TRANSMIT parameters cannot be changed and are therefore not displayed. When DATA MODE is set to FILE PLAY, none of the operating parameters can be changed and therefore none are displayed.

Depending on the type of parameter, settings are either made from a drop-down menu or by numeric entry. If the value of a numeric entry lies outside of the acceptable bounds for a parameter, the error is indicated with a thickened red frame around the parameter. An error message is displayed if an attempt is made to COMMIT an invalid parameter setting.

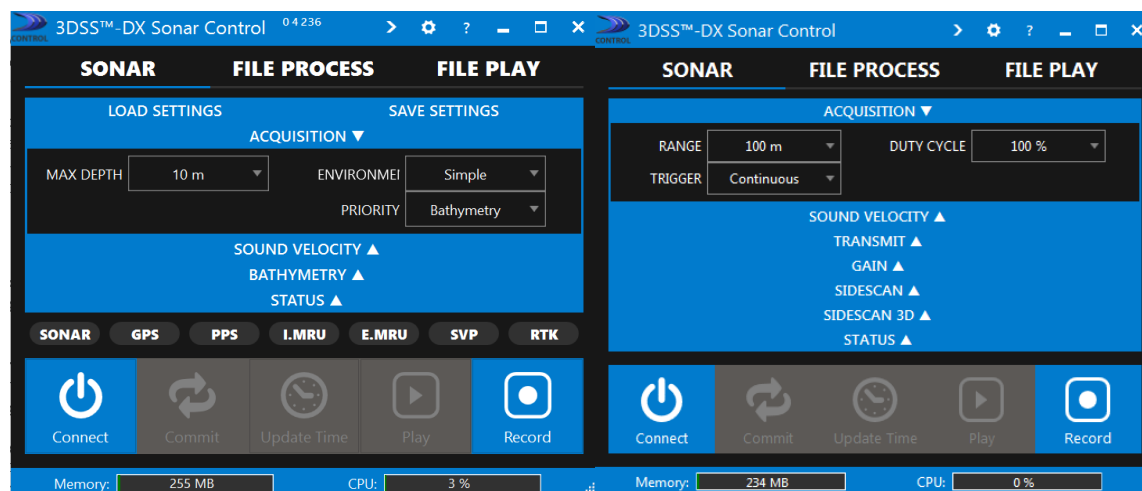


FIGURE 36. OPERATING PARAMETERS UNDER THE ACQUISITION HEADING. DEFAULT ON LEFT. ADVANCED ON RIGHT.

ACQUISITION PARAMETERS

Figure 36 shows both **3DSS-DX Sonar Control** program GUIs with all OPERATING PARAMETER headings collapsed except ACQUISITION parameters. These parameters are only available when the DATA MODE is set to SONAR.

SIMPLE GUI PARAMETERS

MAX DEPTH

Specifies the estimated expected maximum water depth at which the sonar will be operated. This may need to be adjusted if the depth range is large. Setting the maximum depth will automatically set the operating range based on what the PRIORITY is. Below is a table displaying these settings. To customise the range further, please use the advanced GUI.

BATHYMETRY

DEPTH (M)	RANGE (M)
3	20
5	25
7.5	37
10	50
15	75
25	100
35	125
50	125
75	125

WIDE AREA IMAGING

DEPTH (M)	RANGE (M)
3	37
5	50
7.5	75
10	100
15	125
25	150
35	150
50	150
75	200

HIGH-RES IMAGING

DEPTH (M)	RANGE (M)
3	15
5	20
7.5	25
10	37
15	50
25	75
35	100
50	125
75	125



ENVIRONMENT

Specifies the complexity of the environment being worked in. More specifically it will set the number of angles used in the Sidescan3D calculations. Simple will set it to 3, Complex will set it to 4 and should be used for vertical structure mapping or full water column mapping.

PRIORITY

Specifies the priority. The three options include Bathymetry, Wide Area Imagery, and High-Res Imagery. Each of will automatically adjust the Range, Transmit, and Sidescan3D parameters to best suit the desired option.

ADVANCED GUI PARAMETERS

RANGE

Specifies the data acquisition range of the sonar in meters. After transmit, the sonar acquires and processes data out to the operating range. The following settings are available on a drop down menu:

15m
20m
25m
37m
50m
75m
100m
125m
150m
200m
250m

To help achieve the widest useable swath width at the highest possible ping rate, Table 1 shows the ratio of sonar altitude to the swath width (port or starboard) at the different sonar range settings. As a general rule of thumb, good bathymetry results can typically be achieved out to 6 times the sonar altitude to port and to starboard (i.e. 12 times altitude in total swath width) up to a maximum of approximately 150m in total range (i.e. 150m range on port and starboard).

The highlighted entries Figure 37 indicate the recommended **RANGE** settings to achieve a balanced combination of reliable 3D sidescan or bathymetry, wide area coverage and maximum ping rate. The values in each cell show the cross-track distance for each side as a multiple of sonar altitude and the highlighted cells then indicate achievable ratios as a guide for setting the sonar **RANGE** at different operating altitudes. Choose the sonar altitude along the top of Figure 37 and then follow down the column to the highlighted cell that to find the recommended **RANGE** setting(s) in the leftmost column. For example, at an altitude of 12m, the recommended Range



setting is either 50m or 75m and the achievable swath width on each side is expected to lie between 4.0 and 6.2 times the 12m altitude for a total swath width of between 96m and 148.8m.

The actual achieved swath width is dependent on several factors including the acceptable 3D sidescan or bathymetric uncertainty, the seafloor geometry (e.g. sloping versus flat) and the seafloor type and scattering conditions (e.g. sand versus mud versus gravel etc).

	Altitude (m)																								
	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25		
Range=25m	8.3	6.2	4.9	4.0	3.4	3.0	2.6	2.3	2.0	1.8	1.6	1.5	1.3	1.2	1.1	1.0	0.9	0.8	0.6	0.5	0.4	0.3	0.2	0.1	0.0
Range=35m	12	8.7	6.9	5.4	4.4	3.8	3.3	3.0	2.7	2.4	2.2	2.0	1.8	1.6	1.5	1.3	1.2	1.1	1.0	0.8	0.7	0.6	0.5	0.4	0.3
Range=50m	17	12	10	8.3	7.1	6.2	5.4	4.8	4.3	3.9	3.6	3.3	3.0	2.8	2.5	2.3	2.1	2.0	1.8	1.6	1.5	1.3	1.2	1.1	1.0
Range=75m	25	19	15	12	11	9.9	8.8	7.9	7.1	6.4	5.8	5.3	4.8	4.4	4.0	3.7	3.4	3.1	2.9	2.6	2.4	2.2	2.0	1.9	1.7
Range=100m	33	25	20	17	14	12	11	10	9.1	8.2	7.4	6.7	6.1	5.6	5.1	4.7	4.3	4.0	3.7	3.4	3.1	2.9	2.7	2.5	2.3
Range=150m	50	37	30	25	21	19	17	15	14	12	11	10	9.1	8.2	7.4	6.7	6.1	5.6	5.1	4.7	4.3	4.0	3.7	3.4	3.1

FIGURE 37. SUGGESTED RANGE SETTINGS BY SONAR ALTITUDE.

It should also be noted that the achievable swath width for **3DSS-DX-450** 2D sidescan imagery is typically greater than for 3D sidescan and bathymetry because 2D sidescan does not require bathymetry measurements. 2D sidescan imagery for the **3DSS-DX-450** sonar can extend to more than 180m per side.

DUTY CYCLE

Specifies how quickly the sonar retransmits after the data has been acquired for each ping. The following settings are available on a drop down menu:

- 1%
- 10%
- 25%
- 50%
- 75%
- 100%

When the DUTY CYCLE is set to 100%, each transmit occurs as soon as possible after data has been acquired for the current operating range. For example, if the RANGE is set to 150m and DUTY CYCLE is set to 100%, then the ping rate will be approximately equal to 5Hz (e.g. (1500m/s)/(150m)/2). The exact ping rate will be slower since the sonar requires a brief quiescent interval between the



end of one ping and the start of the next ping. The exact ping rate is displayed in the status section of the **3DSS-DX Sonar Control** program (please see below).

A DUTY CYCLE of 100% sets the ping rate to maximum for the current RANGE setting. Setting the DUTY CYCLE to less than 100% reduces the ping rate proportionally. For example, a 50% duty cycle corresponds to a ping rate of one half the maximum achievable for the current RANGE setting.

In general, DUTY CYCLE is set to 100% unless operating in an environment that is highly reverberant. For example, the echoes in an enclosed environment may not decay sufficiently quickly and will overlap echoes from the previous ping for a DUTY CYCLE set to 100%.

TRIGGER

Specifies the trigger source for the sonar. The following settings are available from a drop down menu:

CONTINUOUS - the sonar continuously triggers automatically based on the RANGE and DUTY CYCLE parameter settings. CONTINUOUS triggering is the default setting.

EXTERNAL - the sonar is triggered by an external trigger signal input to the Topside Box (please see Chapter 1).

SOUND VELOCITY PARAMETERS

Figure 38 shows the **3DSS-DX Sonar Control** program GUI with all OPERATING PARAMETER headings collapsed except SOUND VELOCITY. These parameters are available when the DATA MODE is set to SONAR or FILE PROCESS

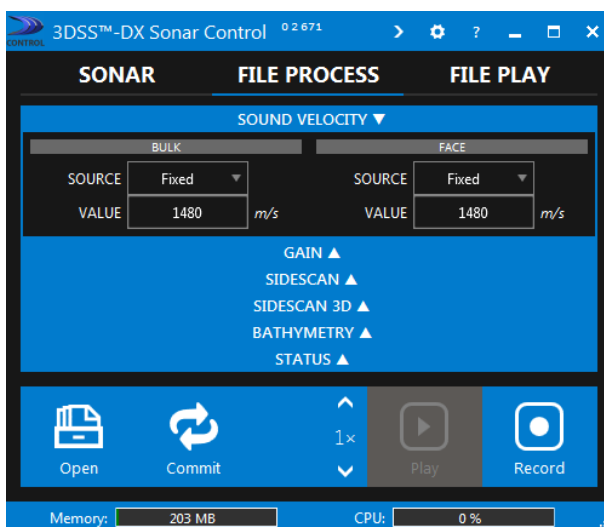


FIGURE 38. OPERATING PARAMETERS UNDER THE SOUND VELOCITY HEADING.

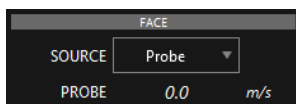
BULK

Specifies the average sound velocity over the path of the sonar pulse from transmit to received echo. Post processing corrections to this fixed value can be made using FILE PROCESS for recorded data or in third party software using a measured sound velocity profile. The BULK sound velocity setting is entered numerically after clicking on the current value. The allowable range of values is 1300m/s to 2500m/s. Currently only a fixed sound velocity is supported.

FACE

Specifies the sound velocity at the face of the sonar transducer. Corrections to this fixed value can be made using FILE PROCESS for recorded data or in third party software. A very close approximation of the FACE sound velocity can be obtained using the water temperature from the sonar (accurate to better than 0.1°C) together with an estimate of water salinity. Sound velocity can be computed to within 1% when salinity is known to within 5ppt. The FACE sound velocity setting is entered numerically. The allowable range of values is 1300m/s to 2500m/s.

The face sound velocity can also come from an external sound velocity probe, such as the AML Micro-X SV. To use an external sound velocity the SOURCE needs to be set to Probe and will show the sound velocity received from the SV probe:



Once COMMITTED the **3DSS-DX Sonar Control** program will listen for sound velocity updates from the **3DSS-DX Sound Velocity Broadcast** program which is discussed later in this manual.

TRANSMIT PARAMETERS

Figure 39 shows the **3DSS-DX Sonar Control** program GUI with all OPERATING PARAMETER headings collapsed except TRANSMIT parameters. These parameters are only available when the DATA MODE is set to SONAR.

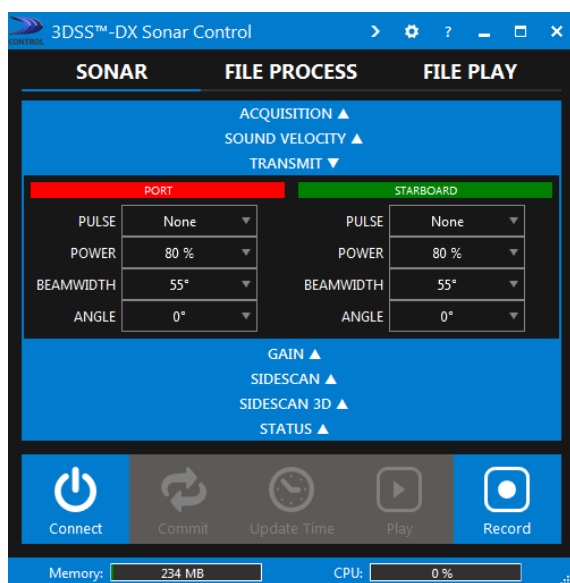


FIGURE 39. OPERATING PARAMETERS UNDER THE TRANSMIT HEADING.

PULSE

Specifies the sonar transmit pulse waveform. Waveforms are specified according to their type, NARROWBAND or BROADBAND together with a suffix indicating the approximate length of the pulse in cycles at 450kHz. For example, the NARROWBAND-15 setting specifies a continuous wave (CW) pulse of 15 cycles or approximately 33μs. The available settings are listed in a drop down menu and the following settings are possible:

NONE
NARROWBAND-10
NARROWBAND-15
NARROWBAND-25
NARROWBAND-50
BROADBAND-80
BROADBAND-200



However, not all PULSE settings are available at each RANGE setting. Figure 40 shows the available PULSE settings by range.

Pulse / Range	25 m	50 m	75 m	100 m	150 m	200 m	250 m
Narrowband-10	Available	Available	Available	Available	Available	Available	Available
Narrowband-15	Available	Available	Available	Available	Available	Available	Available
Narrowband-25	Available	Available	Available	Available	Available	Available	Available
Narrowband-50	Available	Available	Available	Available	Available	Available	Available
Broadband-80	No	Available	Available	Available	Available	Available	Available
Broadband-200	No	No	Available	Available	Available	Available	Available

FIGURE 40. AVAILABLE PULSE SETTINGS BY RANGE.

In general, the longer broadband pulses achieve greater range than the narrowband pulses, but also consume more power, are less effective in complex environments and have lower maximum ping rates. Waveforms of both types however, provide excellent spatial resolution. The NONE setting disables the transmitter and is useful for checking the acoustic noise environment.

	Operating Range	Spatial Resolution	Maximum Ping Rate	Complex Environments	Power Savings
NARROWBAND-10	C-	A+	A+	A+	A+
NARROWBAND-15	C-	B+	A+	A	A+
NARROWBAND-25	C	C+	A	A-	A



NARROWBAND-50	C+	C	B+	B-	B+
BROADBAND-80	B+	B+	B+	B-	B+
BROADBAND-200	A+	B+	C+	C	C+

FIGURE 41. PULSE WAVEFORM PERFORMANCE IN FIVE OPERATIONAL CATEGORIES.

POWER

Specifies the sonar transmit power as a percentage of maximum. The following settings are available on a drop down menu:

1%
2%
5%
10%
20%
50%
80%
100%

Typically transmit power is set to 80% or 100%. However, in very shallow water reducing the transmit power can be used to avoid clipping in the sonar receiver. Reducing transmit power also decreases the quiescent period between pings and therefore increases the maximum achievable ping rate at each range setting.

BEAMWIDTH

Specifies the sonar transmit beamwidth in degrees as shown in Figure 42. The following settings are available on a drop down menu:

19°
22°
30°
44°
55°
90°
125°

By default, the transmit **BEAMWIDTH** is set to 55°. In conjunction with the Angle parameter setting, the transmit **BEAMWIDTH** can be used to direct energy towards regions of interest and away from



other regions. For example, a narrower transmit **BEAMWIDTH** and a downward **ANGLE** setting can be used to direct more of the transmit energy towards the seafloor and away from the surface in order to minimize surface multipaths and to maximize the seafloor backscatter signal strength.

ANGLE

Specifies the sonar transmit beam angle in degrees up or down relative to the fixed transducer tilt angle of 20° as shown in Figure 42. The following settings are available on a drop down menu:

- UP 45°
- UP 30°
- UP 20°
- UP 15°
- UP 10°
- 0°
- DOWN 10°
- DOWN 15°
- DOWN 20°
- DOWN 30°
- DOWN 45°

Typically, the transmit **ANGLE** is set to Down 10° (i.e. the 20° installed transducer angle plus an extra 10° of electronic downward tilt is in effect). Steering the beam up or down from this setting can be used in conjunction with the transmit **BEAMWIDTH** setting to optimize system operation in different environments. For example, for a seafloor that is sloping down on port and up on starboard, the transmit beam **ANGLE** can be steered downward on port and upward on starboard in order to optimize acoustic energy distribution and maximize swath coverage.

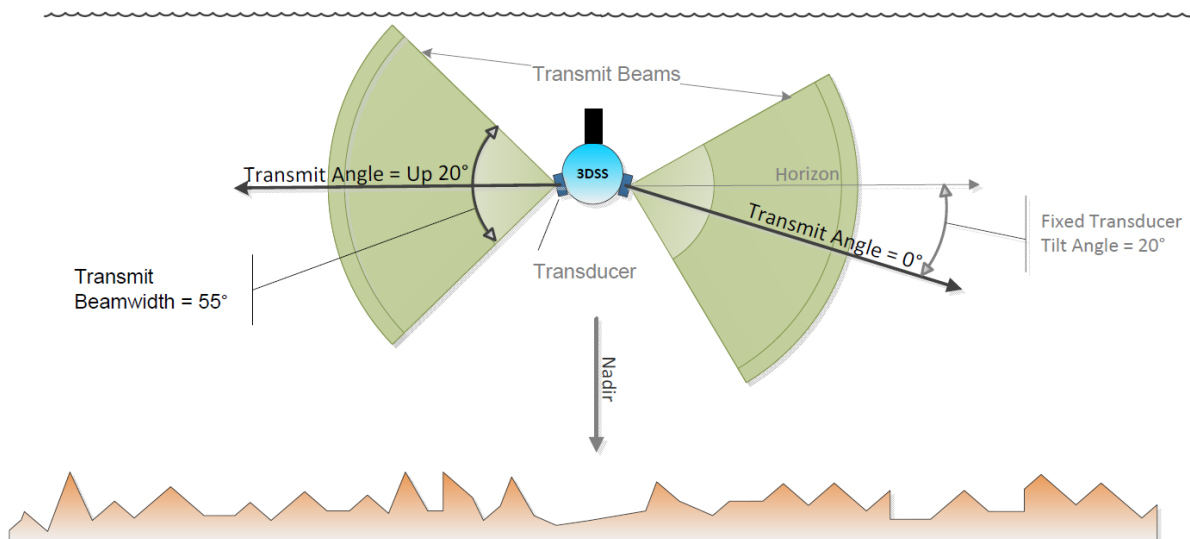


FIGURE 42. TRANSMIT BEAM ANGLE GEOMETRY.

GAIN PARAMETERS

Figure 43 shows the **3DSS-DX Control Sonar** program GUI with all OPERATING PARAMETER headings collapsed except GAIN parameters. These parameters are available when the DATA MODE is set to SONAR or to FILE PROCESS.

The applied gain for a data point at range, R meters, is computed according to the formula:

$$\text{GAIN}(R) = \text{LOG} * \text{Log10}(R) + \text{LINEAR} * R + \text{CONSTANT} \quad ; R \geq 1\text{m}$$

$$\text{GAIN}(R) = \text{CONSTANT} \quad ; R < 1\text{m}$$

Where:

LOG, LINEAR, and CONSTANT are gain coefficients from the GUI

Log10 is the base 10 logarithm

Sonar amplitude values are in floating point format and before gain is applied, the amplitude values lie between 0 and 8,388,608.



FIGURE 43. OPERATING PARAMETERS UNDER THE GAIN HEADING.

LOG

Specifies the gain coefficient applied to the base 10 logarithm of the range of each sample. The LOG coefficient is typically used to compensate for acoustic spreading loss as the transmit pulse propagates to and from the seafloor or a target. Different spreading laws such as spherical (LOG=40), cylindrical (LOG=20) or something in between (e.g. LOG=30) can therefore easily be accommodated.

After initial LOG coefficient values have been set, further fine adjustments can be made together with adjustments to the LINEAR coefficient in order to normalize seafloor echo strength across the swath. These additional adjustments are applied in order to compensate for backscatter versus angle roll-off across the swath (e.g. Lambert's Law).

The LOG gain coefficient setting is entered numerically and the allowable range of values is 0dB/Log10(m) to 60dB/Log10(m).

LINEAR

Specifies the gain coefficient applied to the range of each sample. The LINEAR coefficient is typically used to compensate for acoustic attenuation or absorption loss as the transmit pulse propagates to and from the seafloor or a target. Acoustic attenuation varies with temperature and salinity and are measured in dB/m. For 450 kHz sonar operation and 20°C seawater at 35ppt, the theoretical acoustic absorption coefficient is between 0.12 and 0.14 dB/m.

Note however that a computed acoustic attenuation value should be scaled by a factor of 2 for entry as the LINEAR coefficient since the LINEAR value is applied to the one-way range and not the range to and from a target or the seafloor.



After initial LINEAR coefficient values have been set, further fine adjustments can be made together with adjustments to the LOG coefficient in order to normalize seafloor echo strength across the swath. These additional adjustments are applied in order to compensate for backscatter versus angle roll-off across the swath (e.g. Lambert's Law).

The LINEAR gain coefficient setting is entered numerically and the allowable range of values is 0dB/m to 3.5dB/m.

CONSTANT

Specifies the fixed gain coefficient applied to every sample. The CONSTANT coefficient is typically used to raise or lower the signal levels by a fixed amount for display reasons (e.g. for comparing data sets).

The CONSTANT gain coefficient setting is entered numerically and the allowable range of values is -60dB to +60dB.

SIDECAN PARAMETERS

Figure 44 shows the **3DSS-DX Sonar Control** program GUI with all OPERATING PARAMETER headings collapsed except SIDECAN parameters. These parameters determine the mode of sidescan operation as well as the processing applied in that mode. SIDECAN parameters are available when the DATA MODE is set to SONAR or to FILE PROCESS.

MODE

Specifies one of two possible modes for sidescan operation. Since the **3DSS-DX-450** sonar includes multiple sidescan array elements, these elements can be used to enhance sidescan image quality in two possible ways. The following settings are available on a drop down menu:

INCOHERENT
COHERENT

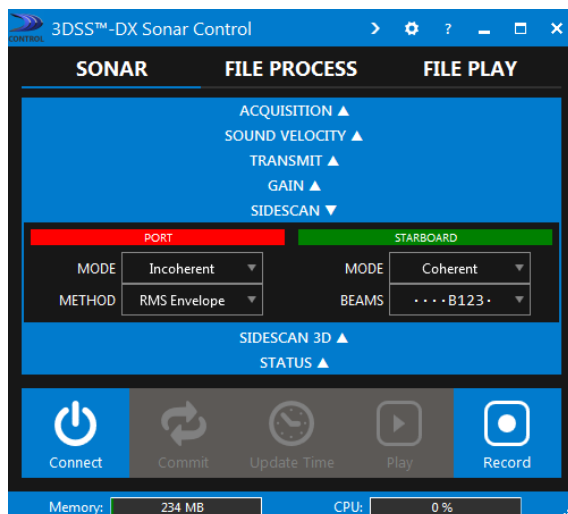


FIGURE 44. OPERATING PARAMETERS UNDER THE SIDESCAN HEADING.

When MODE is set to INCOHERENT, sidescan operation is identical to standard sidescan with the added benefit that the array includes 8 individual sidescan transducer elements that provide independent “looks” at the same portion of the seafloor and the same water-column targets. Each of the separate sidescan data sets can then be combined using one of two different METHOD parameters, RMS Envelope or Max Envelope.

RMS ENVELOPE

Specifies that the sidescan data from each array element is combined by computing the rms value across the elements at each sample time.

MAX ENVELOPE

Specifies that the sidescan data from each array element is combined by computing the maximum value across the elements at each sample time.

When MODE is set to COHERENT, sidescan operation the vertical sidescan receive beam is divided into 8 sectors that can each be enabled or disabled to allow selection or exclusion of a portion of the vertical coverage. Each of the sub-beams is enabled or disabled with a checkbox on a drop down menu when in COHERENT mode.

The vertical sidescan receive beam spans -90° to $+90^{\circ}$ relative to the broadside direction of the transducer array and the 8 sectors are formed as 8 sub-beams overlapped at their -3dB (half power) points. The directions from broadside, along with the approximate beamwidths, of the 8 sub-beams are:



ENDFIRE 90°	(beamwidth ~27 ° up and ~27 ° down)
UP 50°	(beamwidth ~25 °)
UP 30°	(beamwidth ~15 °)
UP 15°	(beamwidth ~15 °)
BROADSIDE 0°	(beamwidth ~15 °)
DOWN 15°	(beamwidth ~15 °)
DOWN 30°	(beamwidth ~15 °)
DOWN 50°	(beamwidth ~25 °)

Any combination of the available sub-beams can be enabled or disabled to tailor the sidescan receive beam to a region of interest. For example, sidescan returns from surface wakes and near surface propeller wash can be substantially reduced by disabling the majority of the upward looking beams (i.e. disabling Endfire 90°, Up 50°, and Up 30°).

Note that the endfire sub-beam includes energy from both the up and down directions and therefore cannot be used to exclude arrivals from above or below the sonar. For example, disabling the endfire sub-beam will eliminate sea surface arrivals from a large sector above the sonar but will also have an effect at nadir. Since the **3DSS-DX-450** transducer array is physically tilted by 20°, disabling the endfire beam will leave an reduced signal region at nadir of approximately $27^\circ - 20^\circ = 7^\circ$ or 13% of altitude. The reduced signal region will only be on the side that has the endfire beam disabled.

SIDESCAN 3D PARAMETERS

Figure 45 shows the **3DSS-DX Sonar Control** program GUI with all OPERATING PARAMETER headings collapsed except SIDESCAN 3D parameters. These parameters determine the mode of sidescan operation as well as the processing applied in that mode. SIDESCAN 3D parameters are available when the DATA MODE is set to SONAR or to FILE PROCESS.

ANGLES

Specifies the maximum number of independent angles-of-arrival to be resolved for each receive sample. The following settings are available on a drop down menu:

1 ANGLE

- 2 ANGLES
- 3 ANGLES
- 4 ANGLES



FIGURE 45. OPERATING PARAMETERS UNDER THE SIDESCAN 3D HEADING.

Typically **ANGLES** is set to its maximum of 4, and need not be adjusted often. However, when the likelihood of multiple arrivals is low (e.g. a simple, flat seafloor in deeper water), the **ANGLES** setting may be reduced for redundant computation and averaging of the reduced number of arrival angles. In simple environments this redundancy and averaging can further reduce bathymetric variance and therefore increase the achievable swath width for a specified uncertainty.

Setting **ANGLES** to 1 is equivalent to conventional interferometry where received backscatter is assumed to arrive from one direction only. In this case all 8 elements in the transducer array are used to compute the best estimate of one arrival direction at each sample time. As shown in Figure 46 however, a problem occurs when multiple arrivals from different directions interfere (e.g. from the seafloor and a water-column target, the sea surface, or a multipath). The single arrival direction estimated using interferometry becomes a combination of multiple arrivals and does not accurately provide the direction for any of the arrivals, including the direct path to the seafloor.

The **3DSS-DX-450** sonar extends conventional interferometry by introducing a new array signal processing methodology called “*Computed Angle-of-Arrival Transient Imaging*”. CAATI does not rely on phase differencing to resolve angles-of-arrival but instead directly computes as many as 4 arrival angles at each sample time using both phase and amplitude. The benefit is particularly

significant in environments prone to multipath such as shallow water and in environments with vertical geometries or water column targets. In addition to providing 3D imaging capabilities that cannot be achieved with interferometry, CAATI also reduces bathymetric variance and uncertainty and extends the achievable swath widths under standard hydrographic specifications such as the IHO Special Order Standard.

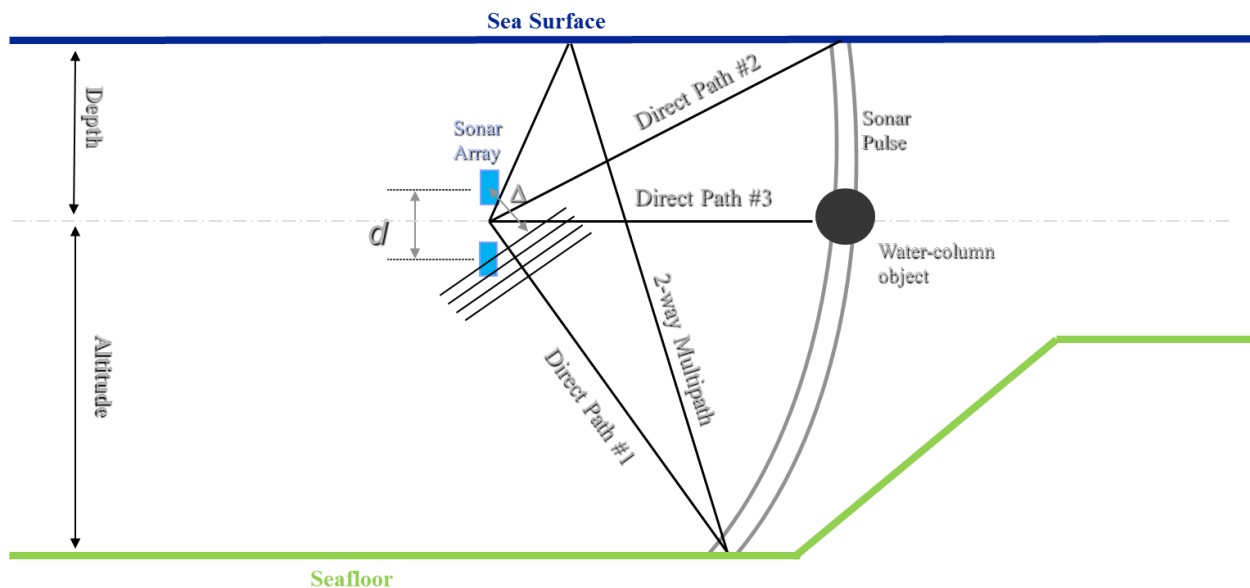


FIGURE 46. THE DIFFICULTY WITH INTERFEROMETRY.

SMOOTHING

Specifies the number of samples to include in each CAATI™ angle computation. The following settings are available on a drop down menu:

- NONE
- 2 SAMPLES
- 3 SAMPLES
- 5 SAMPLES
- 10 SAMPLES
- 15 SAMPLES
- 20 SAMPLES
- 25 SAMPLES
- 30 SAMPLES



40 SAMPLES
50 SAMPLES
60 SAMPLES

Typically **SMOOTHING** is set between 10 **SAMPLES** and 20 **SAMPLES** and need not be adjusted often. In some cases however where resolution is less important than achievable swath width or bathymetric uncertainty, **SMOOTHING** may be increased. Note however that excessive smoothing can lead to artifacts as the environment becomes increasingly complex.

Each 3DSS-DX receive sample represents approximately 1.3cm in range and CAATI™ processing outputs as many as 4 angles for each sample. The angles are computed using samples from both before and after the current sample. The number of samples used in the angle computation is specified with the **SMOOTHING** setting. For example, with a **SMOOTHING** setting of 15 **SAMPLES**, 7 samples before and 7 samples after the current sample are included in each CAATI computation. After the computation, the sample index is advanced by one and the process is repeated for the next sample. Thus, the angle-of-arrival computation is always performed every 1.3cm and is smoothed across the number of samples specified by the **SMOOTHING** parameter. For a setting of 15 **SAMPLES**, smoothing is performed over approximately 20cm (i.e. 15 times 1.3cm).

Note that although the angle computations are performed over a moving window, the amplitude values associated with each angle-of-arrival are computed without smoothing in order to preserve the full resolution of the 3D Sidescan imagery (i.e. same resolution as the 2D sidescan image data).

THRESHOLD

Specifies a receive signal level threshold for angle-of-arrival processing. The **THRESHOLD** setting is entered numerically and the allowable range of values is -150dB to 0dB.

Received signals below the **THRESHOLD** value are not included in the angle-of-arrival computations. The specified value is given in dB below an internal reference value and is independent of the gain setting (i.e. increasing the gain will not result in more of the received signal exceeding the **THRESHOLD** value).

Typically **THRESHOLD** is set to -100dB and need not be adjusted often. Increasing Threshold above -100dB will reduce weak signal contributions to the angle computations but may also reduce data points in the 3D imagery. Decreasing Threshold below -100dB will increase the 3D image data density but may also introduce data points with increased spatial variance.



TOLERANCE

Specifies an acceptable tolerance for how closely the angle-of-arrival solutions must conform to a post-solution integrity test in order to be retained. The TOLERANCE setting is a relative value and is entered numerically and the allowable range of values is 0.001 to 0.999.

Typically TOLERANCE is set to 0.05 and need not be adjusted often. A low value for TOLERANCE specifies a tight tolerance and means that fewer angle-of-arrival solutions will be retained and so fewer 3D Sidescan points will be output. A higher TOLERANCE value allows a greater margin in the post-solution integrity test and therefore generates more 3D Sidescan points, although these may also include a greater number of outliers and greater spatial variance.

BATHYMETRY PARAMETERS

The bathymetry output is a processed version of the Sidescan 3D points where the raw 3D points have been filtered and binned using one of two possible operating modes, "Legacy Mode" or "Multibeam Mode".

Figure 47 shows the **3DSS-DX Sonar Control** program GUI with all OPERATING PARAMETER headings collapsed except BATHYMETRY parameters. These parameters determine the mode of bathymetry operation as well as the processing applied in that mode.

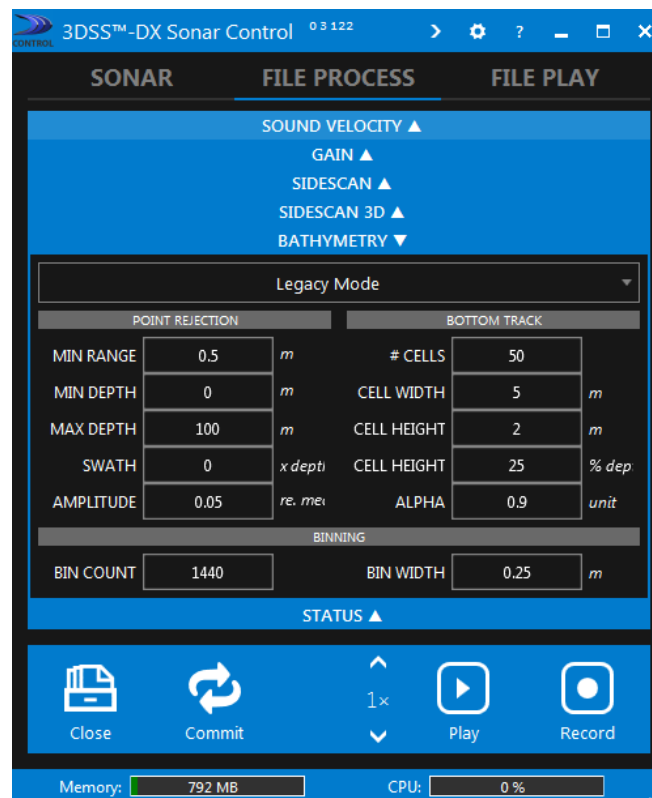


FIGURE 47. OPERATING PARAMETERS UNDER THE BATHYMETRY HEADING IN LEGACY MODE

LEGACY MODE

LEGACY MODE is the method of generating bathymetry from 3D sidescan points through point rejection filtering, bottom track filtering, and equidistant binning. It was the only mode of operation prior to version 0.3 of the **3DSS Sonar Control** program. The mode is called legacy because of this, and currently the MULTIBEAM MODE is preferred. This section describes the LEGACY MODE.

Note: MULTIBEAM MODE is the preferred BATHYMETRY operating mode and LEGACY MODE is maintained for backward compatibility only.

POINT REJECTION

Specifies a minimum and maximum depth window along with a minimum range and maximum swath width for rejecting points. Typically, MIN DEPTH is set to just above the sonar (-0.5m) or just below the sonar (0.5m). Just below the sonar is preferred in deeper water, however in very shallow water the setting should be set to the draft of the sonar (a negative value).



Swath point rejection removes 3D sidescan points outside of X times the sonar altitude. For example, setting the SWATH value to 8 when the nadir depth is 10m will accept points that are within 40m cross-track distance of the sonar to either side producing an 80m swath. Specifying a value of 0 disables the swath width filter (all points within the sonar RANGE setting are included).

Amplitude point rejection removes points whose amplitude falls below a threshold value computed for each ping. The amplitude threshold is computed on a ping by ping basis by multiplying the AMPLITUDE parameter by the average backscatter amplitude for the ping. The recommended value for this setting is between 0.2 and 0.3, or 0 to disable amplitude rejection.

BOTTOM TRACK

The bottom track filter creates a set of cells distributed across the swath (i.e. in rectangular coordinates) and within which the 3D sidescan points must fall in order to be included in bathymetry processing. The cells are automatically moved vertically by the bottom track filter to find and track the seabed. The number of cells (# CELLS), their width (CELL WIDTH) and height (CELL HEIGHT) must be specified. The cell height can be specified as either a constant value in meters or as a percentage of the depth, or as a combination of the two. Finally, the ALPHA value determines how quickly the bottom track filter responds to changes in the bottom. The ALPHA value must be between 0.01 and 0.99 where 0.01 is very slow response and 0.99 is very fast response. The performance of the bottom track filter can be observed and monitored in the **3DSS-DX Profile Display** application as seen in Figure 48. The next figure shows a close up of the bottom track filter cell.

BINNING

After data has been filtered by the basic point rejection options and the bottom track filter, it is binned into equidistant cross-track points. There are two parameters that control this binning step. The BIN COUNT specifies the maximum number of bins and hence the maximum number of points the bathymetry output will have. Typically, the BIN COUNT is set to 1440 or lower. The BIN WIDTH specifies the width of each bin and hence the spacing of the binned bathymetry points. Typically, the BIN WIDTH is set to a value between 0.1m/bin and 0.25m/bin. The product of BIN WIDTH and BIN COUNT specifies the maximum output swath. For example, 1440 bins x 0.1m/bin results in a total swath of 144m (72m port and 72m starboard).



FIGURE 48. 3DSS-DX PROFILE DISPLAY APPLICATION SHOWING THE BOTTOM TRACK FILTER PERFORMANCE.

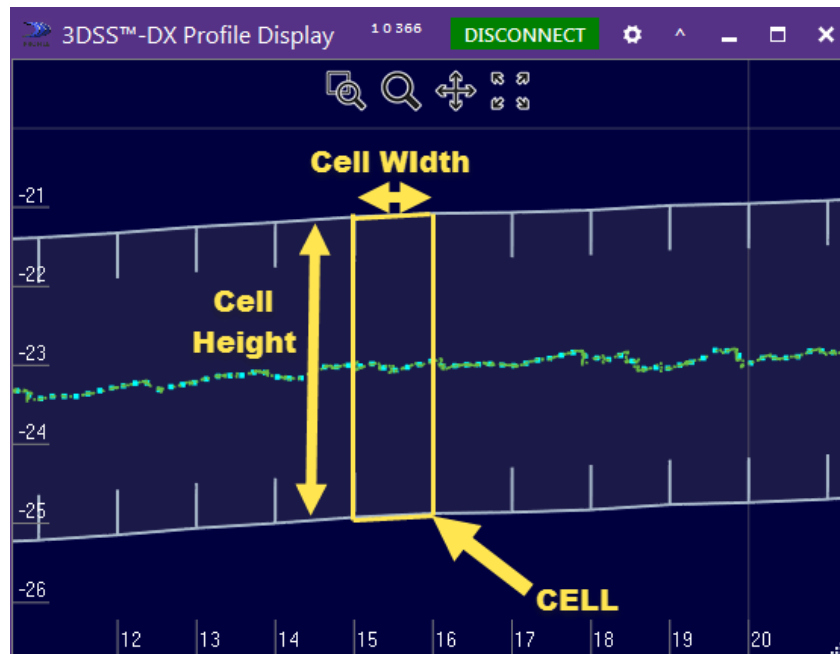


FIGURE 49. CLOSE UP VIEW OF BOTTOM TRACK CELL.

MULTIBEAM MODE

MULTIBEAM MODE emulates multibeam echosounder operation while also providing ultrawide swath coverage capability. New signal detection methods developed take advantage of the unique information available in 3D Sidescan imagery and provide precise and accurate soundings of the seabed and vertical extensions such as pilings or walls or masts.

Figure 38 shows the parameters available when bathymetry processing is set to MULTIBEAM MODE.

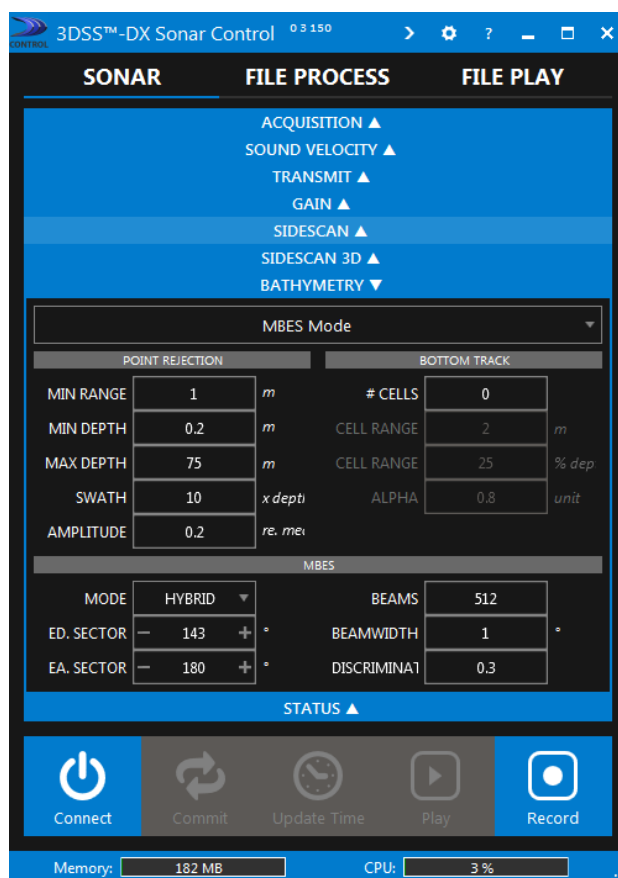


FIGURE 39. OPERATING PARAMETERS UNDER THE BATHYMETRY HEADING IN MULTIBEAM MODE.

POINT REJECTION

Specifies a minimum and maximum depth window outside of which points are rejected, and a maximum cross-track distance (a multiple of the depth) beyond which points are also rejected. Typically, MIN DEPTH is set to just above the sonar (-0.5M) or just below the sonar (0.5M). Just below the sonar is preferred in deeper water, however in very shallow water the setting should be set to the draft of the sonar (a negative value).



Swath point rejection removes bathymetry points outside of X times the sonar altitude. For example, setting the SWATH value to 8 when the nadir depth is 10m will accept point that are within 40m cross-track distance of the sonar to either side producing an 80m swath. Specifying a value of 0 disables the swath width filter.

Amplitude point rejection removes points whose amplitude falls below a threshold value computed for each ping. The amplitude threshold is computed on a ping by ping basis by multiplying the AMPLITUDE parameter by the average backscatter amplitude for the ping. The recommended value for this setting is between 0.2 and 0.3, or 0 to disable amplitude rejection.

BOTTOM TRACK

The bottom track filter creates a set of cells distributed in range and angle (i.e. polar coordinates) within which the 3D sidescan points must fall in order to be included in bathymetry processing. The number of cells (# CELLS) and their extent in range (CELL RANGE) must be specified. The angular size and spacing of the bottom track cells is determined by dividing the MBES sector size by the number of cells specified. The cell range can be specified as either a constant value in meters or as a percentage of range, or as a combination of the two. Setting # CELLS to zero disables the bottom track filter.

Finally, the ALPHA value determines how quickly the bottom track filter responds to changes in the seabed. The ALPHA value must be between 0.01 and 0.99 where 0.01 is very slow response and 0.99 is very fast response. The performance of the bottom track filter can be observed and monitored in the **3DSS-DX Profile Display** application.

Note: The current recommendation is to disable the bottom-track filter in **MULTIBEAM MODE** by setting the number of cells to 0, and to rely of the **MBES DISCRIMINATION** setting to determine the first arrival.

MBES OPTIONS

The remainder of the settings in MBES Mode are parameters used to formulate multibeam equivalent binning. The MODE option can be set to EQUANGULAR, EQUIDISTANT, or HYBRID.

EQUANGULAR AND EQUIDISTANT MODES

In both of these modes, a number of beams (specified by the BEAMS setting) are formed with equiangular or equidistant spacing. The beams are formed over an annular SECTOR specified in degrees, and the width of these beams is specified by the BEAMWIDTH setting. The beams are used to bin the 3D sidescan data, and result in one bathymetric output per beam. The DISCRIMINATION setting is a unit-less (0-1) value used to determine the first arrival in each beam.



Note: The current recommendation is to use DISCRIMINATION values of 0.3 to 0.5 with the bottom-track filter disabled. Disabling the bottom track removes its lag and allows the MBES mode to respond quickly to vertical targets.

HYBRID MODE

Ping DSP's unique hybrid mode is a combined equidistant and equiangular mode that features all the benefits of equidistant binning for typical bathymetry with the benefits of equiangular binning to quantize vertical targets all the way to the surface. In Hybrid mode, the equidistant beams are formed to the ED sector extent, and equiangular beams are formed where between the EA and ED sectors. As an example, with 512 beams, an ED sector setting of 157° will form 446 equidistant beams out to 10 times the water depth, and an EA sector setting of 180° will form an additional 66 equiangular beams between 157° and 180° (horizontal). An illustration of this can be seen in Figure 50. This ensures that the entire water-column is covered.

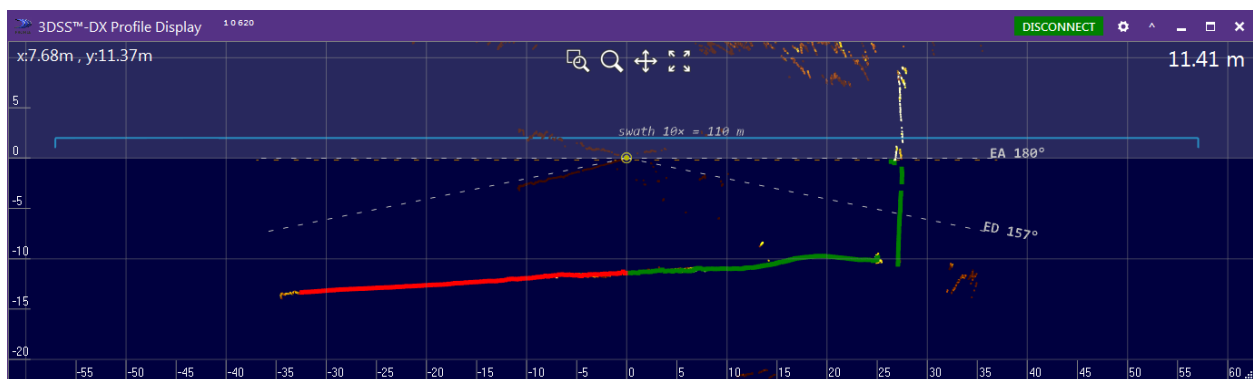


FIGURE 50. 3DSS PROFILE DISPLAY SHOWING HYBRID MULTIBEAM MODE

STATUS INDICATORS

The STATUS INDICATORS section of the **3DSS-DX Sonar Control** program GUI is organized into four different groups as shown in Figure 51.

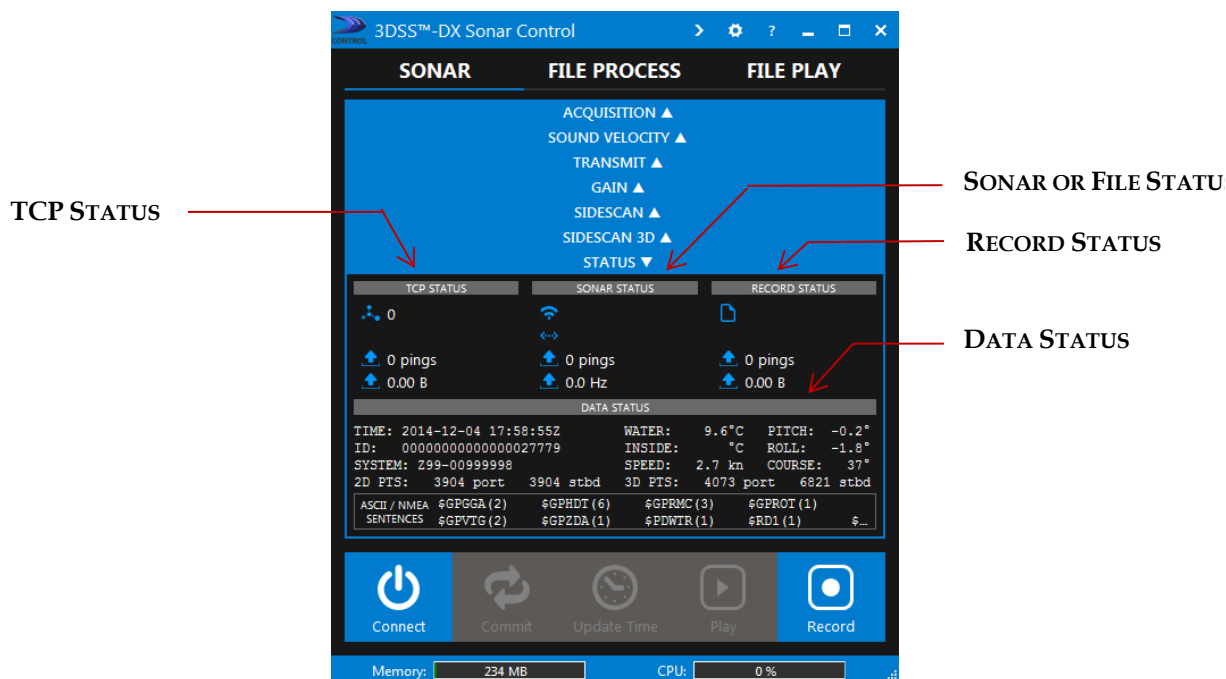


FIGURE 51. STATUS INDICATORS BY GROUP.

TCP STATUS

Indicates the number of TCP data clients, the total number of pings that have been output and the total number of bytes that have been output as shown in Figure 52. It also shows the number of bytes and pings dropped because a TCP client was not able to keep up. These dropped pings are only not sent to the client that cannot keep up, but are still distributed to other clients and written to disk.

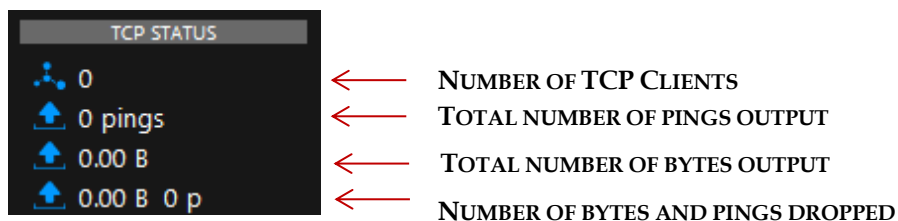


FIGURE 52. TCP STATUS INDICATORS.

SONAR STATUS / FILE STATUS

When DATA MODE is set to SONAR this heading reads SONAR STATUS and indicates the sonar communication status. The display shows the number pings received from the sonar and how many were bad (i.e. corrupted) and dropped. The bad and dropped values should be 0 during normal operation. The display also shows similar information for Ethernet packets from the sonar. The sonar ping rate in Hz is also provided at the bottom of the display. Figure 53 shows an example of the FILE STATUS display.

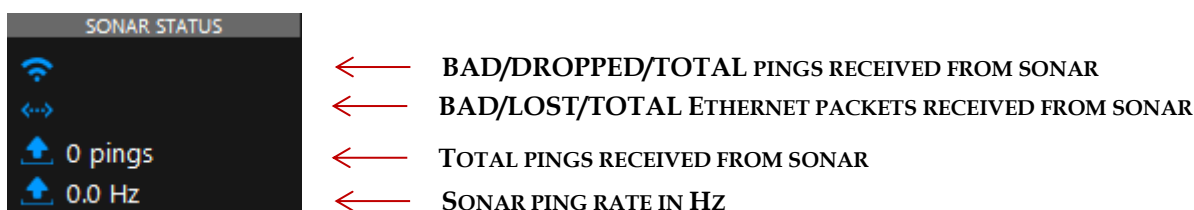


FIGURE 53. SONAR STATUS INDICATORS.

When DATA MODE is set to File PROCESS or FILE PLAY this heading reads FILE STATUS and indicates the file readback status. The display includes the filename, the percentage of the file that has been read, the equivalent ping rate in both Hz and as a multiple of the ping rate in effect during real-time recording, and the total number of bytes readback from file. Figure 54 shows an example of the FILE STATUS display.

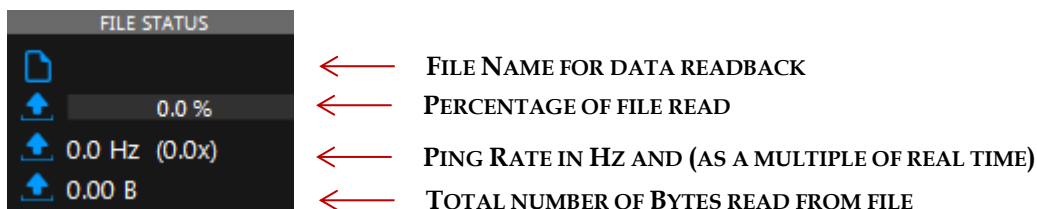


FIGURE 54. FILE STATUS INDICATORS.

RECORD STATUS

Indicates the filename for recording, the total number of pings written to file, and the total number of bytes written to file as shown in Figure 55.

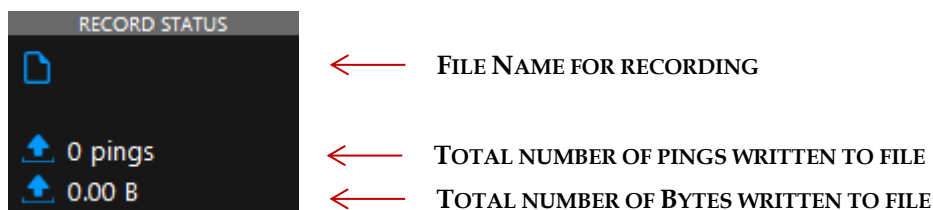


FIGURE 55. RECORD STATUS INDICATORS.



DATA STATUS

Summarizes the system status and the NMEA/ASCII sentences for the current ping as shown in Figure 56.

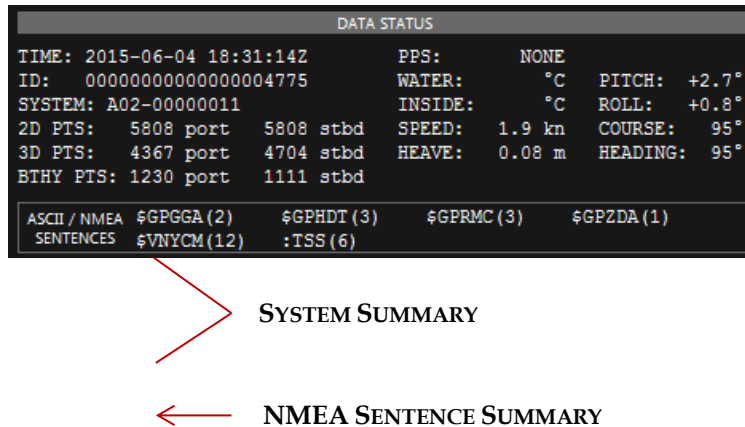


FIGURE 56. DATA STATUS INDICATORS.

The information included in the DATA STATUS display for the current ping is:

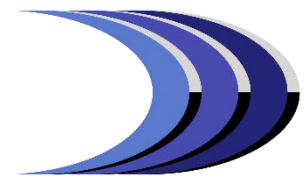
- SONAR CLOCK TIME (Zulu) (if a valid PPS signal is present and the sonar clock is locked, the clock accuracy is indicated as +/-1, +/-5, or +/-50 microseconds)
- PING ID (i.e. ping number since "CONNECT")
- SYSTEM ID
- WATER TEMPERATURE
- INTERNAL TEMPERATURE
- SPEED (from GPS if connected)
- PITCH (from internal MRU)
- ROLL (from internal MRU)
- COURSE (from GPS-GPRMC if connected)
- HEADING (from GPS-GPHDT if connected)
- HEAVE (from MRU-TSS1 if connected)
- 2D Pts (number of 2D Sidescan data points, port and stbd)
- 3D Pts (number of 3D Sidescan data points, port and stbd)
- BATHY Pts (number of Bathymetry data points, port and stbd)
- ASCII/NMEA (lists the ASCII/NMEA sentence types and (number), received during the ping.

COMPUTER RESOURCE USAGE



Reports the **3DSS-DX Sonar Control** program memory usage and CPU load at the bottom of the application window in the status bar:





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SIDESCAN3D DISPLAY PROGRAM

The **Sidescan3D™** display program is provided with the **3DSS-DX-450** sonar system for both real-time and post-time 3D sidescan and 2D sidescan data display. Data received from the **3DSS-DX Sonar Control** program over TCP can be displayed as it arrives or data can be played back from previously recorded ***.3dss-dx** files. 3D sidescan data and 2D sidescan data can be displayed simultaneously in separate windows. Both displays are waterfall type displays that show the most recently acquired data over a specified number of pings and therefore they do not incorporate navigation information for mosaicing or map creation.

For area surveying and map creation, the **3DSS-DX-450** sonar system is compatible with an ever growing number of third party survey software packages, For the current software compatibility list please contact Ping DSP Inc at support@pingdsp.com.

3D SIDESCAN DATA DISPLAY

The **Sidescan3D™** real-time display program main window and its functional parts are shown in Figure 57.

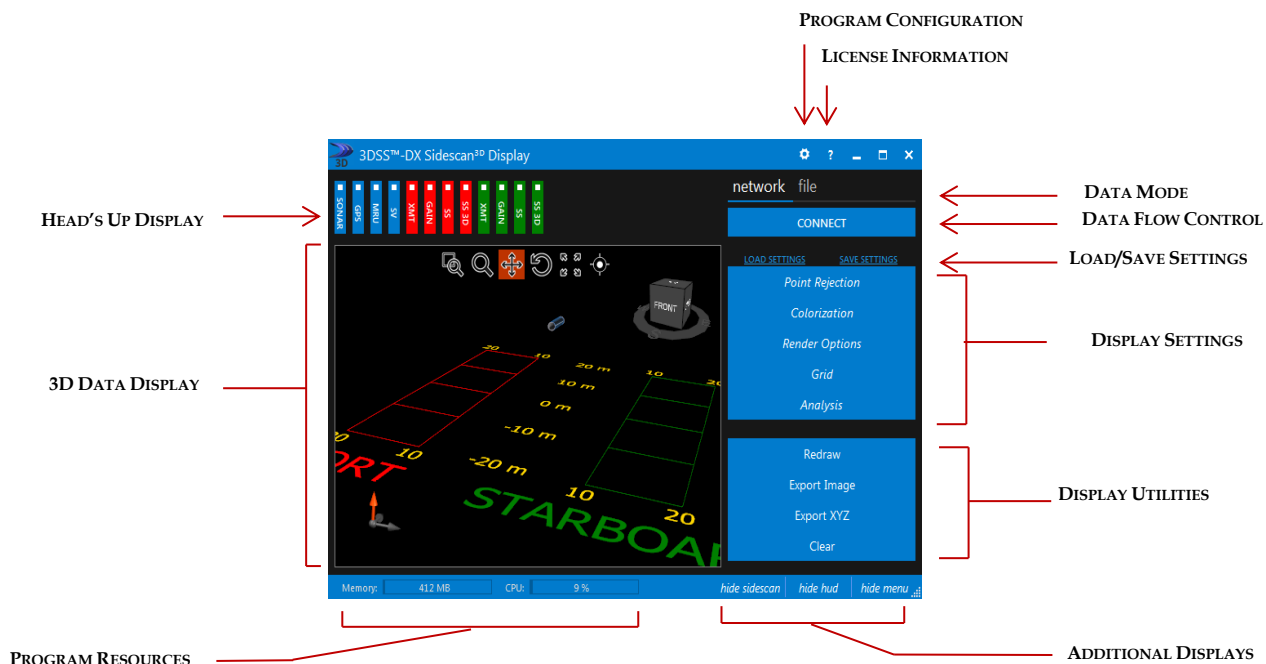


FIGURE 57. SIDESCAN3D™ MAIN DISPLAY WINDOW LAYOUT.

At the top of the main display window on the right hand side of the title bar is the PROGRAM CONFIGURATION icon and also the LICENSE INFORMATION icon. On the right hand side of the window is the sidebar main menu which includes the DATA MODE and DATA STREAM buttons, the DISPLAY SETTINGS drop down menus, and the DISPLAY UTILITIES buttons.

To the left of the sidebar menu is the 3D DATA DISPLAY which includes several quick access tools for rotating, panning, zooming and point picking as well as a view cube to quickly set specific data views. A small icon in the 3D DATA DISPLAY represents the **3DSS-DX-450** sonar position and a rectangular grid provides both cross-track and along track dimensioning. The top left corner of the 3D DATA DISPLAY also provides a direct readout display for the sonar altitude (i.e. water depth directly below the sonar).

Just above the 3D DATA DISPLAY is the HEADS UP DISPLAY which resembles a bookshelf with tagged entries that can be expanded to show sonar and peripheral information associated with the most recent ping of displayed data. At the bottom of the window on the right side are several buttons for showing or hiding different displays, including the heads up display, the sidebar menu and a separate 2D sidescan display window. And finally, at the bottom of the window on the left is the computer resource usage display.

Figure 58 shows the main display window with the sidebar drop down menus fully expanded, the HEADS UP DISPLAY tags partially expanded and a **3DSS-DX-450** data set in the 3D DATA DISPLAY.

For the case without any applied motion corrections, the XYZ location of each data point on the 3D data display is computed by converting the **3DSS-DX-450** range and angles values to X and Z coordinates for each ping, with X representing the cross-track dimension and Z representing depth. The Y coordinate is based on a default 1m/s forward velocity. If a GPS with \$GPRMC output sentences is connected to the sonar (please see Chapter 1), the forward velocity is derived from the GPS data. If PITCH and ROLL correction is enabled as described later in this chapter, the XYZ coordinate of each data point is modified to compensate for sonar motion.

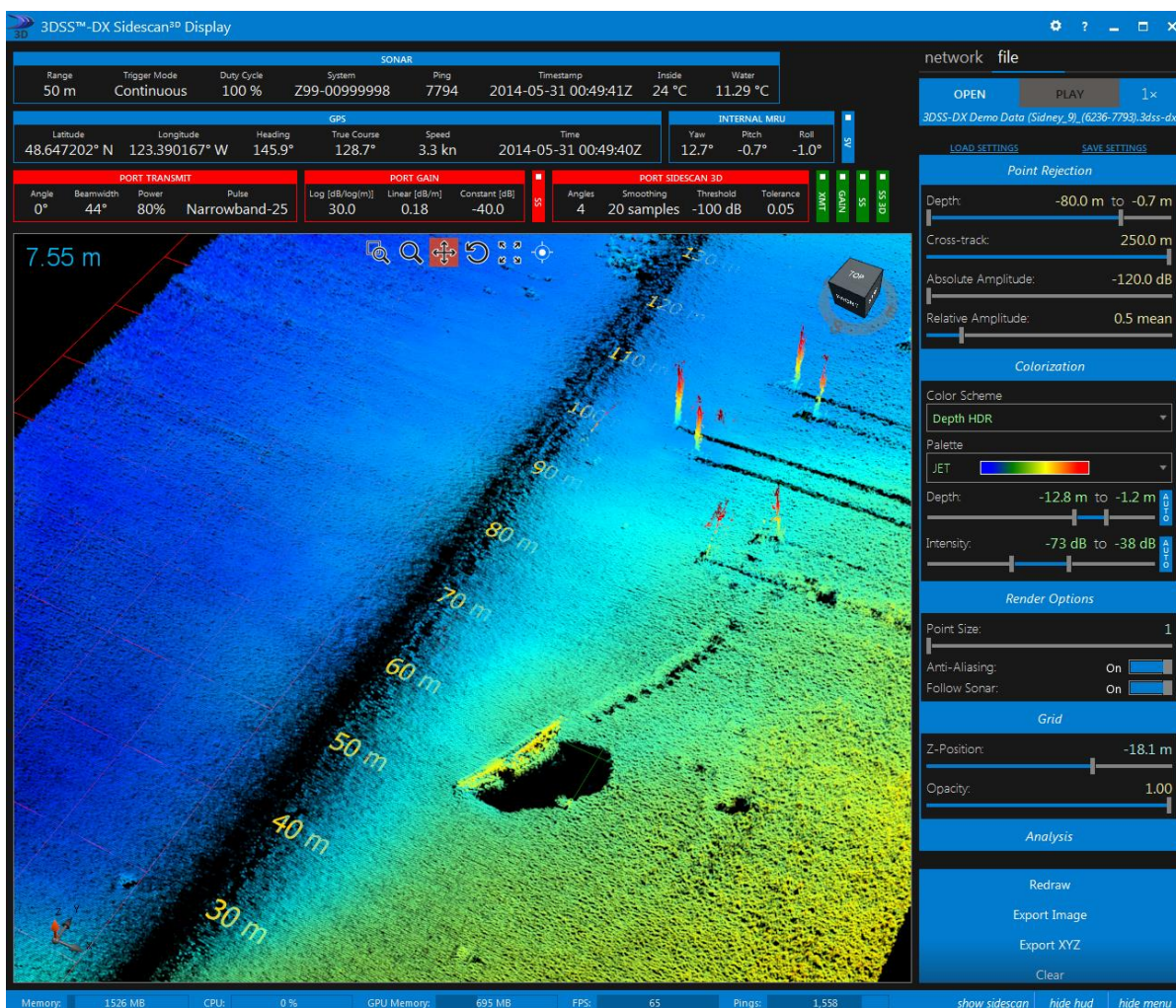


FIGURE 58. MAIN DISPLAY WINDOW SHOWING DATA, EXPANDED SIDEBAR MENU AND HEADS UP DISPLAY.

LICENSE INFORMATION AND APPLICATION CONFIGURATION

Clicking on the question mark in the title bar displays the LICENSE INFORMATION dialog box showing the version of the software as well as the license status and terms. If a license has not yet been obtained the dialog box provides a button for requesting a license as described in Chapter 1.

Clicking on the APPLICATION CONFIGURATION symbol in the title bar opens the APPLICATION CONFIGURATION dialog box as shown in Figure 59.

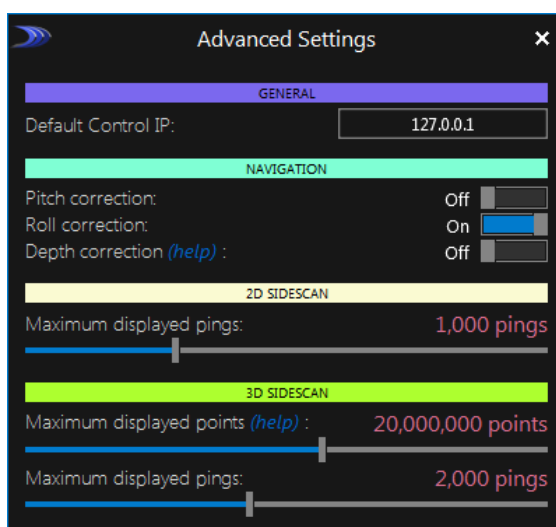


FIGURE 59. SIDESCAN3D DISPLAY PROGRAM CONFIGURATION DIALOG BOX.

DEFAULT CONTROL IP

This setting is the IP address of the 3DSS-DX data server computer (i.e. the Host computer running the **3DSS-DX Sonar Control** program). If the **Sidescan3D™** display program is also running on the Host computer this IP address should be set to 127.0.0.1.

NAVIGATION SETTINGS

The **Sidescan3D™** data display can be optionally be corrected for PITCH, ROLL and DEPTH. Pitch and roll corrections are applied using the **3DSS-DX-450** internal MRU. Depth corrections are currently intended for AUV operation only and should not be confused with heave corrections which are not currently supported in the **Sidescan3D™** program.



2D SIDESCAN SETTINGS

The maximum number of displayed pings provides control of the memory allocation for 2D sidescan image data. Increasing the maximum number of pings displayed provides a longer sidescan record but also consumes more computer memory and GPU resources.

3D SIDESCAN SETTINGS

The maximum number of displayed pings provides control of the memory allocation for 3D sidescan image data. Increasing the maximum number of pings displayed provides a longer 3D sidescan record but also consumes more computer memory and GPU resources. In addition to limiting the maximum number of pings, the maximum number of data points displayed can also be limited. For 3D sidescan data, the data is referred to as a point cloud and is analogous to terrestrial lidar and laser scanning point clouds. A **3DSS-DX-450** point cloud can grow to several million data points and depending on the computing and GPU capabilities of the display computer, the display can start to become jerky and slow if too many points are displayed. By adjusting the maximum number of displayed data points, this slow down in display performance can be avoided.

DATA MODE

DATA MODE allows selection of the data source as either NETWORK or FILE. A selection is made by clicking on the desired DATA MODE and the current DATA MODE is indicated with a blue underline.

When NETWORK is selected DATA FLOW CONTROL button is provided by a single CONNECT button, and when the DATA MODE is set to file, DATA FLOW CONTROL is provided by an OPEN button followed by a PLAY button and a playback speed control (please see the DATA FLOW CONTROL section below).

Note that the DATA MODE cannot be changed without either first releasing the network connection or closing an open data file.

DATA FLOW CONTROL

Allows connection to the **3DSS-DX Sonar Control** program for data input or controls data playback from file. The function and availability of the data flow control buttons changes with the DATA MODE.

NETWORK DATA FLOW CONTROL BUTTONS

Figure 59 shows the Sidescan3D program with the CONNECT button that is available when the DATA MODE is set to NETWORK.

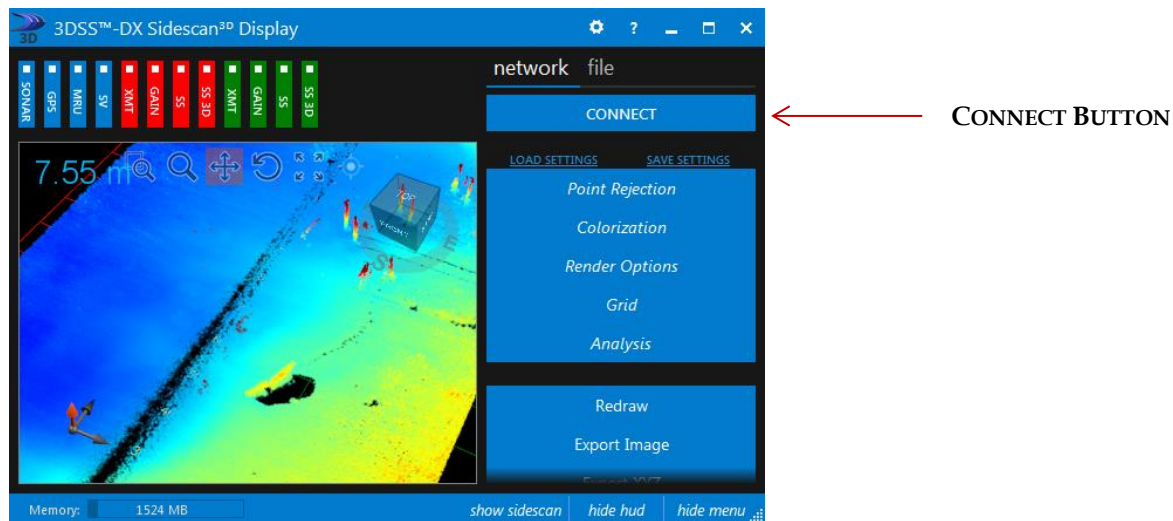


FIGURE 60. DATA FLOW BUTTON FOR NETWORK OPERATION.

CONNECT

Establishes a connection between the **Sidescan3D™** program and the **3DSS-DX Sonar Control** program running on a computer at the IP address specified in the **Sidescan3D™** program configuration. After the connection is established the button label changes to DISCONNECT to indicate that the connection was successful.

FILE DATA FLOW CONTROL BUTTONS

Figure 61 shows the **Sidescan3D™** program with the DATA FLOW buttons that are available when the DATA MODE is set to FILE.

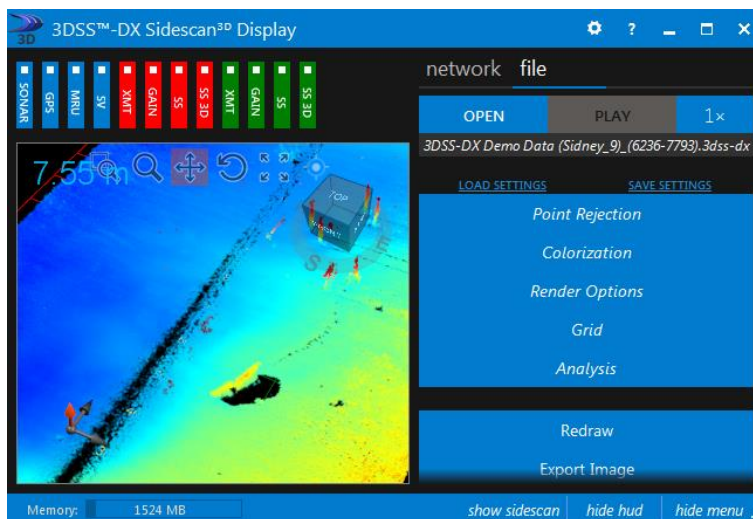
OPEN

Opens a data file dialog box to select the file for playback. After the file has been opened the button label changes to CLOSE to indicate that the open operation was successful.

Pressing the CLOSE button closes the data file. The file must be closed before the DATA MODE setting can be changed.

PLAY

Starts reading data from file at the rate specified by the playback rate setting. Once pressed, the PLAY button changes to STOP to indicate that data playback is underway. Pressing the STOP button pauses playback.



← FILE DATA FLOW BUTTONS

← DATA FILENAME

FIGURE 61. DATA FLOW BUTTONS AND FILENAME DISPLAY FOR FILE PLAYBACK.

PLAYBACK RATE

This button includes both an up and a down arrow and allows the data playback rate from file to be increased or decreased relative to the ping rate at which it was recorded. For example, a setting of 1x indicates that the playback rate from file is equal to the real time ping rate at the time of recording. Values higher than 1x indicate faster than real time playback rates and values less than 1x indicate playback rates slower than real time.

DATA FILENAME

When the Data Mode is set to File, the name of the currently open data file is shown below the buttons.

3D SIDESCAN DISPLAY CONTROLS

Along the top of the 3D DATA DISPLAY portion of the main **Sidescan3D™** program window are the 3D SIDESCAN DISPLAY CONTROLS. The controls are denoted by icons and include several tools for manipulating the display both during and after data collection. For example, during data collection or during data playback, the display viewpoint can be manipulated to change the viewing angle or zoom in on a feature of interest.

Once selected, the display controls are operated with the left mouse button. However, there are two controls that can always be operated regardless of which tool is currently selected:

1. Zoom can always be operated using the mouse wheel
2. Rotate can always be operated by clicking and dragging with the right mouse button.

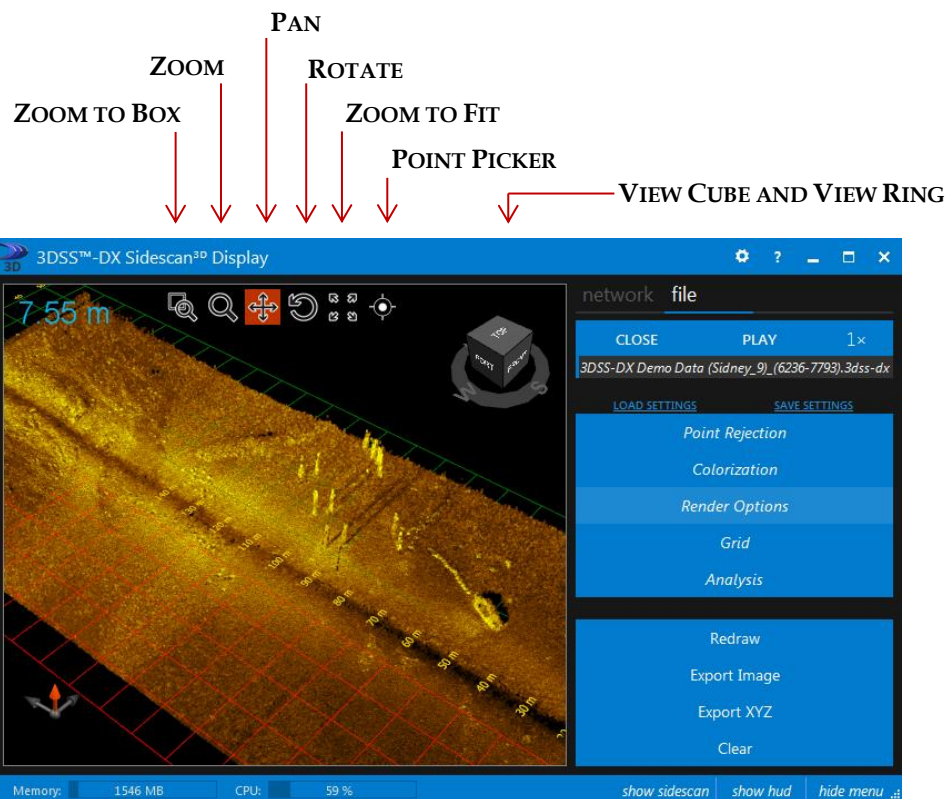


FIGURE 62. 3D SIDESCAN DISPLAY CONTROLS



PAN

Moves the display vertically or horizontally by clicking and dragging on the display.

ZOOM

Zooms the display in or out around the data point closest to the cursor location by clicking and dragging on the display. The Zoom tool is also always available using the mouse wheel.

Note that the display is drawn at full resolution at every zoom level for any window size and therefore continuous zoom all the way down to individual samples is always available.

ZOOM BOX

Zooms the display to the area enclosed by a box drawn by clicking and dragging on the display.

ROTATE

Rotates the display around the data point closest to the cursor location by clicking and dragging on the display. The Rotate tool is also always available by clicking and dragging on the display using the right mouse button.

FIT TO DATA

Zooms the display to show all available data.

POINT PICKER

Enables the POINT PICKER tool for selecting the center of the analysis volume as specified by the ANALYSIS settings on the sidebar menu. For example, the size of the analysis volume can be set for bathymetry information over a section of the seafloor, or the analysis volume can be set small to isolate the coordinates of very few data points such as the shallowest point of a hazardous object (please see below for information regarding ANALYSIS settings).

VIEW CUBE AND VIEW RING

The edges, sides and corners of the VIEW CUBE provide a set of preset viewing angles for quickly orienting the display view. Note however that the VIEW CUBE does not alter the zoom level or pan position when re-orienting the display.

The VIEW RING is a circle surrounding the VIEW CUBE that allows the display to be rotated in a plane by clicking and dragging anywhere on the VIEW RING.

DISPLAY SETTINGS

The Display Settings are located on the sidebar menu as shown in Figure 57 and provide parameters for both selecting and presenting the 3D sidescan data.

POINT REJECTION

Specifies which data points will be displayed. Data points can be excluded based on three rejection criteria:

1. Depth window – data outside a specified window is not displayed
2. Cross-track – data beyond a maximum cross-track distance is not displayed
3. Amplitude – data below either an absolute or a relative backscatter amplitude is not displayed

Figure 63 shows the expanded POINT REJECTION slider controls.

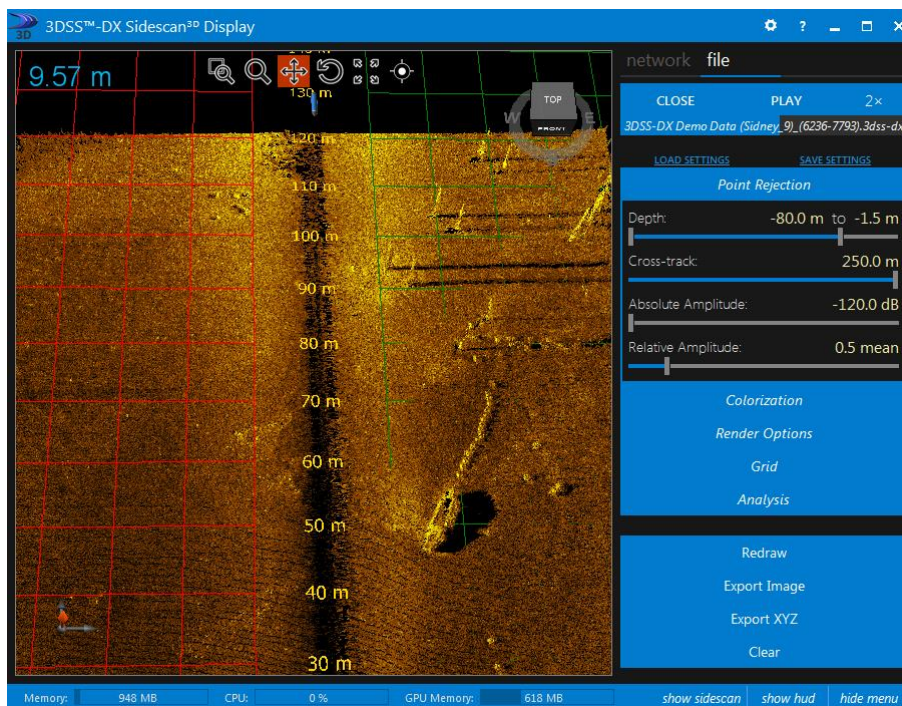


FIGURE 63. POINT REJECTION SLIDER CONTROLS.



DEPTH WINDOW

The DEPTH sliders specify the upper and lower limits for including data on the display. Data outside of the DEPTH WINDOW specification are not displayed but are also not discarded. For example, if the lower DEPTH slider is set to -10m any data deeper than 10m below the sonar will not be displayed. However if the slider is then moved to -20m and the REDRAW button is clicked, the display will be updated with any available data down to 20m depth.

CROSS-TRACK

The CROSS-TRACK slider specifies the outer limit of the cross-track distance for data on either side of the sonar. Data outside of the CROSS-TRACK specification are not displayed but are also not discarded. For example, if the CROSS-TRACK slider is set to 25m any data at a cross-track distance of more than 25m will not be displayed. However if the slider is then moved to 50m and the REDRAW button is clicked, the display will be updated with data any available data out to a cross-track distance of 50m on either side of the sonar.

AMPLITUDE

Two types of amplitude rejection are provided, ABSOLUTE AMPLITUDE and RELATIVE AMPLITUDE. The ABSOLUTE AMPLITUDE slider specifies minimum absolute backscatter amplitude below which data is not displayed. The ABSOLUTE AMPLITUDE threshold is the same for the entire display and the units are dB relative to an internal reference which is not calibrated in anyway.

The RELATIVE AMPLITUDE slider specifies a minimum backscatter as a fraction of the mean amplitude for each ping. Thus the RELATIVE AMPLITUDE threshold varies from ping to ping and is calculated on the fly. For example, a RELATIVE AMPLITUDE threshold of 0.3 specifies that data points for each ping that fall below 0.3 of the mean amplitude for the ping will not be displayed. And as for other display settings, a change to the RELATIVE AMPLITUDE threshold will change the display when a REDRAW is executed.

COLORIZATION

Specifies how the 3D data is to be colorized and allows selection of the color palette. Figure 65 shows the COLORIZATION settings expanded on the sidebar menu.

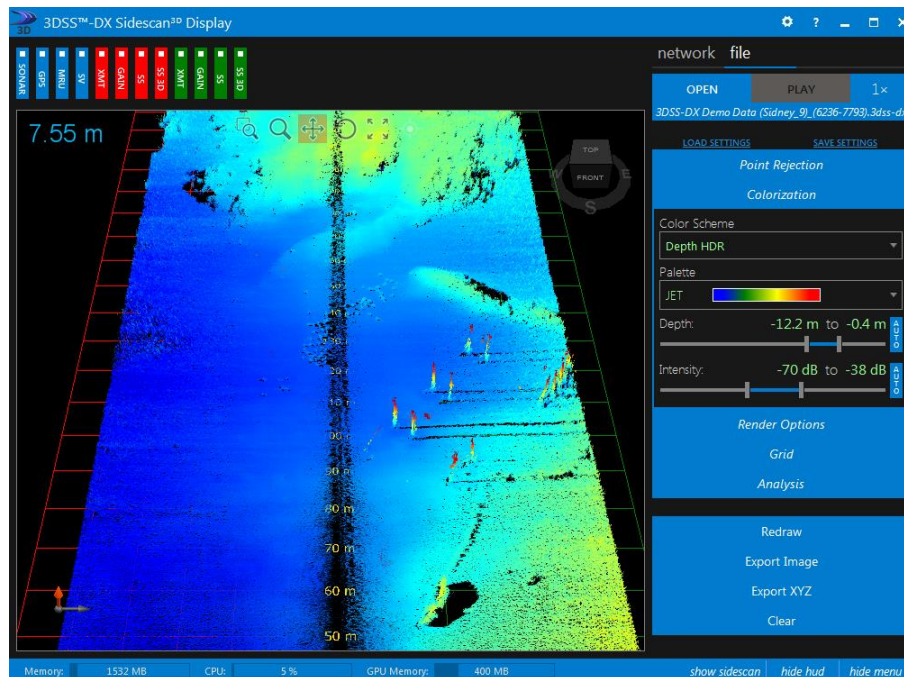


FIGURE 64. 3D SIDESCAN COLORIZATION SETTINGS, 3D BATHYMETRY EXAMPLE.

COLOR SCHEME

Colorization is divided into two primary types with three alternatives for each type:

1. 3D Bathymetry as shown in Figure 64 : Data is colorized by depth with the following variations:
 - a. DEPTH: Color is chosen from the color palette linearly according to Z value.
 - b. DEPTH PLUS: Same as DEPTH but data density modulates color saturation
 - c. DEPTH HDR: Same as DEPTH but backscatter intensity and data density both modulate color saturation.
2. 3D Imagery as shown in Figure 65 : Data is colorized by backscatter intensity with the following variations:
 - a. INTENSITY: Color is chosen from the palette linearly according to intensity value.
 - b. INTENSITY PLUS: Same as INTENSITY but data density modulates color saturation
 - c. INTENSITY HDR: Same as INTENSITY but backscatter intensity and data density both modulate color saturation

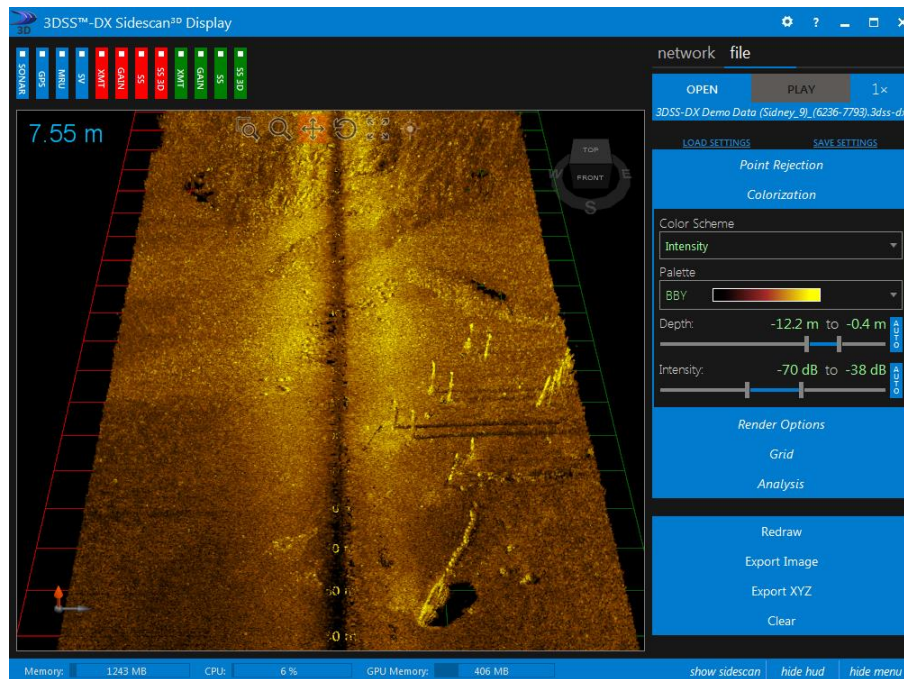


FIGURE 65. 3D SIDESCAN COLORIZATION SETTINGS, 3D IMAGERY EXAMPLE.

PALETTE

The linear distribution of colors used to span the depth range specified by the DEPTH slider and/or the intensity range specified by the INTENSITY slider. The PALETTE is selected from a drop down menu with the options shown in Figure 66.

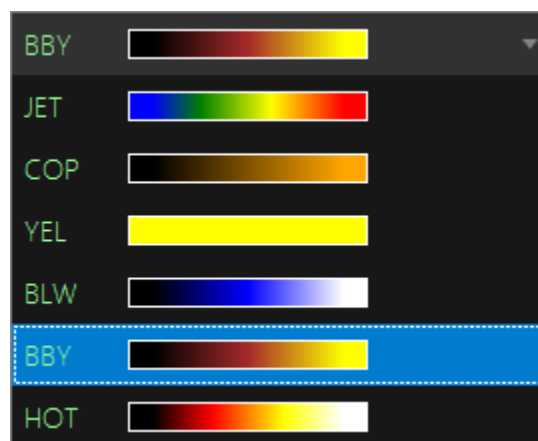


FIGURE 66. COLOR SCHEME SELECTION OPTIONS.



3D Bathymetry examples in this manual use the JET color palette and 3D Imaging examples use the BBY color palette.

DEPTH SLIDER

Sets the depth range spanned by the color PALETTE. Data points with Z values below the left hand slider value are displayed using the left most color of the color PALETTE. Data points with Z values above the right hand slider value are displayed using the right most color of the color PALETTE. Data points with Z values between the slider values are displayed using linear mapping of their Z value to the PALETTE color distribution .

The AUTO button beside the DEPTH SLIDER computes a pair of DEPTH SLIDER values to span the depth range of the currently displayed data.

A change to the DEPTH SLIDER settings will apply instantly to the entire display.

INTENSITY SLIDER

Sets the intensity range spanned by the color PALETTE. Data points with intensity values below the left hand slider value are displayed using the left most color of the color PALETTE. Data points with intensity values above the right hand slider value are displayed using the right most color of the color PALETTE. Data points with intensity values between the slider values are displayed using a logarithmic mapping of their intensity value to the PALETTE color distribution .

The AUTO button beside the INTENSITY SLIDER computes a pair of INTENSITY SLIDER values to span the intensity range of the currently displayed data.

A change to the INTENSITY SLIDER settings will apply instantly to the entire display.

RENDER OPTIONS

Specifies how the display is rendered. Figure 67 shows the Render Options settings expanded in the sidebar menu.

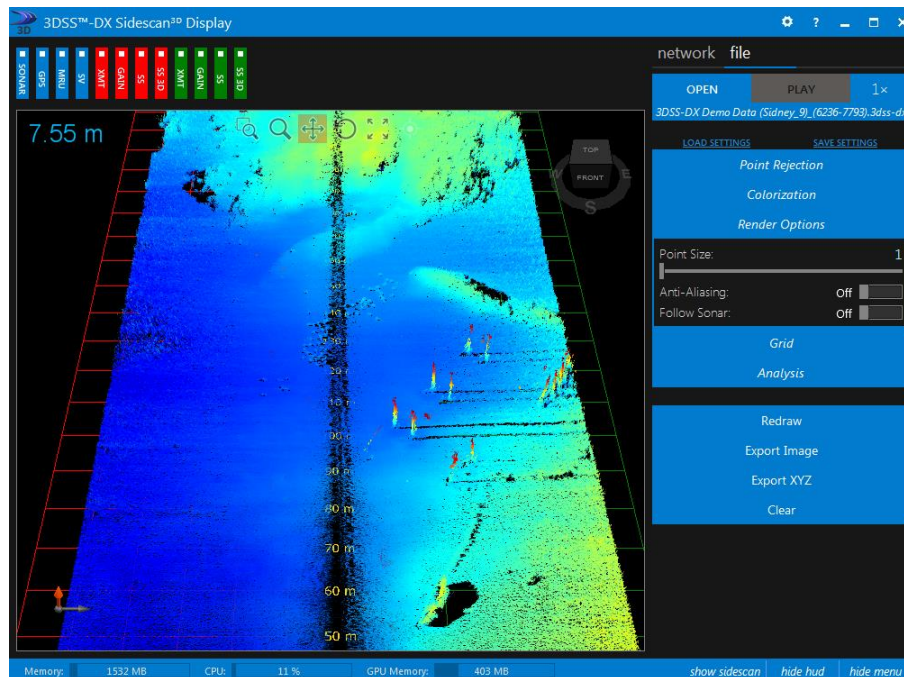


FIGURE 67. 3D SIDESCAN RENDER OPTIONS.

POINT SIZE

Specifies how large each data point is drawn on the display. In general a POINT SIZE of 1 is preferred. However, at high levels of zoom, increasing POINT SIZE will help make data visible when fewer and fewer data points make up the display.

ANTI-ALIASING

Prevents display pattern artifacts from arising due to aliasing. For example, during real-time display, ANTI-ALIASING prevents the display from twinkling as the data is scrolled.

FOLLOW SONAR

Specifies that the display will scroll as new pings are added. When switched off, the display will remain stationary and new pings will continue to be added but will not be visible once the display area is filled.

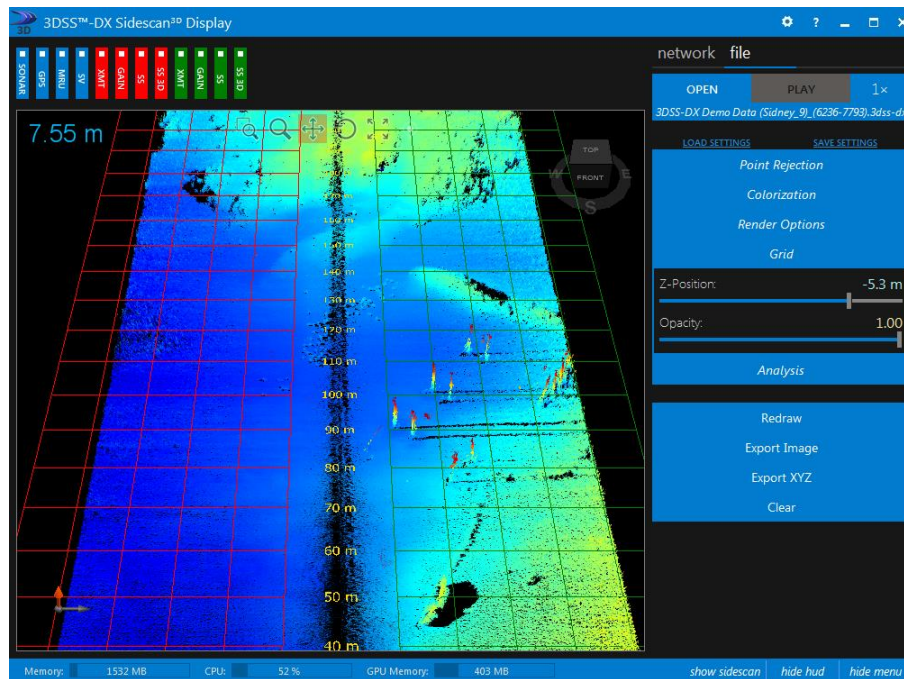


FIGURE 68. GRID Z POSITION AND OPACITY SETTINGS.

GRID

The 3D data display includes a rectangular 10m grid that can be superimposed on the display at any depth setting. The GRID settings specify the vertical plane and opacity for drawing the grid as shown in Figure 68. The grid can be removed from the display by either setting its OPACITY to 0.00 or by setting its Z position to -100.0 . The grid spacing is fixed at 10m in both the X (cross-track) and Y (along-track) directions.

ANALYSIS

Specifies the BOX AREA, BOX HEIGHT and BOX OPACITY of the bounding box used in conjunction with the POINT PICKER tool for simple analysis of 3D sidescan data.

Figure 70 and Figure 70 show the expanded ANALYSIS section of the sidebar menu along with the resulting bounding box positioned on the 3D display using the POINT PICKER tool. When selected, the POINT PICKER tool will position a new bounding box around the data point nearest to the cursor where the left mouse button is pressed.

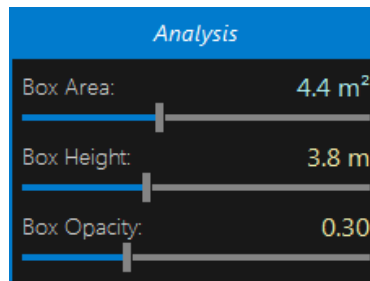


FIGURE 69. BOUNDING BOX ANALYSIS OPTIONS.

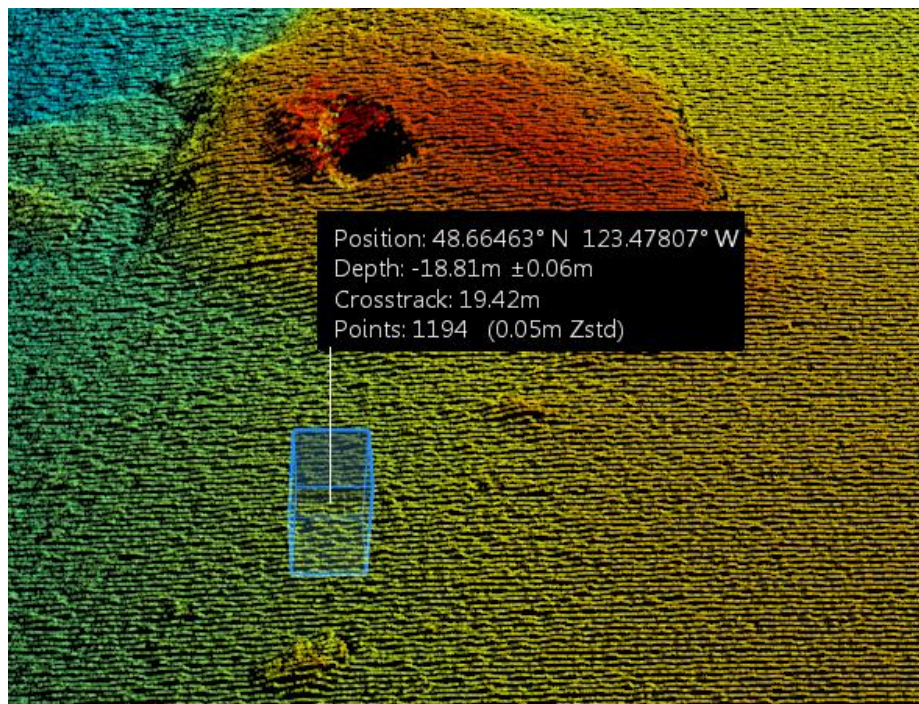


FIGURE 70. BOUNDING BOX DIMENSIONING FOR SIMPLE ANALYSIS OF 3D SIDESCAN DATA.

Each bounding box reports the following 4 lines of information about the data points contained within its boundaries.

1. Approximate latitude and longitude of the box.
Note: This position is not corrected and does not include the GPS to sonar lever arm.
2. Mean depth and depth uncertainty to 95% confidence for the region within the box.



3. The cross-track distance of the box. Also defined as the distance from sonar's nadir to the box in meters.
4. Number of data points and the standard deviation of the Z location for the points within the box.

Note that the depth uncertainty and the standard deviation of the Z location for the points within the box are not the same thing. The standard deviation of the Z location is a measure of the point to point variation over the entire set of data points within the box. The depth uncertainty on the other hand is a measure of how close the mean depth within the box will be to the reported mean 95% of the time if the measurement is independently repeated again and again.

By sizing the bounding box area to represent a specific grid cell size (e.g. 1m²) spot analysis measurements can be made to evaluate depth uncertainties on the fly. For example, spot analysis measurements can be made to directly determine the useable swath width for a specified grid resolution and bathymetric uncertainty.

DISPLAY UTILITIES

The Display Utilities are located on the sidebar menu as shown in Figure 57 and provide functions for updating, clearing and exporting the display.

REDRAW

Redraws the display with the current display settings. If changes are made to the display settings a REDRAW is only necessary if data already displayed is to be updated with the new settings. New display settings are applied to new data being displayed immediately after the settings have been changed.

EXPORT IMAGE

Renders the currently visible display to a PNG graphics file and provides multiple choices for image output resolution.

EXPORT XYZ

Exports XYZ and AMPLITUDE data for all currently displayed data in the display buffer. Data that is in the display buffer but rejected by the POINT REJECTION settings will not be exported, however data that is not rejected but is not present in the current view will be exported.

The output format is ascii csv as shown below:



-45.106,1.211,-10.768,9042
-45.125,1.211,-10.745,11395
-45.140,1.211,-10.738,11077

CLEAR

Clears the display and the data buffer.

SAVING AND LOADING SETTINGS

Stores and retrieves the display settings such as slider positions for point rejection and colorization.

HEADS UP DISPLAY

When enabled, shows the **3DSS-DX-450** sonar, MRU and GPS information for the current ping using a group of color coded collapsible tags. Figure 71 shows an example HEADS UP DISPLAY (HUD) with a mix of collapsed and expanded tags. GPS information is only available if a GPS is connected to the Topside Box as described in Chapter 1.

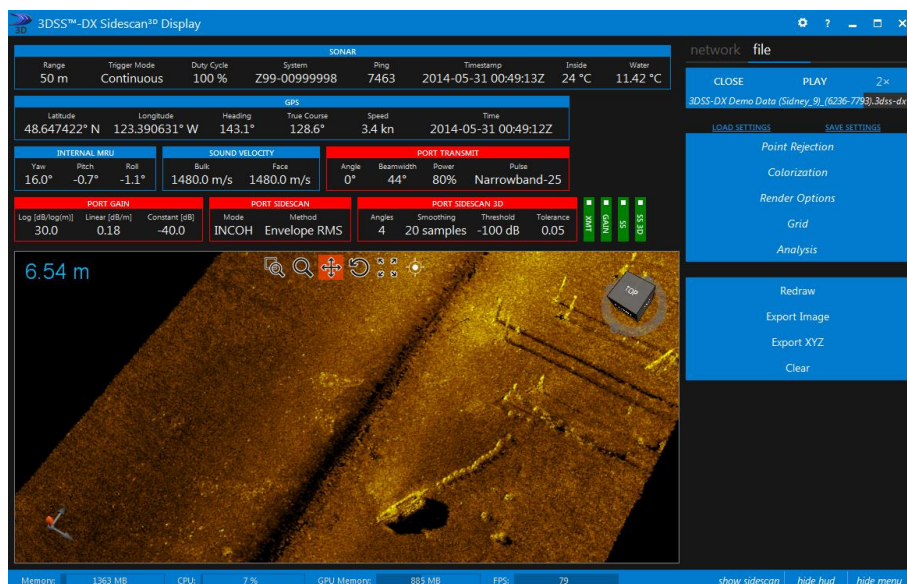


FIGURE 71. PARTIALLY EXPANDED HEADS UP DISPLAY (HUD).

The HUD display INFORMATION TAGS and their color codes are as follows:

1. Sonar – blue.



2. GPS – blue.
3. Internal Motion Sensor (I.MRU) – blue.
4. External Motion Sensor (E.MRU) – blue.
5. SBG Motion Sensor – blue (for 3DSS-iDX only).
6. Sound Velocity (SV) – blue.
7. Port Transmit Settings (Xmit) – red
8. Port Gain settings (Gain) – red
9. Port 2D Sidescan settings (SS) – red
10. Port 3D Sidescan settings (SS 3D) – red
11. Starboard Transmit Settings (Xmit) – green
12. Starboard Gain settings (Gain) – green
13. Starboard 2D Sidescan settings (SS) – green
14. Starboard 3D Sidescan settings (SS 3D) – green

Also note that the HUD can be hidden from view using the **HIDE HUD** button along the bottom of the window on the right side and as described below.

GOOGLE EARTH POSITION LOOK-UP

When running the **Sidescan3D™** display program while also connected to the internet (e.g. during data replay or reprocessing), GPS locations present on the HUD display can be automatically located on Google Maps. With the GPS HUD tag expanded, a right mouse click on either the latitude or longitude will open the dialog box shown in Figure 72. Selecting **GOTO (Google Maps)** opens a new default browser window or tab and pinpoints the GPS position on Google Maps.

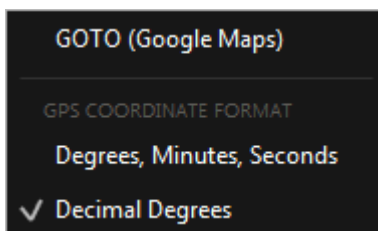


FIGURE 72. GPS TAG DIALOG BOX FOR GOOGLE MAPS LOCATION.

COMPUTER RESOURCE INFORMATION

Reports the resource usage of the Sidescan3D program. Resources included in the report are:

1. Memory useage
2. CPU load
3. GPU memory
4. Achievable frames per second (FPS)
5. Pings displayed
6. Points displayed

As the main window is reduced in size, fewer resources are included on the display with the priority for display being the order given above.

SHOW OR HIDE OPTIONS

Enables or disables viewing of the sidebar menu and the heads up display in the main window and enables or disables the 2D sidescan window. The 2D sidescan window is a second program window with its own display controls and settings (please see the *2D Sidescan Display* section below).

2D SIDESCAN DATA DISPLAY

When enabled shows the **3DSS-DX-450** 2D sidescan data as a waterfall image display in a separate window. Figure 73 shows an example 2D sidescan display. The cross-track axis is derived from the range to each 2D sidescan sample and the along-track axis is derived from GPS velocity of it is available (please see Chapter 1). If GPS velocity is not available, a default 1m/s forward velocity is assumed.

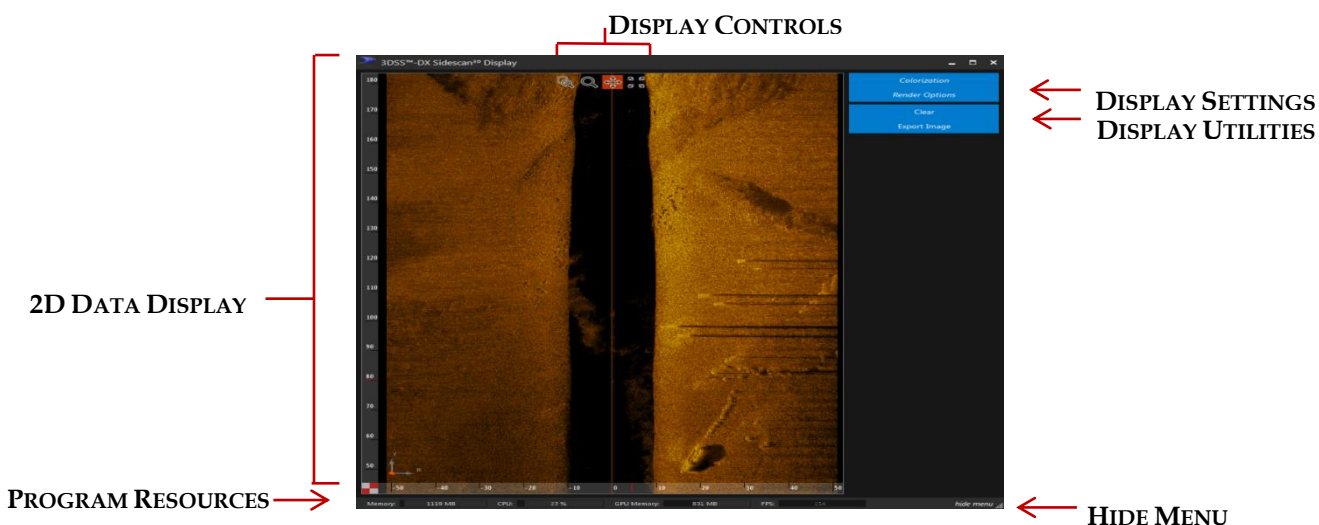


FIGURE 73. 2D SIDESCAN DISPLAY WINDOW LAYOUT.

2D SIDESCAN DISPLAY CONTROLS

Figure 74 shows the real-time controls for modifying the view of the 2D sidescan display. The tools are activated and deactivated with a mouse click and the active tool is highlighted.

PAN

Moves the display vertically or horizontally by clicking and dragging on the display.

ZOOM

Zooms the display in or out around the data point closest to the cursor location by clicking and dragging on the display. The Zoom tool is also always available using the mouse wheel.

Note that the display is drawn at full resolution at every zoom level for any window size and therefore continuous zoom down all the way down to individual samples is always available.

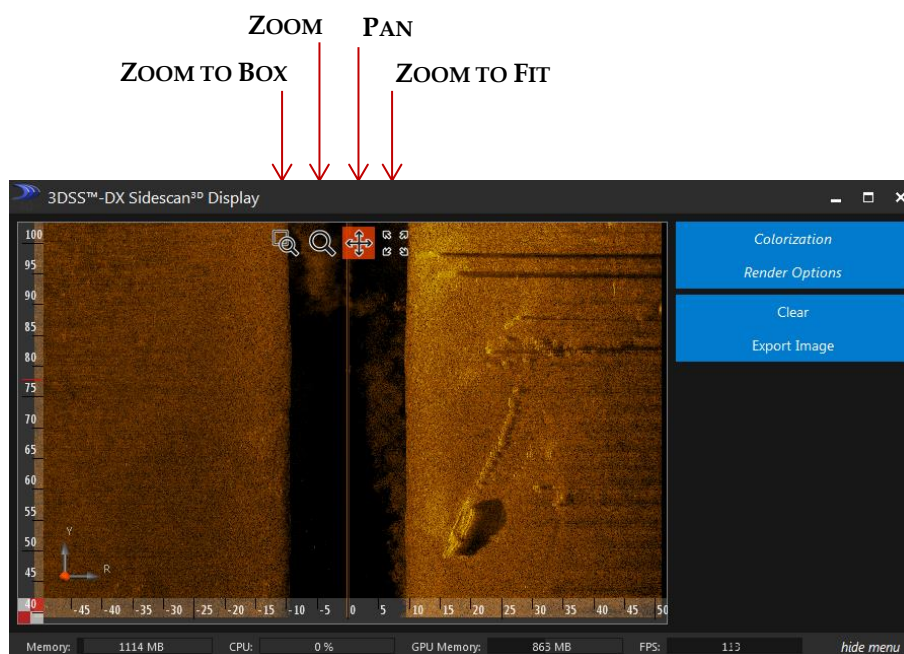


FIGURE 74. 2D SIDESCAN DISPLAY CONTROLS.

ZOOM BOX

Zooms the display to the area enclosed by a box drawn by clicking and dragging on the display.

FIT TO DATA

Zooms the display to show all available data.

2D SIDESCAN DISPLAY SETTINGS

The DISPLAY SETTINGS are located on the sidebar menu as shown in Figure 73 and provide parameters for both selecting and presenting the 2D sidescan data.

COLORIZATION

Specifies how the 2D sidescan data is to be colorized along with the color palette. Figure 75 shows the COLORIZATION settings expanded on the sidebar menu.

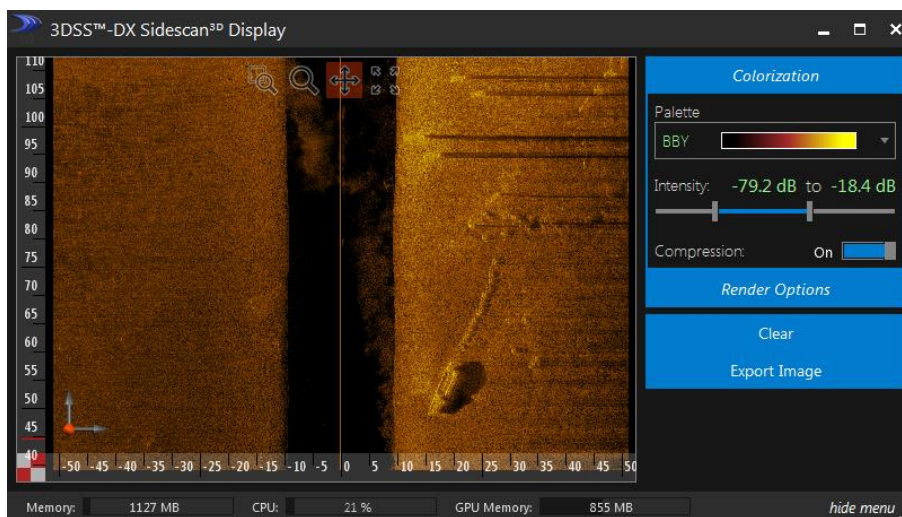


FIGURE 75. 2D SIDESCAN DISPLAY COLORIZATION SETTINGS.

PALETTE

The linear distribution of colors used to span the intensity range specified by the INTENSITY slider. The PALETTE is selected from a drop down menu with the available options shown in Figure 76.

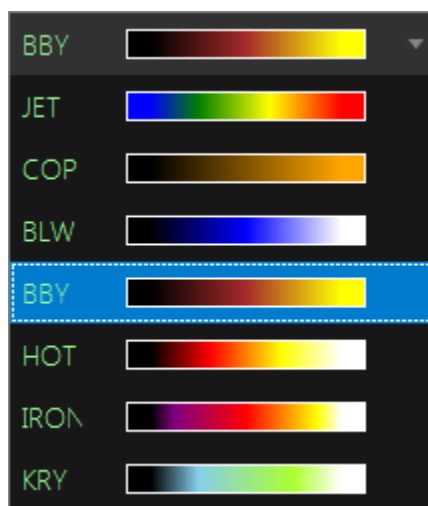


FIGURE 76. 2D SIDESCAN COLOR PALETTE SELECTION.

A change to the INTENSITY SLIDER settings will immediately apply to the entire display. The examples shown in this manual utilize the BBY color palette.

INTENSITY SLIDER

Shown in the expanded section of the sidebar in Figure 75 and used to set the intensity range spanned by the color PALETTE. Data points with intensity values below the left hand slider value are displayed using the left most color of the color PALETTE. Data points with intensity values above the right hand slider value are displayed using the right most color of the color PALETTE. Data points with intensity values between the slider values are displayed using a logarithmic mapping of their intensity value to the PALETTE color distribution .

A change to the INTENSITY SLIDER settings will immediately apply to the entire display.

COMPRESSION

Shown in Figure 75 and specifies whether the displayed 2D sidescan intensity values should be linearly or logarithmically mapped to the color PALETTE. COMPRESSION on indicates logarithmic mapping.

RENDER OPTIONS

The 2D sidescan display rendering options as shown in the expanded section of the sidebar menu in Figure 77.

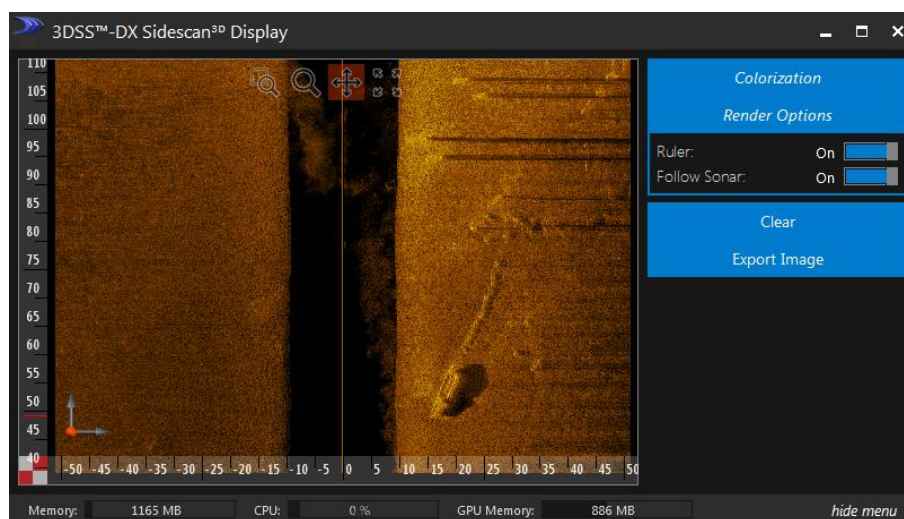


FIGURE 77. 2D SIDESCAN DISPLAY RENDERING OPTIONS.



RULER

Enables or disables the **R** (cross-track range) and **Y** (along-track distance) rulers superimposed on the display. Both rulers are dimensioned in meters.

FOLLOW SONAR

Specifies that the display will scroll as new pings are added. When switched off, the display will remain stationary and new pings will continue to be added but will not be visible once the display area is filled.

2D SIDESCAN DISPLAY UTILITIES

EXPORT IMAGE

Renders the currently visible display to a PNG graphics file and provides multiple choices for image output resolution.

CLEAR

Clears the display and the data buffer.

PROGRAM RESOURCES

Repeats the resource usage report for the **Sidescan3D™** display program. Resources included in the report are:

- Memory usage
- CPU load
- GPU memory
- Achievable frames per second (FPS)
- Pings displayed
- Points displayed

As the sidescan window is reduced in size, fewer resources are included on the display with the priority for display being the order given above.

3DSS TARGET ACQUISITION

Version 1.0 of the 3D Display software adds the ability to capture targets in 3D. Using the acquire target tool in the toolbar a box can be drawn around a target in-order to clip it. This process can be seen in Figure 78. Target clipping produces the best results when the target is viewed from the top. Once clipped out the target is sent to the 3DSS Target Logger window.

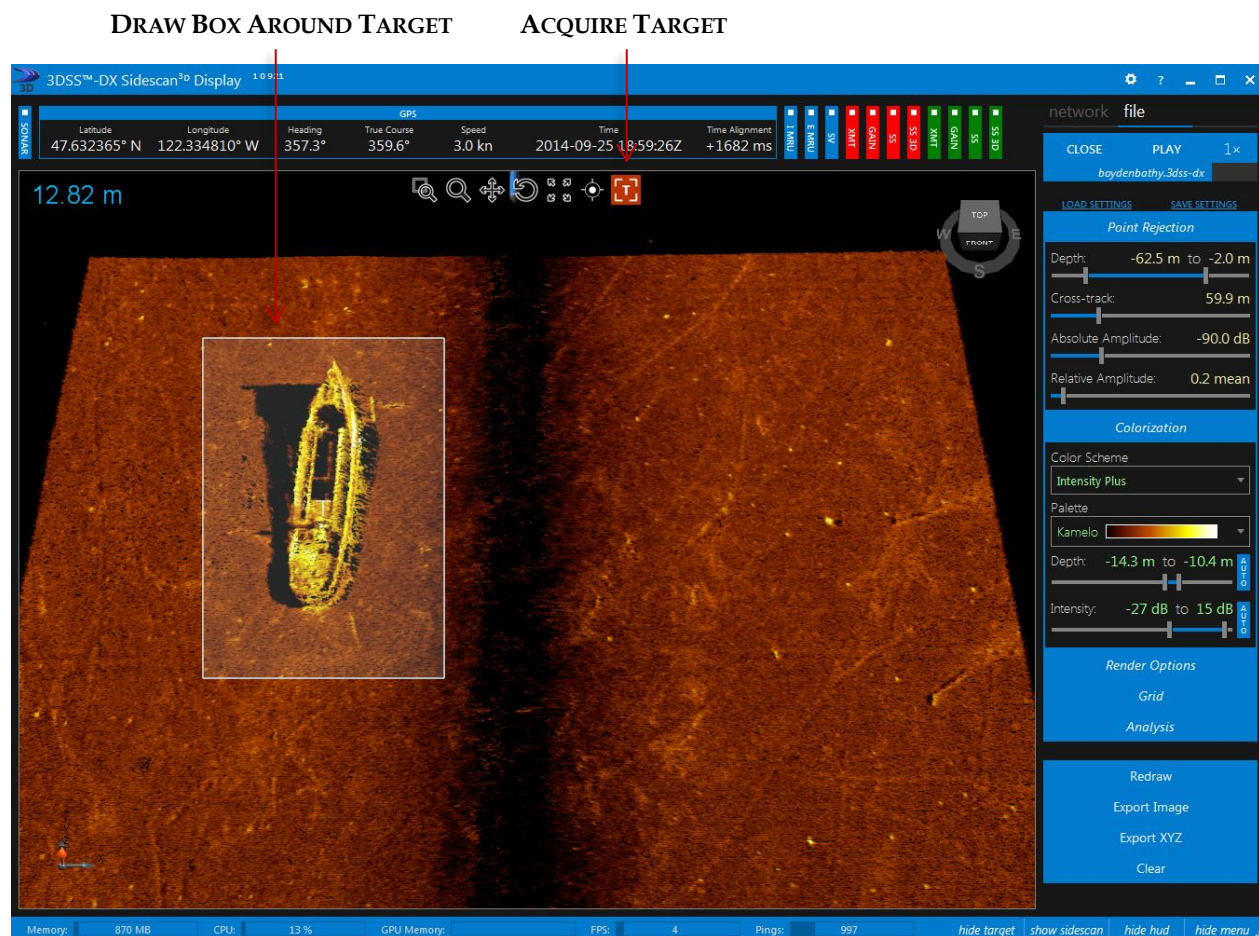


FIGURE 78. CLIPPING OUT A TARGET USING THE TARGET ACQUISITION TOOL

3DSS TARGET LOGGER

With version 1.0 of the 3D Display software a new target logger window has been added and is used to view targets, load and save targets, and annotate and measure targets.

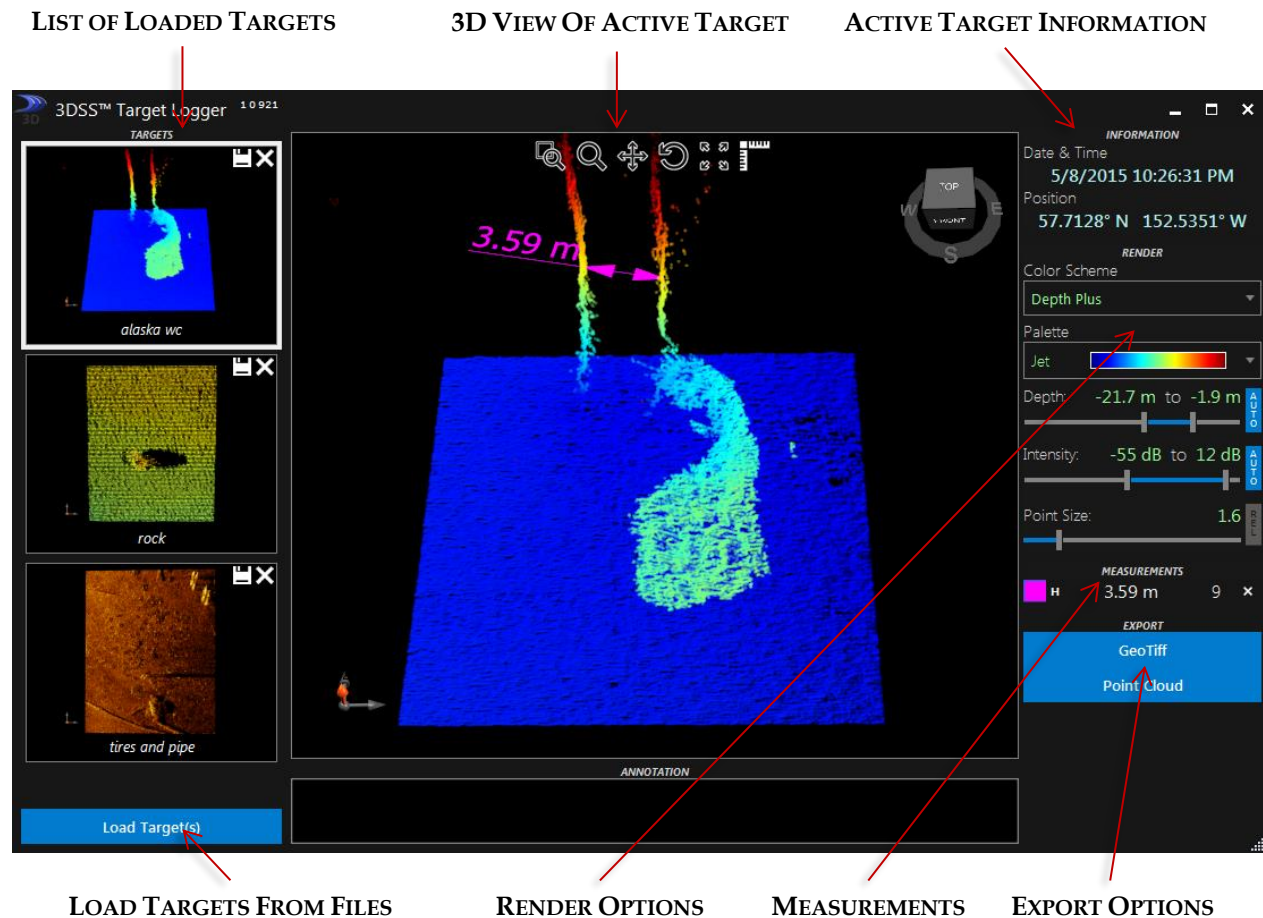


FIGURE 79. 3DSS TARGET LOGGER

LOADED TARGET LIST

The very left side of 3DSS Target Logger is a list of icons of loaded targets. Each icon has a preview image of the target, the filename of the target if it has been saved, a close button and a save target button (Figure 80). Clicking on the target icon loads it into main target view in the center of the window. At the very bottom of the list is a button to load previously saved targets from 3DSS Target files (.3dss-target).



FIGURE 80. TARGET PREVIEW ICON

INFORMATION

DATE & TIME

Displays the UTC date and time indicating when the target was captured by the sonar system.

POSITION

Displays the approximate GPS of the center of the target. This position is not corrected for any offsets between the GPS and the 3DSS sonar; hence it is only accurate to roughly the distance between the GPS and 3DSS sonar.

RENDER OPTIONS

The render options here control how the target is viewed in the center of the window. The controls are identical to those in the main 3D display.

MEASUREMENTS

The 3DSS Target Logger has the ability to make 3 different distance measurement between any two points. By click on the ruler in the toolbar you can cycle through the three measurement modes and perform:

1. Horizontal distance measurements. 
2. Vertical distance measurements. 
3. Euler distance measurements. 

Once a measurement mode is selected (as indicated by a highlighted ruler icon with either H, V, or E) a measurement can be performed by clicking on two points in the 3D point cloud. The measurement can be aborted by pressing the ESC key. Figure 81 shows an example of two horizontal measurements and one vertical measurement performed on a target (J.E. Boyden in Lake Union, WA). The measurement leaders and values are drawn in the 3D target view, and a

list of measurements is also available on the right side panel. The list facilitates the modification of the measurement leader size, text size, color, and also allows for the removal of measurements.

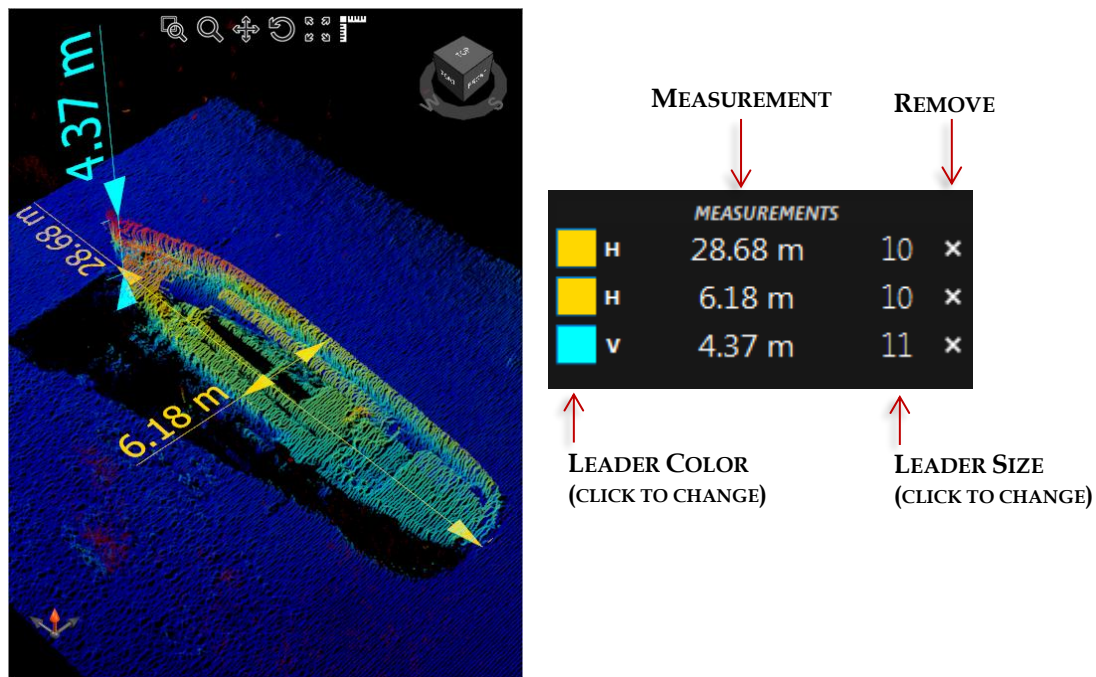
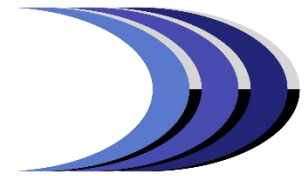


FIGURE 81. EXAMPLE TARGET MEASUREMENTS

EXPORT OPTIONS

Targets can be exported to both GeoTIFF and Point Cloud Data (PCD).

1. GeoTIFF is a public domain standard which defines how to embed georeferencing information within a TIFF image file.
2. Point Cloud Data (.PCD) is an ASCII format which each point in the point cloud on a separate line in the file. Each point has the X, Y, and Z coordinates of the point in meters, as well as an intensity value I in decibels. The PCD point cloud is not georeferenced, however the ASCII header in the file does include latitude and longitude information for the target. More information about the PCD format can be found at http://pointclouds.org/documentation/tutorials/pcd_file_format.php.



CHAPTER 5: PROFILE DISPLAY

PROFILE DISPLAY PROGRAM

The **Profile** display program is provided with the **3DSS-DX-450** sonar system for both real-time and post-time bathymetry cross-section display. Bathymetry data received from the **3DSS-DX Sonar Control** program over TCP is displayed as it arrives, as is the state of the bottom track filter.

The Profile display shows both the raw 3D sidescan profiles and the bottom tracked and equidistant binned bathymetry output. The port and starboard raw 3D sidescan points are colored by the return intensity. The generated bathymetry for port and starboard is colored in red and green respectively. Bathymetry and 3D point sizes can be adjusted in the settings menu.

Bathymetry Mode set to Legacy, the position of the bottom track filter is drawn in white, and any raw 3D sidescan points falling outside of the bottom track cells are discarded. In Mbes Bathymetry Mode, the EA and ED sectors are drawn.

The **Profile** display must be connected to the **Control** program to start displaying profiles. To accomplish this, the **CONNECT** button on the top toolbar must be clicked (see Figure 82). The Sonar is positioned at (0,0) indicated by a yellow icon.

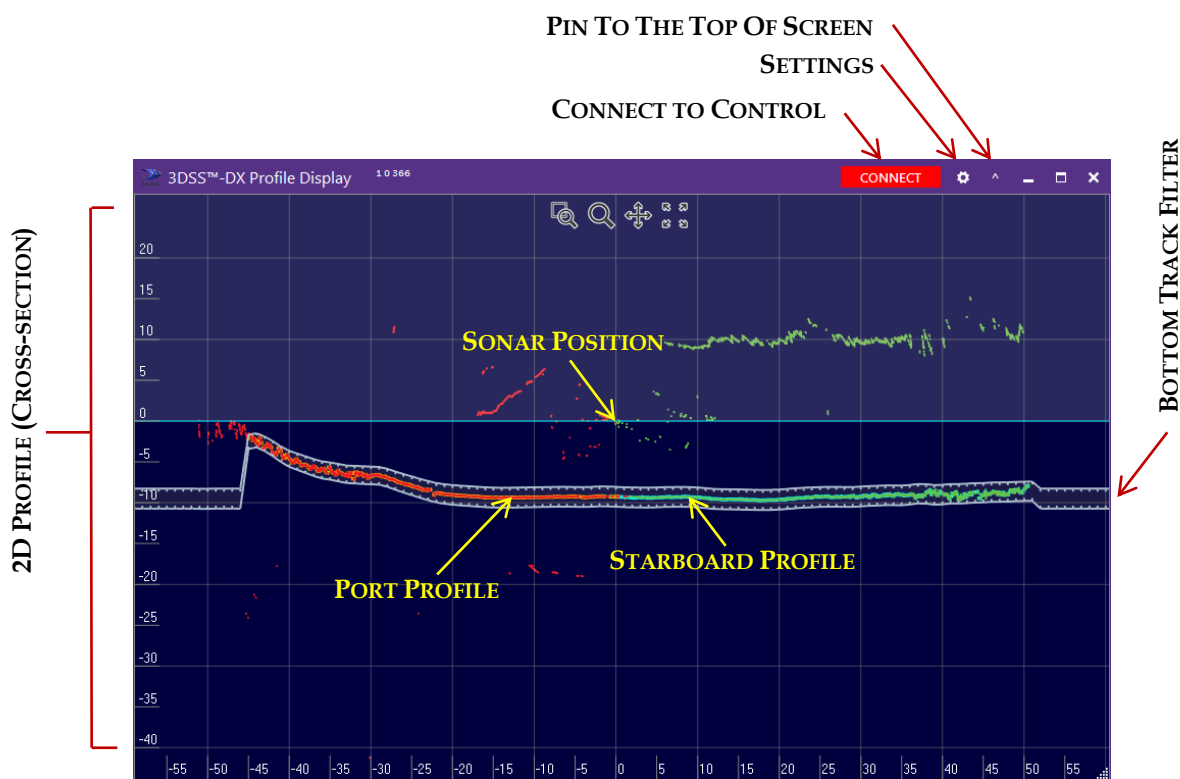


FIGURE 82. THE 3DSS-DX PROFILE DISPLAY IN LEGACY BATHYMETRY MODE

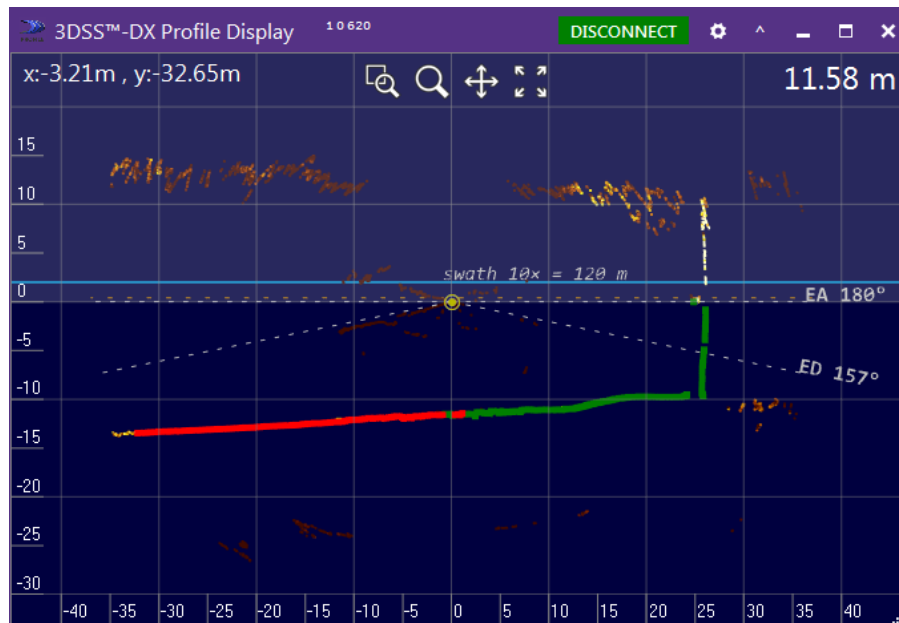


FIGURE 83. THE 3DSS-DX PROFILE DISPLAY IN MBES BATHYMETRY MODE

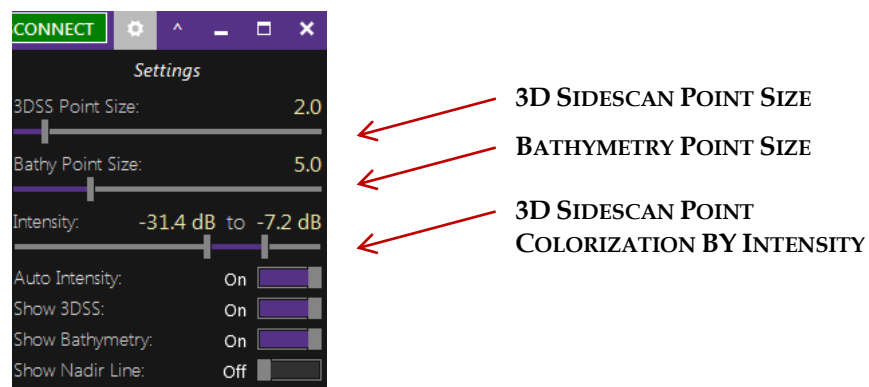


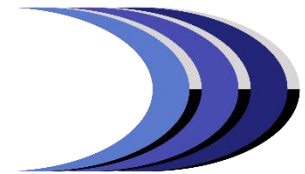
FIGURE 84. 3DSS-DX PROFILE DISPLAY SETTINGS

There are 6 settings which affect how the points in the profile display are drawn:

1. 3DSS Point Size – The size of the 3D sidescan points in pixels. The value can be set to 0.1 to disable drawing the 3D sidescan points.
2. Bathymetry Point Size – The size of the bathymetry points in pixels. The value can be set to 0.1 to disable drawing the bathymetry points.



3. Intensity – The intensity range to use for colorizing the 3D sidescan points. The left slider sets the lowest intensity displayed and colorized blue, while the right slider sets the highest intensity displayed and colorized red. This scale is the same as the scale in the 3D Display.
4. Auto Intensity – Automatically adjusts the Intensity slider on each ping in-order to best display the dynamic range of the 3D sidescan points.
5. Show 3DSS – Enables the drawing of 3D sidescan points.
6. Show Bathymetry – Enables the drawing of bathymetry points.



CHAPTER 6: SOUND VELOCITY BROADCAST

NOTE: This section only applies to the 3DSS-DX. The 3DSS-iDX does not require the installation and operation of the Sound Velocity Broadcast application.

SOUND VELOCITY BROADCAST APPLICATION

The **Sound Velocity Broadcast** application (3DSS™ SVB) is provided with the **3DSS-DX-450** sonar system and is used to transmit the face sound velocity to the sonar. The application connects to a sound velocity probe through a serial port on the computer running the application and receives then broadcasts sound velocity measurements over UDP to the 3DSS-DX Control application and 3DSS-DX sonar.

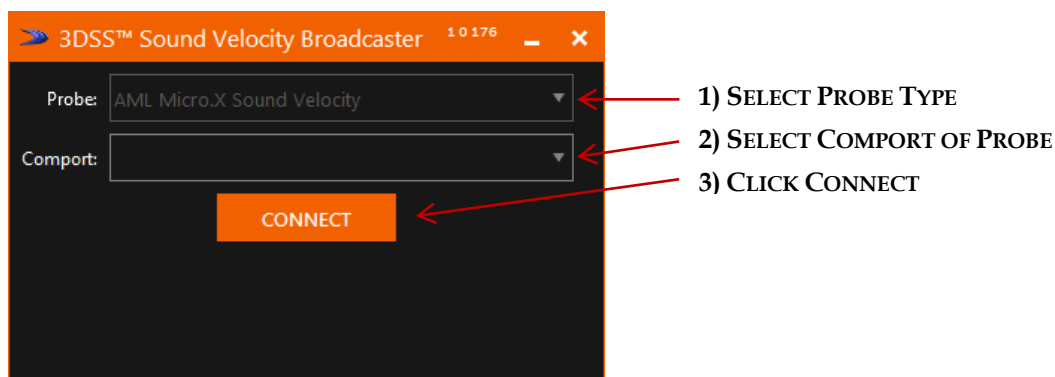
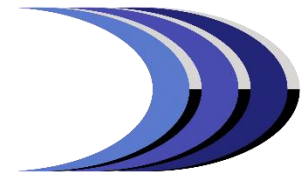


FIGURE 85. THE 3DSS SOUND VELOCITY BROADCAST APPLICATION

To use the application, simply select the **PROBE** type, the **COMPORT** on which the probe is connected, and click **CONNECT**. The application will immediately start displaying and broadcasting the measured sound velocity. To use this sound velocity, the sound velocity Source must be set to *Probe* in the **3DSS-DX Control** program.

Currently, the AML Micro-X Sound Velocity probe is supported by the application, as well as any sound velocity probe that continuously outputs the a decimal (text) sound velocity at 9600 bps. To use other sound velocity probes please contact Ping DSP Inc.

In order for the application to function properly the AML Micro-X Sound Velocity probe must be set to use either auto baud-rate or fixed 9600 bps.



CHAPTER 7: 3DSS TO XTF CONVERTER

3DSS TO XTF CONVERTER PROGRAM

The **3DSS to XTF Converter** program is a step-by-step wizard that converts data from within 3DSS-DX files into the XTF file format. The converter can generate 3 types of XTFs:

1. 2D Sidescan XTFs which contain only 2D backscatter imagery.
2. 3D Sidescan XTFs which contain the full water-column high resolution 3D sidescan.
3. Bathymetry XTFs which contain bottom-tracked and equidistant binned bathymetry.

STEP 1: FILE SELECTION

The first step in using the XTF converter is to choose the 3DSS-DX files you wish to converter. The generated XTF files will be placed in the same directory as the source 3DSS-DX files and will have the same filename but with a .XTF extension instead.

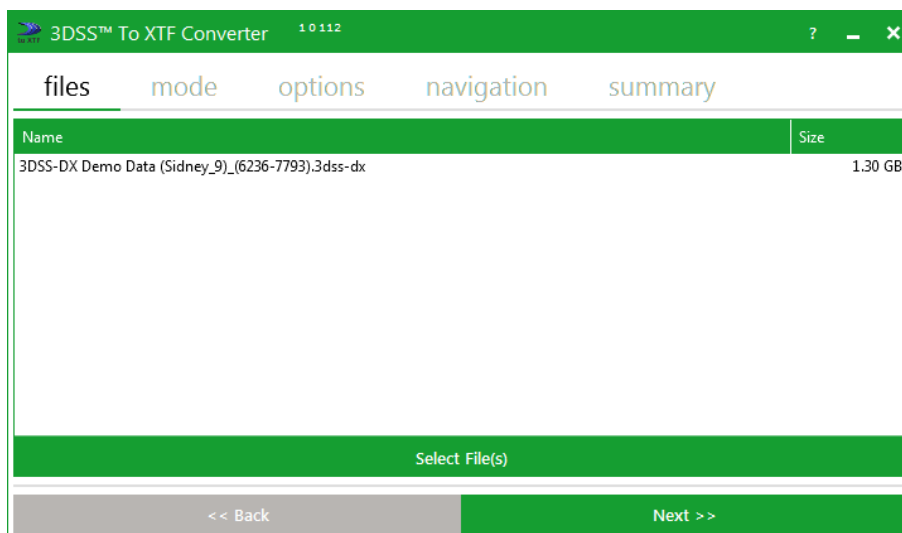


FIGURE 86. XTF CONVERTER STEP 1, FILE SELECTION.

Once the files you wish to convert are selected press the Next button to move to the next step.

STEP 2: CONVERSION MODE

The conversion mode determines what type of data you'd like to place in the XTF. The three choices are 2D Sidescan data, 3D Sidescan data, or Bathymetry data. Select the appropriate mode and click Next.

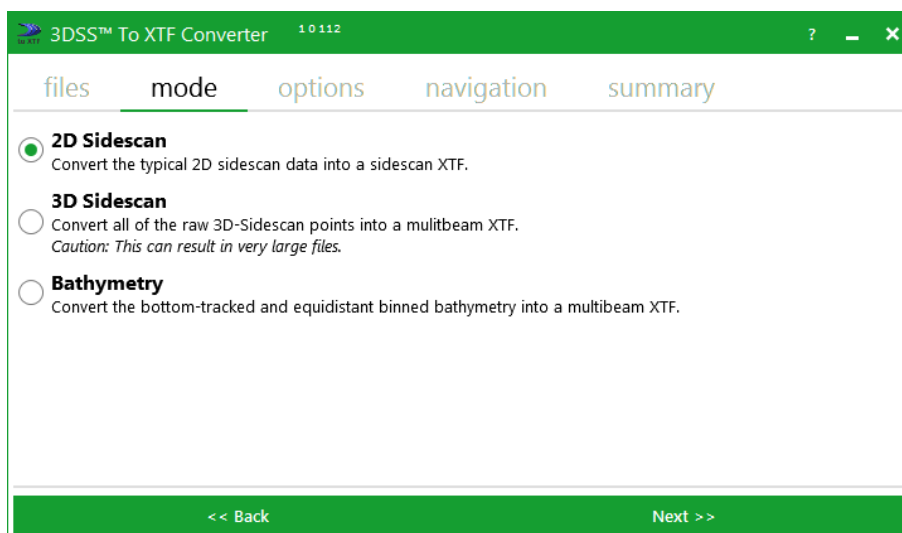


FIGURE 87. XTF CONVERTER STEP 2, MODE.

STEP 3: OPTIONS

The third step is a set of options and varies depending on the converter mode.

For 2D sidescan, the options are:

1. Sample Size – specifies the size in bits of each intensity value in the 2D sidescan XTF. Not all third-party software supports 32-bit sample size, hence 16-bit sample size is recommended as it is most compatible.
2. Scale Samples – specifies the scaling to apply to samples to fit them into a 16-bit sample size. The 3DSS is a 24-bit+ sonar system, meaning that the full dynamic range of the sonar exceeds the 16-bits available in an XTF and hence the 24-bit value must be scaled. There are two types of scaling options:
 - a. Fixed scaling where each sample is divided by either 4, 16, 64, or 256.
 - b. Auto scaling where the backscatter for the first few pings in each file is analyzed and an automatic scaling value is determined. Auto 1% and Auto 0.1%.

The recommended setting is either Auto 1% or Auto 0.1%.

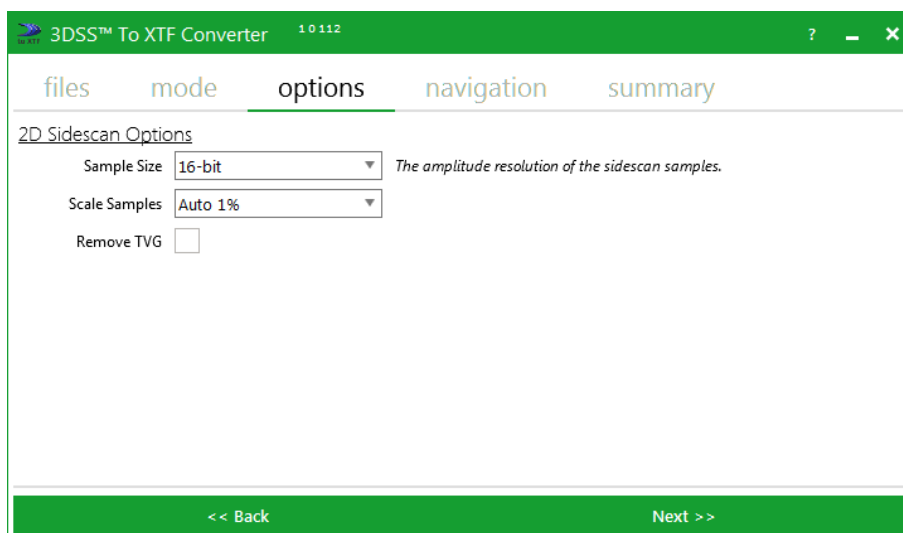


FIGURE 88. XTF CONVERTER STEP 3, 2D SIDESCAN OPTIONS.

For 3D sidescan, the options are:

1. Beam or Point Reference – specifies whether to reference the points to the sonar's nadir direction or each transducer's broadside direction. This option is illustrated in Figure 89.
2. Minimum Depth Filter – specifies a cut-off above which points are rejected. This is used to reject sea surface returns and multipath coming from the sea surface. Typically this setting is set to either 0m to reject above surface points or to 1m to reject points very close to the sea surface.
3. Minimum Amplitude Filter – rejects weak points and provides water reverberation suppression. Amplitude cut-off expressed as a multiple of the average amplitude of the ping. Typically this setting is either 0.2 or 0.3, or 0.0 to disable the filter.
4. Point Decimation – specifies a decimation option to reduce the number of points. The number of points in a 3D sidescan ping can approach over 10,000. Low decimation reduces this number by approximately a factor of 2, medium decimation by a factor of 3, and high decimation by a factor of 4.

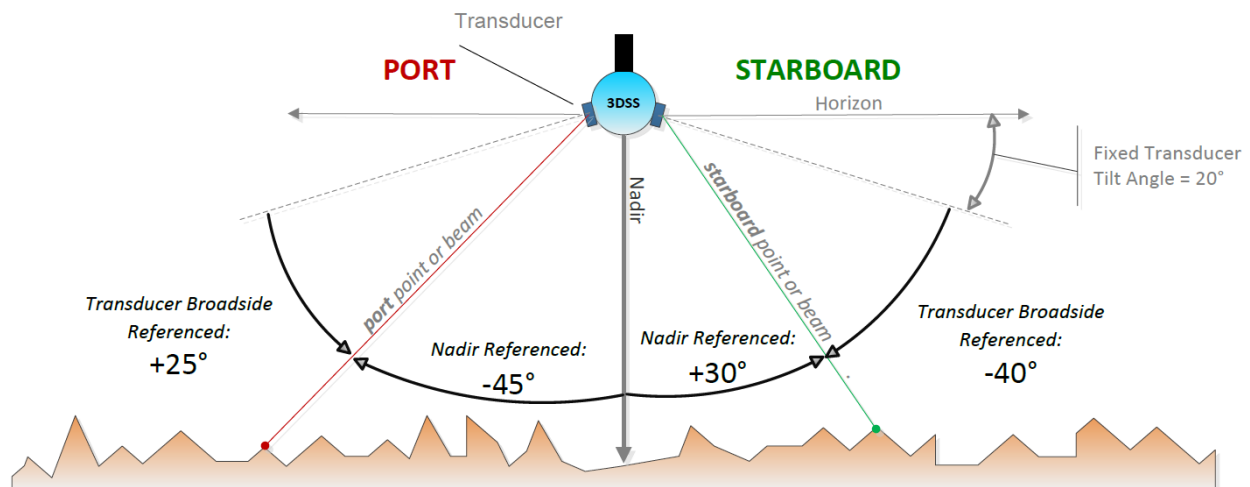


FIGURE 89. XTF CONVERTER BEAM REFERENCE FOR 3D SIDESCAN AND BATHYMETRY

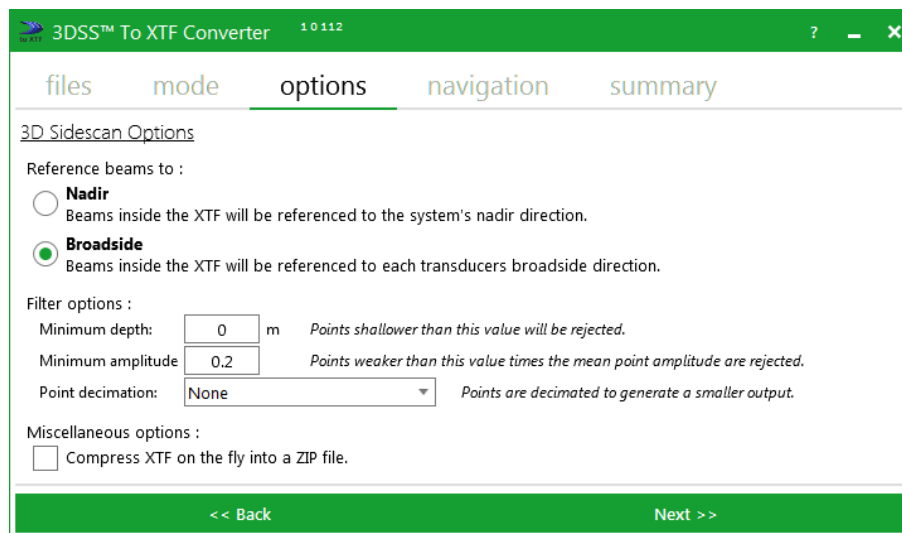


FIGURE 90. XTF CONVERTER STEP 3, 3D SIDESCAN OPTIONS

For Bathymetry, the only option currently is:

1. Beam or Point Reference – specifies whether to reference the points to the sonar's nadir direction or each transducer's broadside direction. This option is illustrated in Figure 89. *It is important to note that this selection affects the required vessel configuration settings in 3rd party software when importing the XTF data.*

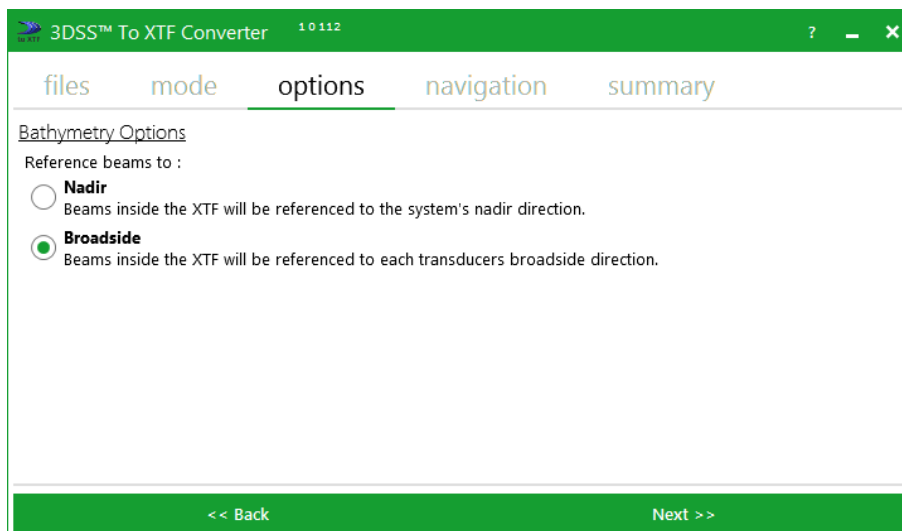


FIGURE 91. XTF CONVERTER STEP 3, BATHYMETRY OPTIONS

STEP 4: NAVIGATION

The fourth step specifies which navigation data sources should be used when converting to XTF.

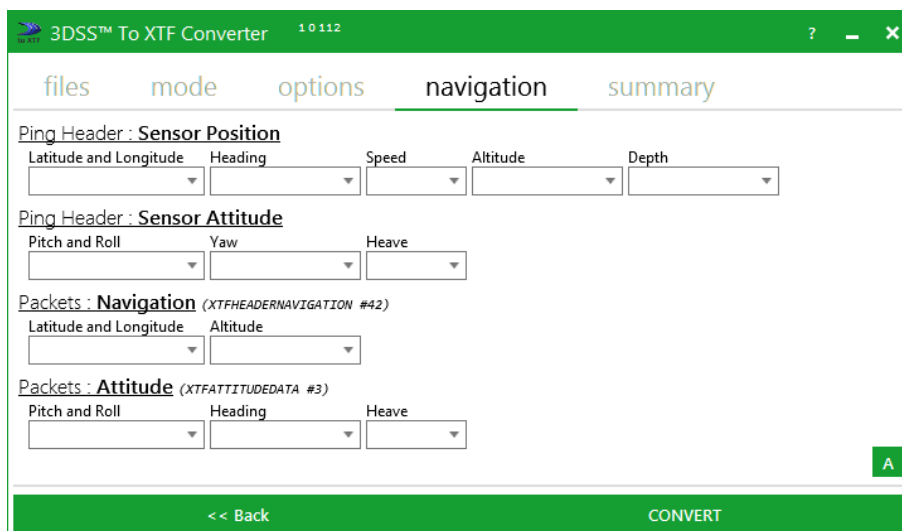


FIGURE 92. XTF CONVERTER STEP 4, NAVIGATION OPTIONS

In XTF, each ping can have a **Sensor Position** and **Attitude** specified. The .3dss-dx file format stores all NMEA sentences received by the sonar head between the start and end of a ping. This means



that if a NMEA position sentences such as \$GPGGA has been received during a ping, the sentence can be parsed and inserted as the sensor position in the XTF. If a position sentence has not been received, the XTF sensor position fields are left as 0. **Sensor Attitude** is typically provided at a rate much faster than the ping rate and will be present in every ping.

Currently the XTF converter is able to parse the following NMEA sentences as well as external TSS1 MRU:

1. For Latitude and Longitude:

\$GPRMC	(Minimum recommended GNSS data)
\$GPGGA	(Detailed GNSS position information)
\$GPGGK	(Detailed GNSS position RTK information)
\$OTPOS	(OceanServer custom position sentence)
SBG.NAV	(SBG Ellipse-2E navigation packets)

2. For Heading:

\$GPRMC	(Course-made good, not true heading)
\$GPVTG	(Course over ground and ground speed)
\$GPHDT	(True heading of the vessel / GPS antenna)
\$OTPOS	(OceanServer custom sentence)
SBG.EULER	(SBG Ellipse-2E Euler packets)

3. For Speed:

\$GPRMC	(Ground speed)
\$GPVTG	(Course over ground and ground speed)
\$OTPOS	(OceanServer custom sentence)
SBG.SHIP	(SBG Ellipse-2E ship motion packets)

4. For Altitude:

\$OTPOS	(OceanServer custom sentence)
\$xxGGA	(Detailed GNSS position information)
SBG.NAV	(SBG Ellipse-2E navigation packets)

5. For Depth:

\$OTPOS	(OceanServer custom sentence)
---------	-------------------------------

6. For Pitch and Roll:

Internal MRU	(MRU internal to the sonar head)
TSS1	(External MRU – TSS1 pitch and roll)
\$OTPOS	(OceanServer custom sentence)
SBG.EULER	(SBG Ellipse-2E Euler packets)



7. Yaw:

\$GPRMC	(Course made good)
\$OTPOS	(OceanServer custom sentence)
SBG.EULER	(SBG Ellipse-2E Euler packets)

8. Heave:

TSS1	(External MRU – TSS1 heave)
SBG.SHIP	(SBG Ellipse-2E ship motion packets)

The XTF file can also contain Navigation and Attitude records independent of a ping. The XTF converter generates a Navigation record (*XTFHEADERNAVIGATION #42*) whenever it encounters the specified NMEA sentence with latitude and longitude. Similarly, the XTF converter also generates Attitude records (*XTFATTITUDEDATA #2*) whenever it encounters the specified NMEA or TSS pitch and roll sentence (the heading is sourced from the nearest sentence which contains the heading).

For each XTF field, a NMEA or TSS source can be selected by clicking on the drop-down box:

Ping Header : Sensor Attitude

Pitch and Roll	Yaw	Heave
None (0) Internal MRU :TSS1		None (0)
	(XTFHEADERNAVIGATION #42)	
	Altitude	

The drop-down box lists only the available sentences in the file to be converted, and orders them in a least to most preferred order. In the example above, the TSS1 is last on the list as it is assumed to be the most reliable pitch and roll data source.

Finally, the window has a small **A** button in the bottom right corner that automatically selects the last entry in every drop-down box.

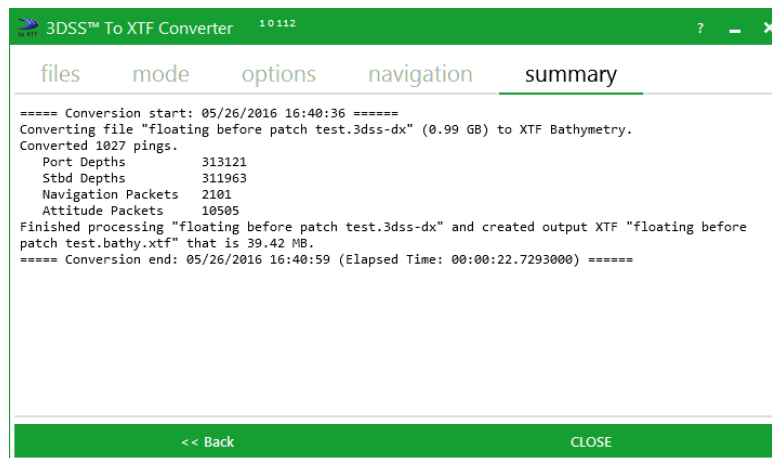


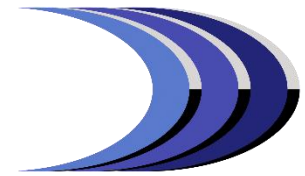
STEP 5: CONVERSION AND SUMMARY

The final step is to click the Convert button and wait. The converter will create an XTF file in the same folder as the original .3dss-dx data file. Depending on the conversion mode the XTF file will be named either:

<original name>.sidescan.xtf
<original name>.sidescan3d.xtf
<original name>.bathy.xtf

A summary report will be displayed at the end of the conversion:





CHAPTER 8: CHART DISPLAY

3DSS™ CHART DISPLAY

The 3DSS Chart Display is an OpenCPN based plugin that overlays bathymetric data from the 3DSS-DX-450 sonar onto a chart.

CHART DISPLAY FEATURES

- Displayed bathymetric data is pitch and roll corrected using the 3DSS-DX-450's internal MRU and is binned prior to being displayed.
- Bathymetry colorization can be adjusted by setting a minimum and maximum depth.
- The 3DSS™ Chart Display supports both real time acquisition and file playback.
- Provides the option to save and load bathymetry overlay for future reference.
- Waypoints are automatically placed when playing back from or recording to a new file; showing where each file begins and ends on the map.
- OpenCPN provides navigational and survey planning features and supports a variety of external plugins.

USING THE 3DSS™ CHART DISPLAY

Installation and configuration of OpenCPN and the Chart Display software is covered in Chapter 1 of this manual under "Software Installation". To run the 3DSS Chart Display, first click on the PingDSP icon  in the OpenCPN toolbar. This brings up the 3DSS™ Chart Display GUI. Pushing the "Start" button will activate the software and 3DSS data it receives will be plotted on the OpenCPN display.

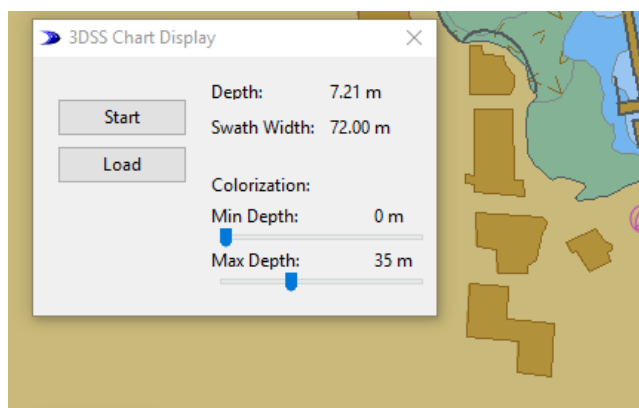


FIGURE 93 - 3DSS™ CHART DISPLAY GUI

For this software to receive and plot sonar data the 3DSS-DX Sonar Control application must be running and processing data (either in real time or from file playback). If running the sonar in real time, the 3DSS™ Chart Display will only plot sonar data if data is being recorded in the 3DSS™ Sonar Control Software. This is to ensure coverage shown is indicative of data captured from the sonar.

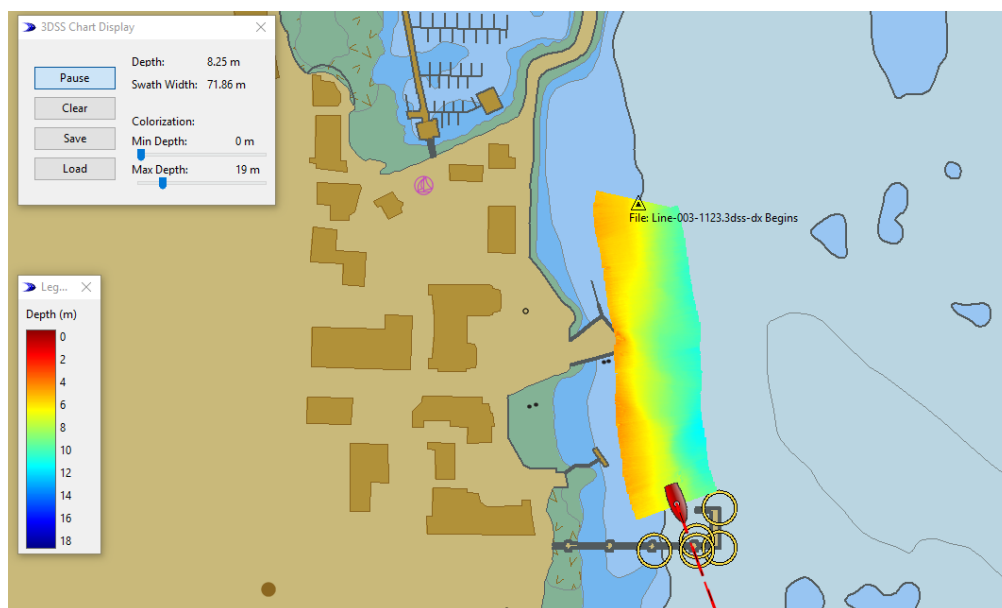


FIGURE 94 - 3DSS™ CHART DISPLAY SHOWING OVERLAY AND WAYPOINTS

When the software is running, a legend will be displayed showing how the colorization correlates to depth (relative to sonar). The minimum and maximum depth ranges can be selected via the “Min Depth” and “Max Depth” sliders. Changing the colorization sliders will cause the overlay to be re-drawn. A shallower depth range will make differences in depth more apparent but limit the range at which meaningful data is displayed.

For OpenCPN to automatically follow the boat, select the “Auto Follow” icon in the OpenCPN tool bar.

SAVING AND LOADING OVERLAY

Saving a 3DSS™ Chart Display overlay may create a large amount of files. These represent the individual tiles that are being overlaid in OpenCPN. It is recommended to save new surveys in separate folders and to save periodically. When loading, it is ok to select any file relating to the specified survey, this will load all the files associated with that survey.

WAYPOINTS

The 3DSS™ Chart Display software will set a waypoint whenever the file that is being played back or recorded to in the 3DSS™ Sonar Control Software changes, or if the Chart Display is started during file play or recording. Waypoints that are set within 3% of the beginning or end of a file will remain after OpenCPN is restarted. This acts as a log showing the start and end locations of all recorded 3DSS files.

Waypoints can be accessed through the OpenCPN “Route and Mark Manager” by clicking on the associated icon in the OpenCPN toolbar. A list of all waypoints is shown in the “waypoints”

tab where you can delete waypoints, center the view on a selected waypoint, or plot a course to a selected waypoint. Waypoints can also be exported to a .gpx file.

Waypoints can be added manually by right clicking a spot on the map and selecting “Drop Mark” or adding them in the “Route and Mark Manager”.

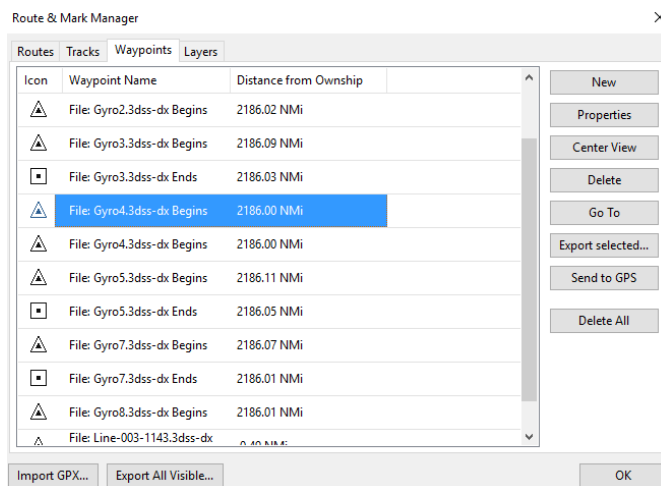


FIGURE 95- ROUTE & MARK MANAGER

ROUTES AND TRACKING

OpenCPN provides tools for creating routes as well as tracking where the vessel has been. Routes can be created by selecting “Create Route” icon in the OpenCPN toolbar. This allows the user to draw a route on the map. The route can then be “activated” in the “Route and Mark Manager” and OpenCPN will provide a navigational aid to assist in following the selected route.

Tracking can be toggled on and off by selecting the “Enable Tracking” Icon in the OpenCPN toolbar. With tracking enabled a red highlighted line will be drawn behind the vessel. Both routes and tracks can be accessed in the “Route and Mark Manager” and be imported and exported as .gpx files.

The figure below shows an example of both tracking and the route tool. In this case the route is “activated” and a navigational aid is shown on the right.

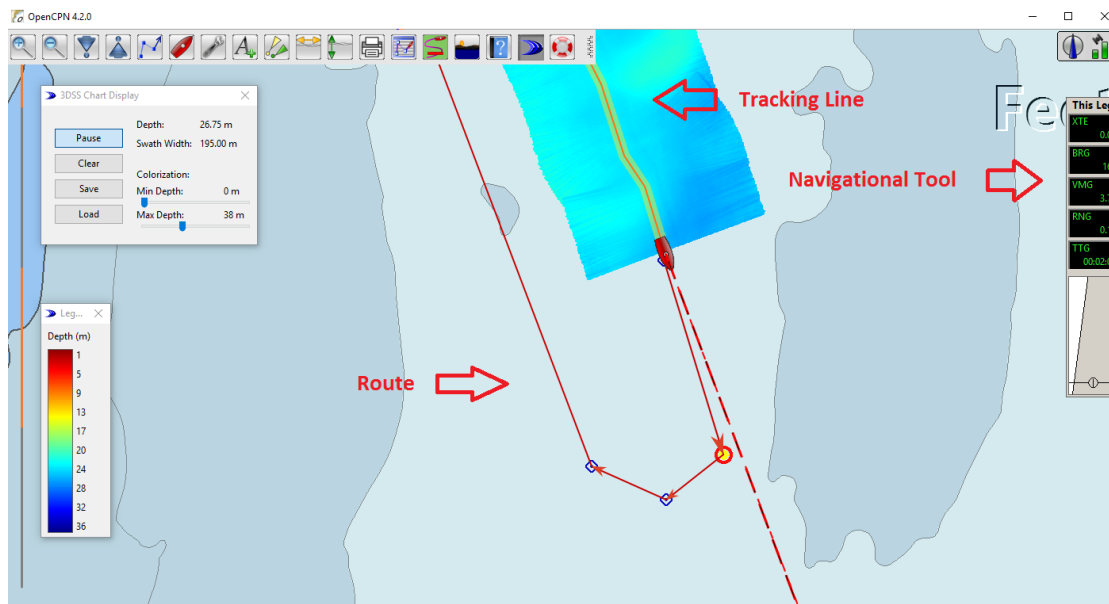


FIGURE 96 - ROUTES AND TRACKING

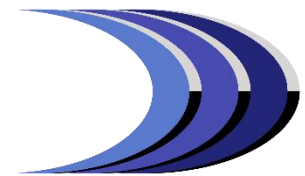
OPENCNP AND LICENSE

The OpenCPN user manual can be found at http://opencpn.org/ocpn/opencpn_manual.

Additional plugins for OpenCPN can be found at <http://opencpn.org/ocpn/downloadplugins>.

OpenCPN source code can be found at <https://github.com/OpenCPN/OpenCPN>.

OpenCPN is open source software and is protected by the GNU General Public License. License information can be found at <http://opencpn.org/ocpn/gpl>.



CHAPTER 9: INTERFACING WITH HYPACK

HYPACK SETUP

To use the 3DSS-DX Sonar with HYPACK 2015 requires:

1. HYPACK 2015
2. HYPACK 2015 July 2015 Supplemental Install (or newer). The update is installed by unzipping it and overriding all the HYPACK files in the C:\HYPACK 2015 folder.
[ftp://ftp.hypack.com/outgoing/support](http://ftp.hypack.com/outgoing/support)
3. The newest HYSWEEP 3DSS-DX driver available at: www.pingdsp.com/files
4. The newest GPS.DLL driver available at: www.pingdsp.com/files

To use the 3DSS-DX Sonar with HYPACK 2016 or newer requires:

1. HYPACK 2016 or newer

Once HYPACK is installed and updated, please open HYPACK and click on **PREPARATION -> HARDWARE SETUP** and add the **PING DSP 3DSS-DX** under the **HYSWEEP SURVEY** tree as seen in Figure 97.

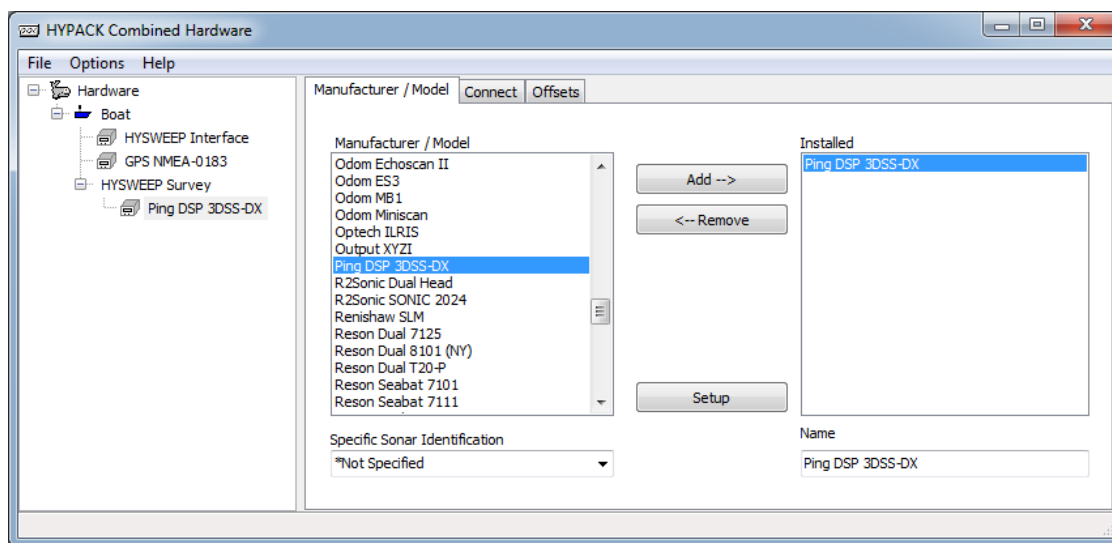


FIGURE 97. HYPACK COMBINED HARDWARE SETUP

Clicking setup while the **PING DSP 3DSS-DX** is selected brings up a menu where the HYSWEEP 3DSS-DX driver can be configured to use specific data sources for attitude. This menu is shown below in Figure 98.

The FORMAT option selects between RAW and BINNED data. The RAW option uses raw Sidescan-3D data and the interferometry processing of HYSWEEP (both filters and binning). If using the RAW mode, please familiarize yourself with the interferometric filters and binning options available in HYSWEEP. **Warning:** The interferometric filters are **not** enabled in HYSWEEP by default.

The BINNED option uses the bottom-tracked and binned bathymetry output of the 3DSS-DX and skips interferometry processing in HYSWEEP. **Important:** The BINNED option is the **preferred** method of capturing data in HYSWEEP.

The MRU option selects between \$VNYCM and TSS for real time pitch, roll, and heave correction. The \$VNYCM setting uses the basic MRU internal to the 3DSS-DX Sonar, while the TSS options uses the data provided by an external MRU (in TSS1 format) and connected to the 3DSS-DX Topside Box MRU input.

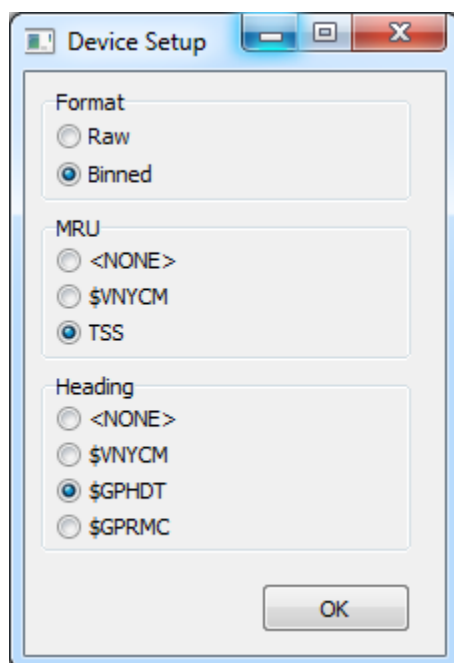


FIGURE 98. HYSWEEP PING DSP 3DSS-DX DRIVER OPTIONS.

The heading option selects between \$GPHDT and \$GPRMC. The \$GPHDT option uses the heading provided by an external GPS connected to the 3DSS-DX Topside Box, while the \$GPRMC option uses the course made good field in \$GPRMC NMEA sentences.

Typically, these two settings are set to TSS and \$GPHDT respectively. However, if the MRU/INS solution used has a HYSWEEP driver (such as Applanix PosMV), then both settings can be set to <NONE> and the MRU/INS driver used under HYSWEEP and Hardware.

Finally, the IP address and port number over which the **HYSWEEP 3DSS-DX Driver** connects to the **3DSS-DX Control** application needs to be specified under the CONNECT tab.

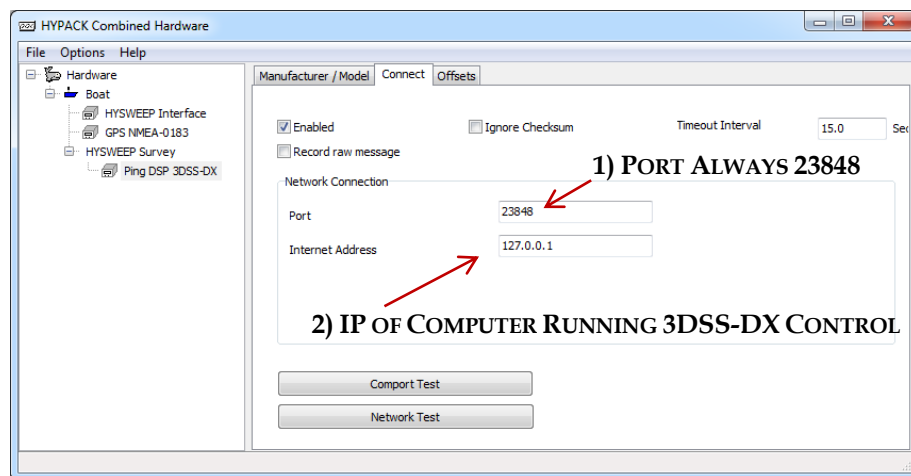


FIGURE 99. HYSWEEP 3DSS-DX DRIVER CONNECT TAB.

The IP address must be set to the IP address of the computer running **3DSS-DX Control**, or to **127.0.0.1** if HYPACK and the **3DSS-DX Control** program are run on the same computer. The port number must always be set to **23848**.

Under the OFFSETS tab, the offset from the port and starboard transducers to the GPS position can be entered. SONAR HEAD 1 refers to the port transducer and SONAR HEAD 2 refers to the starboard transducer. The distance between the acoustic centers of two transducers is 8 cm. **Please see Appendix for drawing showing the acoustic center of the 3DSS sonar.**

HYPACK GPS DRIVER

The **3DSS-DX Control** program has the ability to forward any GPS NMEA sentences received on the GPS input into HYPACK. These NMEA sentences are time-stamped by the sonar and the HYPACK GPS NMEA-0183 driver (GPS.DLL) uses the timestamps when placed in a vendor specific Ping DSP mode.

To use the HYPACK GPS NMEA-0183 driver please add it under the BOAT in the HYPACK COMBINED HARDWARE setup as seen in Figure 99. Once added, select the driver and click on the SURVEY CONNECT tab as seen in Figure 100. In this tab, set the DEVICE CONNECTION to the following settings:

- Connection Type : Network
- Protocol : TCP
- Role : Client
- Host : IP address of computer running 3DSS-DX Control
- Port : 23849

Finally the driver must be placed in vendor specific mode. Under the SURVEY DEVICES tab, click the SETUP button with the GPS NMEA-0183 driver selected. In the GPS SETUP (Figure 101), under the

ADVANCED tab, change THE USER MODIFIED NMEA MESSAGES setting to PING DSP. You may also select which sentence should be accepted in the USED SENTENCES box.

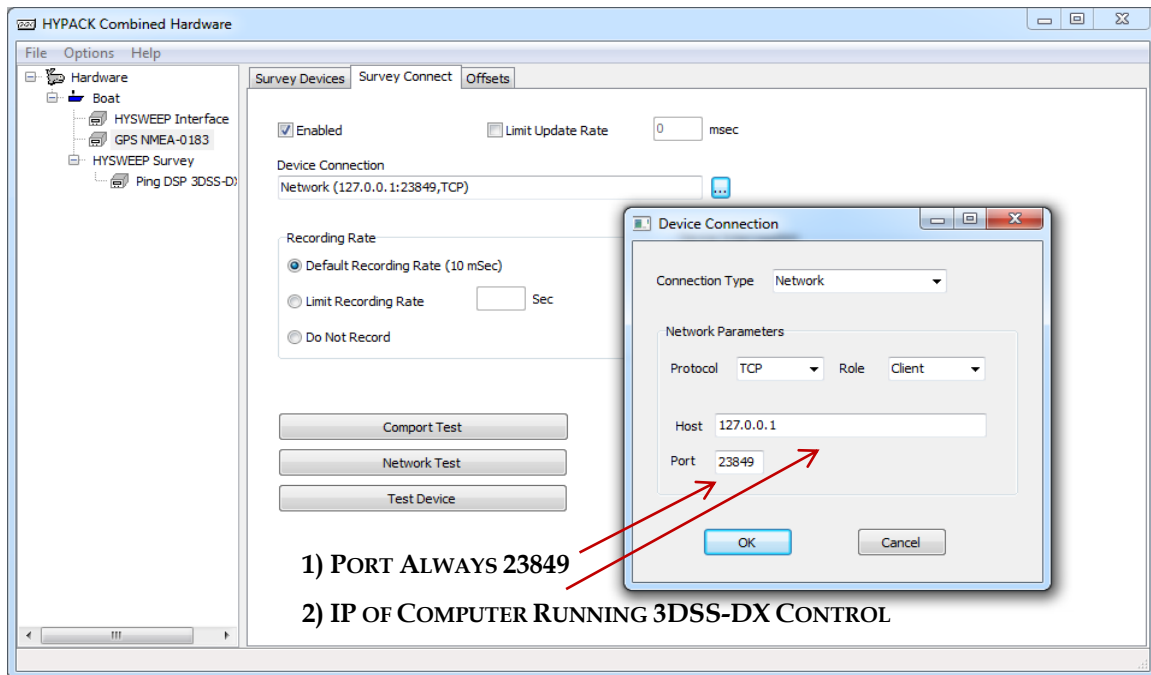


FIGURE 100. HYPACK GPS DRIVER NETWORK CONFIGURATION

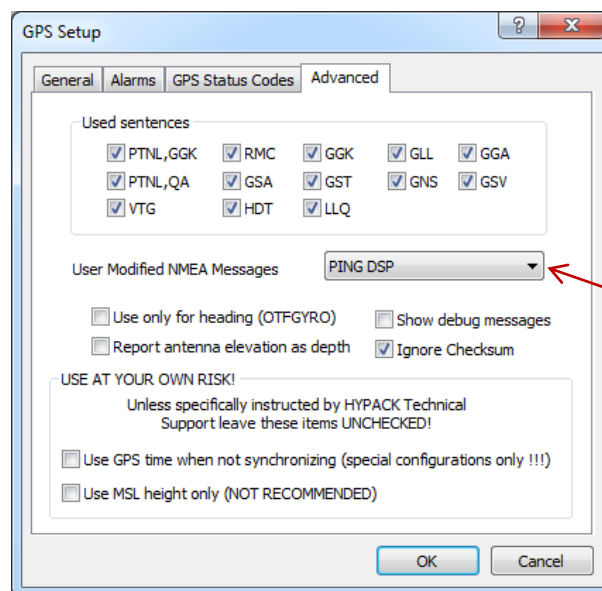


FIGURE 101. HYPACK GPS DRIVER SETUP

HYPACK WITH INTEGRATED SBG INERTIAL NAVIGATION SYSTEM

The 3DSS-iDX variant features an integrated SBG INS that can use HYPACK's SBG drivers to provide additional information to HYSWEEP including error estimates. It is recommended to use the INS sensor specific driver and the instruction below apply to both the 3DSS-iDX-BASE/FULL and the 3DSS-iDX-PRO.

HYPACK SBG INS DRIVER

The SBG Ekinox driver in HYPACK can be used for both the SBG Ellipse and the SBG Ekinox. The driver provides positioning information and connects to both the Ellipse and Ekinox over UDP. To configure HYPACK Combined Hardware Setup for the 3DSS-iDX, open Preparation => Hardware Setup, and add the SBG Ekinox under Boat. Configure the Device Connection with the following parameters:

- Connection Type: Network
- Protocol: UDP
- Role: Client
- Host: 192.168.228.1
- Port: 24333
- Write Port: 0

A screenshot of this configuration is shown below:

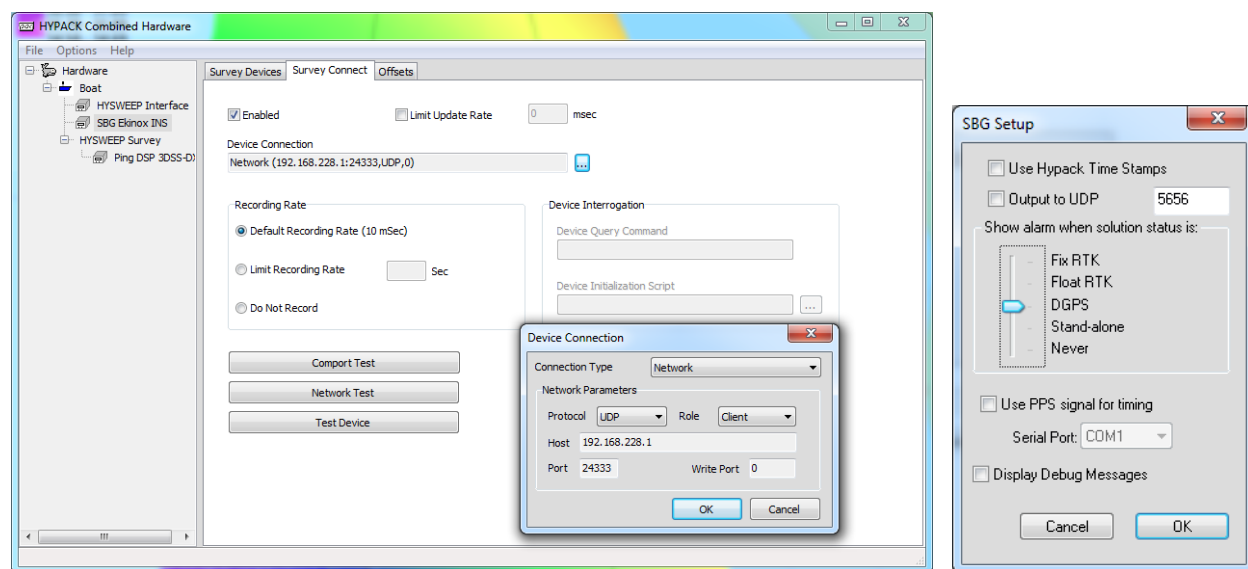


FIGURE 102. SBG INS CONFIGURATION IN HYPACK

The SBG Setup does not need to output a copy of the messages on UDP port 5656, and should not need to use HYPACK Time Stamps since all data (INS and sonar) will be synchronized to GPS time.

HYSWEEP SBG INS DRIVER

A second SNG driver needs to be added under HYSWEEP Survey to provide attitude data to HYSWEEP. The SBG Systems Ekinox driver can be used for both the SBG Ellipse and SBG Ekinox. The driver should be added under HYSWEEP survey and configured as shown in the screenshot below:

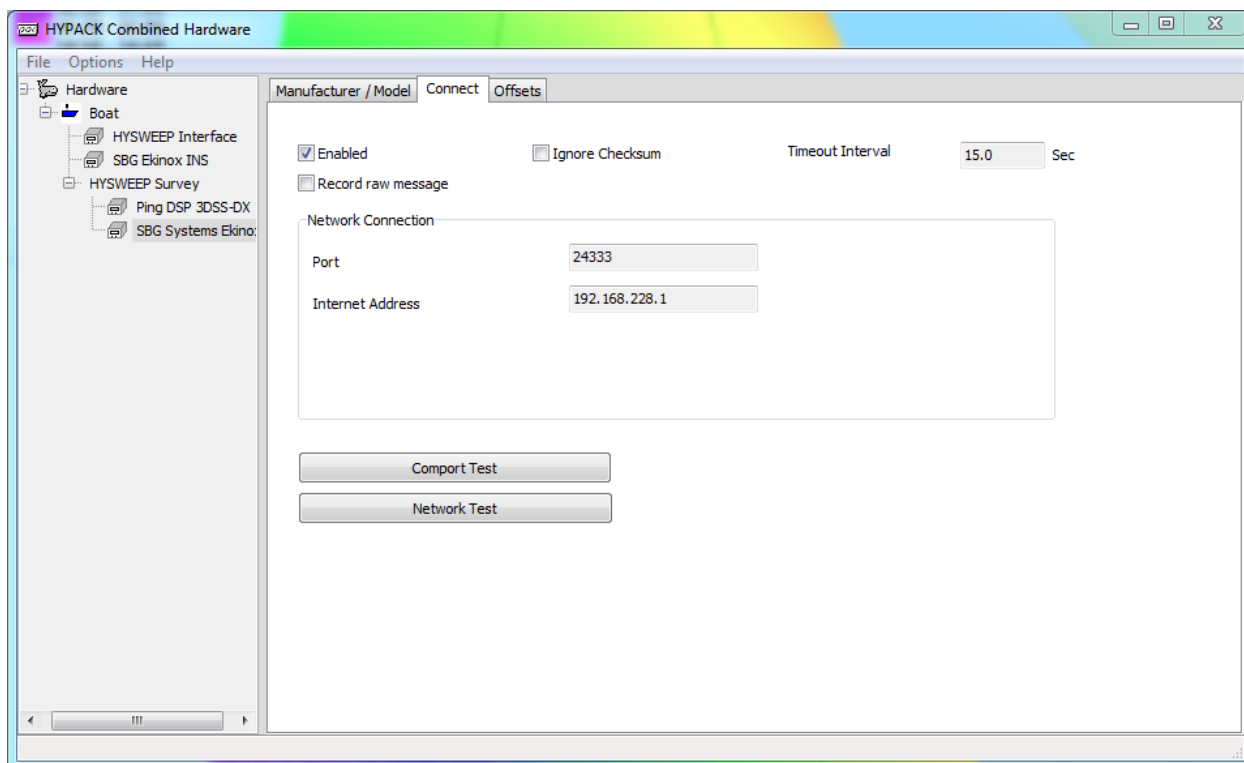


FIGURE 103. SBG INS CONFIGURATION FOR HYSWEEP

The port number and internet address will be the same as in the other driver with the port number being 24333, and the Internet address will be 192.168.228.1.

IMPORTANT NOTE:

Additionally, the Ping DSP driver needs to be reconfigured to NOT provide attitude data to HYSWEEP. Select the Ping DSP 3DSS-DX driver and click Setup, and configure the driver to not provide MRU and Heading information by clicking on **<NONE>** in both cases. This configuration is shown in the screenshot below:

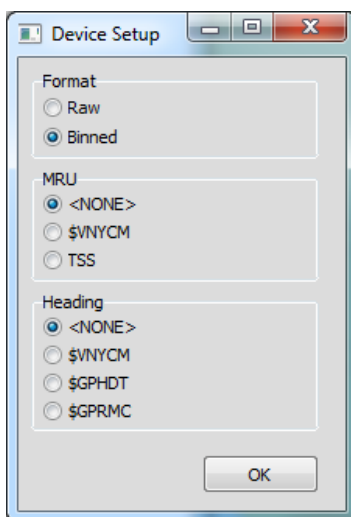


FIGURE 104. HYSWEEP 3DSS CONFIGURATION WITH ATTITUDE DATA DISABLED

That completes the HYPACK hardware configuration for the 3DSS-iDX.

As a final step, the SBG INS needs to be selected as the source of Heading, Heave, and Pitch / Roll information within HYSWEEP. To do this, launch HYSWEEP Survey and in the HYSWEEP Survey window click View => Device Selections..., and select the SBG Systems Ekinox as the source for Heading, Heave, and Pitch/Roll. This is shown in the screenshot below:

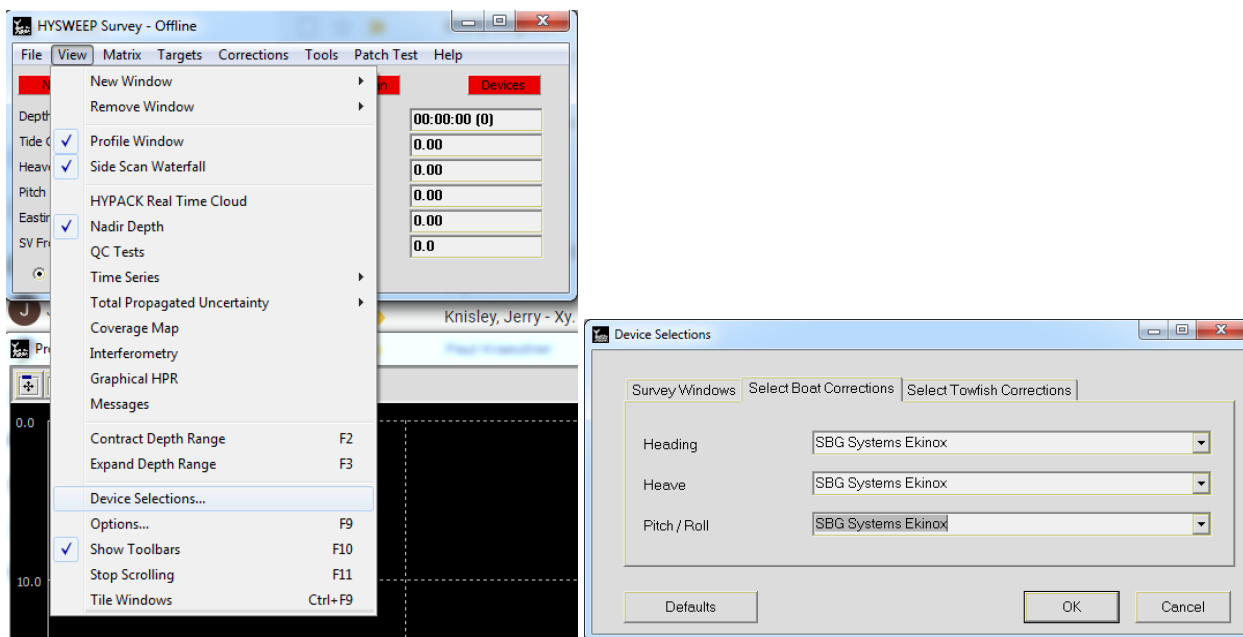
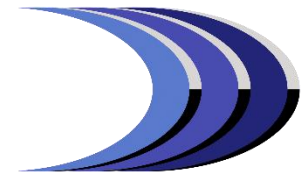


FIGURE 105. HYSWEEP DEVICE SELECTIONS - SBG INS



CHAPTER 10: INTERFACING WITH SONARWIZ

SONARWIZ SETUP

The 3DSS-DX Sonar is compatible with SonarWiz 6 or higher. To use the 3DSS-DX Sonar with SonarWiz requires:

1. Chesapeake SonarWiz 6 or higher.
2. Chesapeake 3DSS-DX Bathymetric Server (V6.03.002 or newer) , available from Chesapeake support website:

<http://www.chestech-support.com/ctisupport/sonarWizServerDownloads.asp>

Once the 3DSS-DX Bathymetric Server is installed, it should be visible under Sonars in SonarWiz:

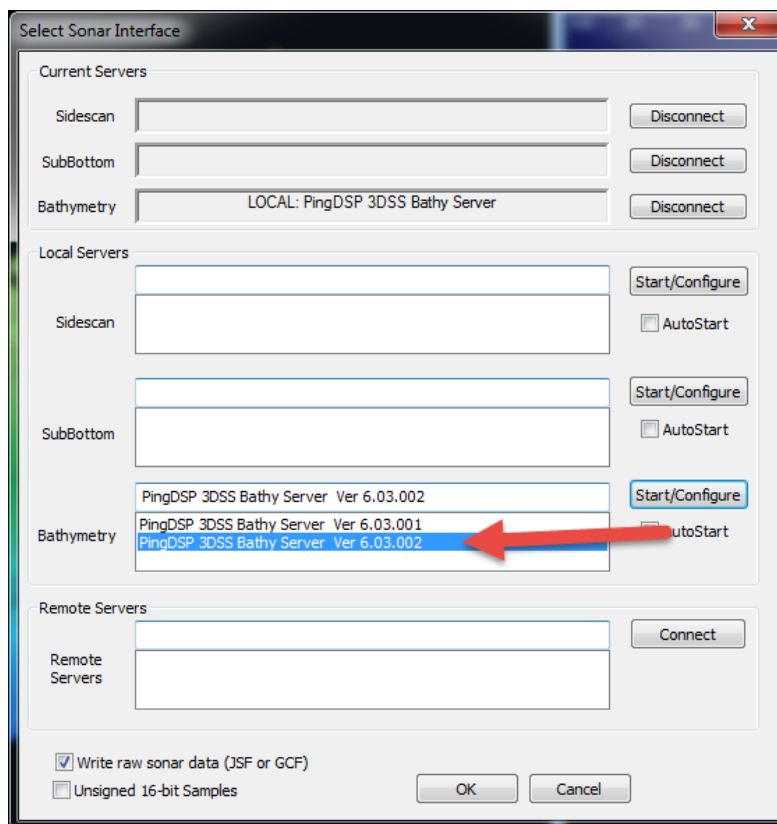


FIGURE 106. SONARWIZ BATHYMETRY SONAR INTERFACE SELECTION

Click START/CONFIGURE to configure the server such that it will connect to the 3DSS-DX-CONTROL application and receive sonar data. In the configuration shown in Figure 107, the IP address of the computer running 3DSS-DX-CONTROL needs to be specified, as well as the port 23848. Once these two are specified, the Connect button can be clicked to connect the bathy server to the 3DSS-DX-CONTROL application.

When operating properly, the bathy server will display navigation data, a 2D sidescan waterfall, and green status next to both 3DSS and SONARWiz to indicate a connection to both 3DSS-DX-CONTROL and SONARWIZ.

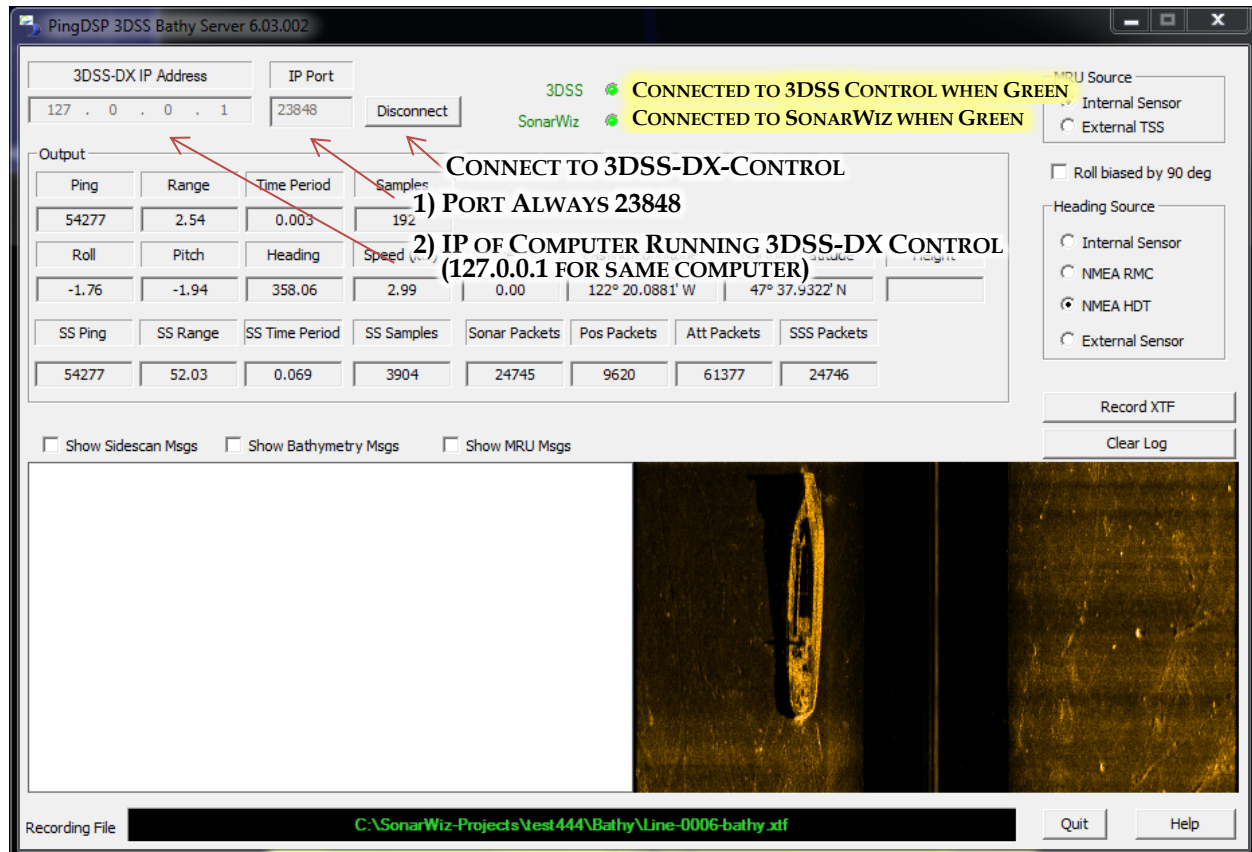


FIGURE 107. SONARWIZ 3DSS SERVER CONFIGURATION

3DSS-DX-CONTROL forwards each ping to the bathy server, and each ping can include multiple sources of attitude data. For example, if an external MRU is connected to the 3DSS-DX top side box MRU input, the External TSS option should be used for the MRU source.

MRU Source

☒ Internal Sensor
☐ External TSS

☐ Roll biased by 90 deg

Heading Source

☐ Internal Sensor
☐ NMEA RMC
☒ NMEA HDT
☐ External Sensor

FIGURE 108. SONARWIZ 3DSS SERVER ATTITUDE CONFIGURATION

Similarly, the external GPS should be used for the heading. Here the two most common settings are either HDT or RMC. HDT is preferred for GPS units that output heading using a dual antenna solution, while RMC is only with basic GPS units that output Course Made Good (noisier than a true dual-antenna heading, and requires forward motion).

Finally, the 3DSS sonar must be added to the vessel configuration used by the SONARWIZ project. This can be seen in Figure 109, where both the port and starboard transducers for the 3DSS have been added.

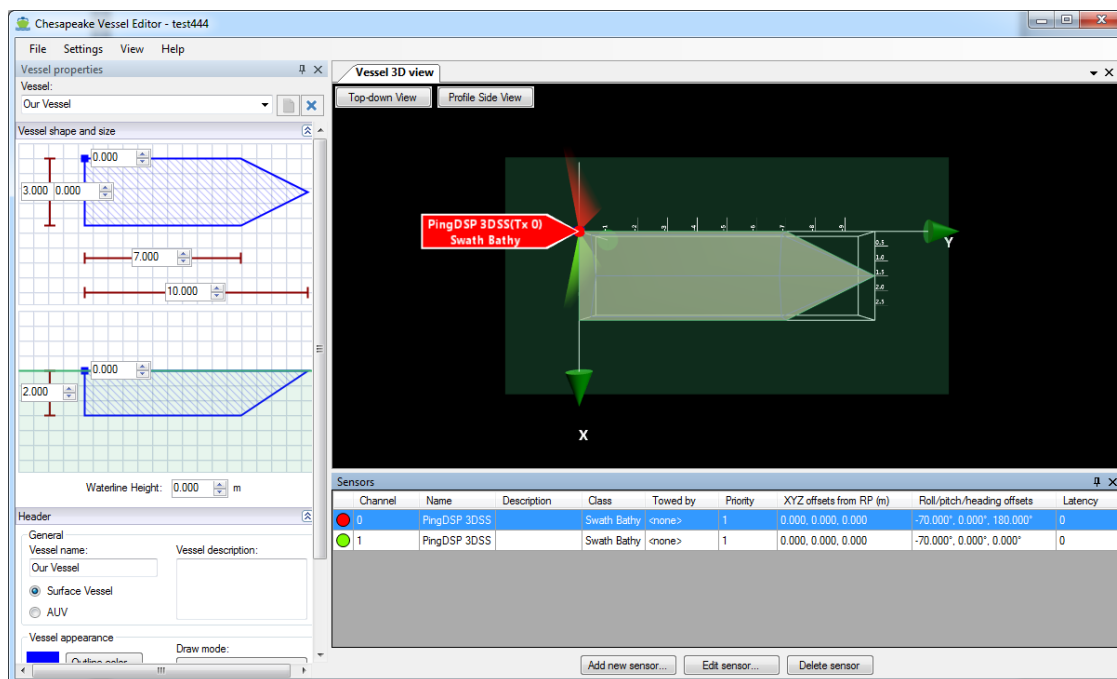


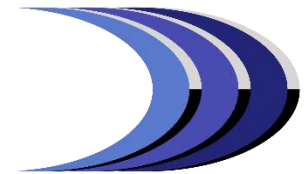
FIGURE 109. SONARWIZ VESSEL CONFIGURATION WITH 3DSS SONAR



The vessel configuration has individual XYZ offsets, as well as, individual roll/pitch/heading offsets for each transducer, and these can be configured in one of three ways:

1. When operating in real-time using the SONARWIZ 3DSS BATHYMETRY SERVER, offsets for the two transducers transducer (prior to patch test correction) must be:
 - PORT: -70° roll, 0° pitch, and 180° heading.
 - STARBOARD: -70° roll, 0° pitch, and 0° heading.
2. When importing XTF data converted with the 3DSS-TO-XTF CONVERTER in **Broadside** referenced beams mode:
 - PORT: +70° roll, 0° pitch, and 0° heading.
 - STARBOARD: -70° roll, 0° pitch, and 0° heading.
3. When importing XTF data converted with the 3DSS-TO-XTF CONVERTER in **Nadir** referenced beams mode:
 - PORT: 0° roll, 0° pitch, and 0° heading.
 - STARBOARD: 0° roll, 0° pitch, and 0° heading.

The XYZ offsets must also be entered, however depend greatly on the installation. Of note is the approximately 8 cm X distance difference between the port and starboard transducers (-4cm for port and +4cm of starboard). **Please see Appendix for drawing showing the acoustic center of the 3DSS sonar.**



CHAPTER 11: IMPORTING INTO CARIS HIPS AND SIPS

CARIS HIPS – XTF CONVERSION

The 3DSS-DX Sonar data is compatible with CARIS 8 or higher but must first be converted to XTF using the the 3DSS-To-XTF CONVERTER. Using the XTF converter:

1. The 3DSS data should be converted to a bathymetry XTF with **Broadside** referenced beams.
2. The navigation and attitude information should be inserted into XTFHEADERNAVIGATION #42 and XTFATTITUDEDATA #3 packets.

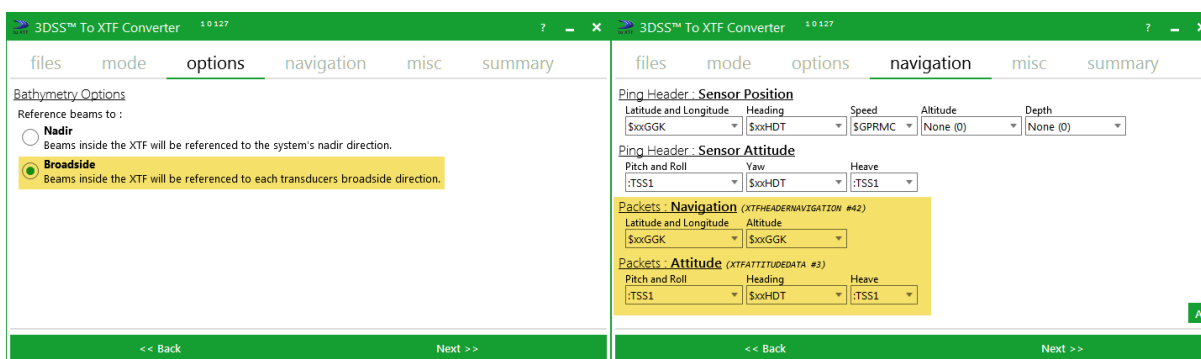


FIGURE 110. RECOMMENDED XTF CONFIGURATION FOR HIPS IMPORT.

CARIS HIPS – VESSEL EDITOR

The vessel configuration in CARIS must be set to contain dual transducers with the proper pitch and roll offsets:

Transducer 1	Date	Time	Time Correctio...	X (m)	Y (m)	Z (m)	Pitch (deg)	Roll (deg)	Yaw (deg)	Manufacturer	Model
Transducer 2	1	2014-001	00:00	0.000	-1.550	0.000	0.800	0.000	70.000	0.000	Unknown
Navigation	2		00:00								
Gyro											
Heave											
Pitch											
Roll											
Transducer 1	Time Correctio...	X (m)	Y (m)	Z (m)	Pitch (deg)	Roll (deg)	Yaw (deg)	Start beam num			
Transducer 2	1	0.000	-1.450	0.000	0.800	0.000	-70.000	0.000	5,000		
Navigation	2								0		
Gyro											
Heave											
Pitch											
Roll											

FIGURE 111. HIPS VESSEL CONFIGIURATION WITH 3DSS SONAR

The starting beam number for the 2nd transducer must be set to a large value such as 5000. This can be seen in Figure 111, where both the port and starboard transducers for the 3DSS have been added.



When importing XTF data converted with the 3DSS-To-XTF CONVERTER in **Broadside** referenced beams mode the roll for each of the transducers must be set:

- PORT: +70° roll (Transducer 1)
- STARBOARD: -70° roll (Transducer 2)

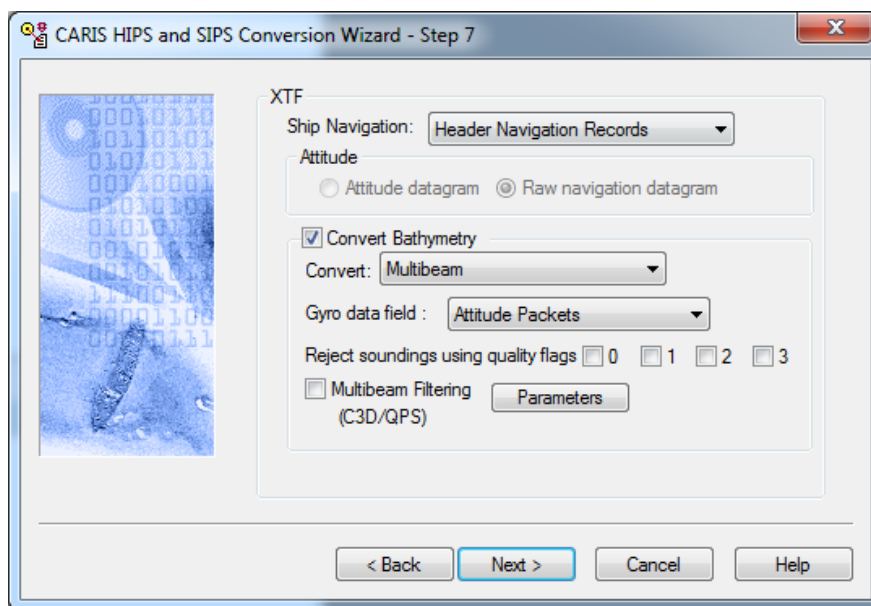
The XYZ offsets must also be entered, however depend greatly on the installation. Of note is the approximately 8 cm X distance difference between the port and starboard transducers (-4cm for port and +4cm of starboard). **Please see Appendix for drawing showing the acoustic center of the 3DSS sonar.**

CARIS HIPS – XTF IMPORT

The converted XTF data is imported into HIPS and SIPS using the Conversion Wizard under File->Import. The import process has 11 steps in the wizard:

1. Selecting XTF as the import format (Triton XTF).
2. Create new or update existing survey lines
3. Select XTF files (Raw Data).
4. Select a project, vessel, and day. The vessel configuration must be dual transducer as described in the previous section.
5. Choose type of horizontal coordinate reference system.
6. If desired, specify extents outside of which data will automatically be rejected. You can also specify a minimum and maximum depth here to reject soundings outside of these values on import.
7. Select XTF specific options:
 - a. XTF TIMESTAMPS – The **Automatic** setting should be used to let HIPS choose the best source of timing.
 - b. USE TIME SYNCHRONIZATION – **Off** by default. Each file will determine its own reference time.
 - c. SHIP NAVIGATION – Select the source of navigation data from the drop-down. Preferred setting is **Header Navigation Records**, this refers to using XTFHEADERNAVIGATION #42 packets.

- d. CONVERT BATHYMETRY – Ensure that the Convert Bathymetric checkbox is **checked** if you are converting an XTF file with either 3D sidescan data or bathymetry data.
- e. CONVERT – Select **Multibeam**.
- f. GYRO DATA FIELD – Select the source of gyro data from the drop-down. Preferred setting is **Attitude Packets**, this refers to using the XTFATTITUDEDATA #3 packets.
- g. REJECT SOUNDINGS USING QUALITY – The 3DSS XTF converter does not specify quality flags, hence **all checkboxes here must be unchecked**.
- h. MULTIBEAM FILTERING – Additional filters can be specified here. The 3DSS XTF format uses QPS multibeam entries internally, hence additional filtering can be applied using this option. Typically the additional filtering is not used and the checkbox is **not checked**.



8. Select XTF 2D sidescan conversion options.
 - a. CONVERT SIDE SCAN – This option is only used with XTF files which contain 2D sidescan. When converting 3D sidescan or bathymetry make sure this option is **not checked**.
9. Select options for converting horizontal tow fish layback or tow cable length distances. This step does not apply to bathymetry data and **all options must be not checked**.



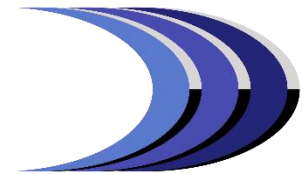
10. Convert – The final step is a summary of the setup in text form. The Convert button can now be clicked to start the conversion.
11. The final dialog is a report summary of the conversion which details how many bathymetric samples, position samples, and attitude samples were accepted.

Please refer to the CARIS HIPS and SIPS Module Reference Guide for further details.

CARIS SIPS – XTF IMPORT

The 2D sidescan converted XTF data can be imported into SIPS using the same set of steps as for the HIPS bathymetry import. The main differences are:

- Step 7 – The CONVERT BATHYMETRY checkbox can be **unchecked**, since sidescan XTF files do not contain bathymetry.
- Step 8 – The CONVERT SIDE SCAN checkbox must be **checked**, and SONAR CHANNELS **1,2** selected. Navigation can be taken from either THE PING HEADER: SENSOR, or HEADER NAVIGATION PACKETS. Gyro data can be taken from either PING HEADER: SENSOR or from ATTITUDE PACKETS.
- Step 9 – The 3DSS is hard mounted and not towed, hence LAYBACK / CABLEOUT must be **unchecked**.



CHAPTER 12: INTERFACING WITH QPS QINSY



QPS QINSY SOFTWARE SETUP

Three types of 3DSS-DX Sonar data can be recorded in the QINSy database:

1. Binned bathymetric data output.
2. Raw 3D point cloud data.
3. 2D sidescan amplitude data.

The QINSy driver can be setup to record any combination of the 3DSS outputs, and does not need to record all of them. Additional navigation, attitude, and timing information are also extracted from the 3DSS data stream provided it is connected to the 3DSS Top-Side Unit.

Additional to the sonar data from the 3DSS-DX, QINSy can also record:

- Position data, when a position source is interfaced with the 3DSS-DX top side unit.
- Heading data, when a position source is interfaced with the 3DSS-DX top side unit.
- Motion data, from either the internal sensor or an external motion sensor interfaced with the 3DSS-DX top side unit.
- Time information.

This additional data is typically only interfaced with QINSy when you want to replay and reprocess the 3DSS-DX data using 3DSS-DX Control. Under real-time circumstances, the position and attitude data is typically provided by the QINSy driver for your INS solution, and the 3DSS-DX must be supplied with GPS and PPS so that it may synchronise to UTC time.

TEMPLATE DATABASE SETUP

The 3DSS-DX has 2 independent transducer, one on each side; it looks like a dual head multibeam system. QINSy handles dual head multibeam by specifying two separate single head drivers. To interface the 3DSS-DX binned bathymetry data with QINSy, 2 separate drivers must be defined; one for the port side and one for the starboard side. Additionally, to interface the 3D point-cloud data, two more separate drivers must be defined (port and starboard). Finally, to interface the 2D sidescan data, a normal sidescan sonar with 2 transducers must be defined.

For each individual sound pulse (ping), the recorded reflections are stored as angles (relative to nadir) and slant ranges (relative to the center of each transducer).

IP ADDRESS AND PORT NUMBER

When the 3DSS-DX Control application and QINSy are running on the same PC, then the driver IP and port number can be configured as below:

Port number:	23848
IP address:	127 . 0 . 0 . 1

When the 3DSS-DX Control application and QINSY are running on separate computers, then the driver IP must be changed to the IP address of the PC running 3DSS-DX Control:

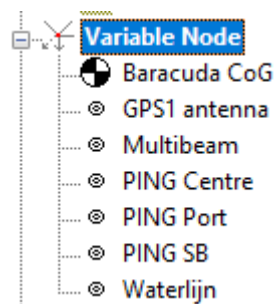
Port number:	23848
IP address:	192 . 168 . 228 . 5

The IP and Port number are the same for all instances of the QINSy 3DSS drivers.

VARIABLE NODE DEFINITION

Create 3 variable nodes for the 3DSS-DX:

1. A node for the center of the 3DSS-DX sonar.
2. A node for the port transducer.
3. A node for the starboard transducer.



The exact offsets of the two transducers can be found on the last page of this manual.

BINNED BATHYMETRY DATA

The 3DSS-DX produced a binned bathymetric output. The newest version of the 3DSS-DX Control software introduces MBES equivalent binning, where the sector, beamwidth, and type of binning (equi-distant or equi-angular) can be specified. When 3DSS-DX Control cannot calculate a value for the bin or beam, then the bin or beam stays empty and no value is output.

An instance of a system with a driver must be defined for each transducer (port and starboard).

On the first page, select the 3DSS-DX BATHYMETRY PORT driver, specify the appropriate IP address and port number (23848), and click NEXT.

On the second page, select the matching node, in this case the node for the *Port* transducer. The PORT transducer node must have a Roll offset of +70°, and the MAX BEAMS PER PING should be set to 2000.

Edit System [MB Ping PORT]

System

Name: MB Ping PORT

Type: Multibeam Echosounder

Socket Settings

Driver: Ping DSP 3DSS-DX-450 Bathymetry Port

Port number: 23848

IP address: 127 . 0 . 0 . 1

Maximum update rate: 0.000 s

Notes

I/O Parameters are only used in online mode and have no effect in replay mode

This driver also supports other system types [More info...](#)

< Back

Next >

Finish

Cancel

Help

Multibeam Echosounder Parameters

Model, Transducer Location and Mounting Angles

Description	Value
Object	Baracuda
Transducer Setup	Assume Common Acoustic Center
Transducer Node	PING Port
Roll offset	70.000°
Pitch offset	0.000°
Heading offset	0.000°

Decoding

Max. beams per ping: 10000

Sound Velocity

☒ Use sound velocity from unit

m/s

☐ Use sound velocity: 1485.000

Echosounder Stabilization

☐ Unit is roll stabilized
☐ Unit is pitch stabilized
☐ Unit is heave compensated

< Back

Next >

Finish

Cancel

Help

Repeat the same for the starboard transducer, selecting 3DSS-DX BATHYMETRY STARBOARD driver, the matching node, and a roll offset of -70°.

After installation, a multibeam calibration must be carried out to determine the correct roll, pitch, and heading offset values. Typically, the pitch and heading offset values for the two transducers will be the same, while the roll value may be different by up to several degrees.

3D SIDESCAN POINT CLOUD DATA

The 3D point cloud data contains all of the points measured / generated by the sonar; no points are filtered out. As a result, this point cloud contains a very large amount of data, but also a great number of hidden gems (or rubies). Select this driver only when you are looking for objects or features.

In-order to use the point cloud data in QINSy, two more instances of a system must be created which use the 3DSS-DX SIDE SCAN 3D PORT and STARBOARD drivers:

Edit System [3D Ping Port]

System

Name:

Type: Multibeam Echosounder

Socket Settings

Driver: Ping DSP 3DSS-DX-450 SideScan3d Port

Port number:

IP address:

Maximum update rate: s

Notes

I/O Parameters are only used in online mode and have no effect in replay mode

< Back **Next >** Finish Cancel Help

Multibeam Echosounder Parameters

Model, Transducer Location and Mounting Angles

Description	Value
Object	Baracuda
Transducer Setup	Assume Common Acoustic Center
Transducer Node	PING Port
Roll offset	70.000°
Pitch offset	0.000°
Heading offset	0.000°

Decoding

Max. beams per ping:

Sound Velocity

☒ Use sound velocity from unit

☐ Use sound velocity:

Echosounder Stabilization

☐ Unit is roll stabilized

☐ Unit is pitch stabilized

☐ Unit is heave compensated

< Back **Next >** Finish Cancel Help

Similar to the process for the bathymetry driver, the matching node must be selected and the matching offsets specified (+70 for port, and -70 for starboard). The MAX BEAMS PER PING must be specified as 32768.

The IP address and port number are the same for both port and starboard, and the same as the values entered in the previous section.

2D SIDESCAN DATA

The 3DSS-DX sidescan driver setup is a normal SSS setup in QINSy:

STEP 1

Edit System [Ping SSS]

System
Name:
Type:

Socket Settings
Driver:

Port number:
IP address:

Maximum update rate: s

Notes
I/O Parameters are only used in online mode and have no effect in replay mode

This driver also supports other system types [More info...](#)

< Back **Next >** Finish Cancel Help

STEP 2

Sidescan Sonar General Parameters

Location
Object:

Associated Multibeam System
System:

Sound Velocity
☒ Use sound velocity from unit:
☐ Use sound velocity: m/s

Properties (Administrative)
Manufacturer:
Model:

STEP 3

Sidescan Sonar Channel Setup

Tx Frequency	Orientation	Node
100	Port	PING Port
100	Starboard	PING SB

Channel Properties
Transducer:
Orientation:
Slot ID:
Number of beams:
Frequency (kHz):

Add a new system instance with a TYPE of SIDESCAN SONAR, and select PING DSP 3DSS-DX-450 as the driver. The IP address and port number are the same. There is no need to create two instances for port and starboard; the one driver handles both.



POSITION DATA - SENT THROUGH 3DSS TOP SIDE

The NMEA-0183 data sent to the 3DSS-DX sonar through the top side unit can be pulled out with hardware timestamping. The 3DSS-DX-450 POSITION driver needs to be added as a POSITION NAVIGATION SYSTEM in QINSy:

The image shows two side-by-side screenshots of the QINSy software interface. The left window is titled 'Edit System [Ping GGA]' and contains the following fields: 'Name' (Ping GGA), 'Type' (Position Navigation System), 'Driver' (Ping DSP 3DSS-DX-450 Position), 'Port number' (23848), 'IP address' (127.0.0.1), and 'Maximum update rate' (0.000 s). The right window is titled 'Position System Parameters' and contains the following fields: 'Object' (Baracuda), 'Antenna' (PING Centre), 'Receiver number' (GGA), 'Horizontal datum' (ETRS89 (EUREF89)), 'Vertical datum' (ETRS89 (EUREF89)), 'Height level' (No Level Correction), 'Height offset' (0.000 m), 'Unit' (Meters), 'SD horizontally' (0.500 m), and 'SD vertically' (1.000 m). Both windows have a 'Next >' button and a 'Finish' button.

The IP address and Port number are the same as for the sonar data. The ANTENNA LOCATION of the position data depends on the input GPS / INS. If the INS/GPS corrects for the lever arm to the 3DSS sonar, then the location should be the center of the 3DSS-DX sonar. If the INS/GPS does not correct for the lever arm, then the location should be the position of GPS antenna 1 on the vessel, or the location of the INS depending on the setup.

To read the correct position data from the 3DSS, define a slot ID in the RECEIVER NUMBER field. This RECEIVER NUMBER should specify which NMEA-0183 message QINSy will decode. Typically, either GGA, or GKG is selected here, but RMC can be used as a fall back.

If the INS manufacturer has their own QINSy driver, it is possible to their driver directly, and skip adding the 3DSS-DX Position driver. If you choose this option, the 3DSS sonar must be running on UTC time, meaning you must connect GPS and PPS to the sonar top side so that it has the time.



HEADING DATA – SENT THROUGH 3DSS TOP SIDE

The 3DSS-DX-450 HEADING driver needs to be added as a GYRO COMPASS to supply QINSy with the heading. The IP and Port numbers should be specified as the same, and the OBSERVATION TYPE should be specified as BEARING (TRUE):

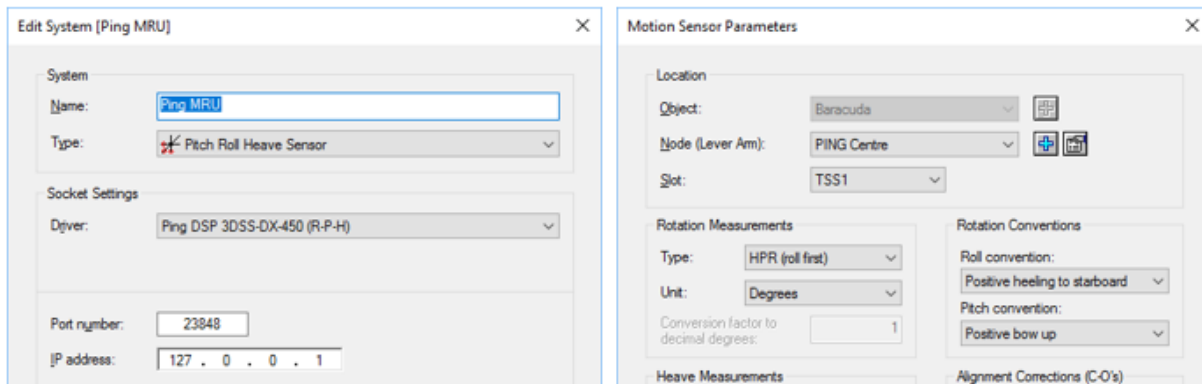
The image shows two side-by-side software configuration windows. The left window, titled 'Edit System [Ping Heading]', contains the following fields: 'Name' (Ping Heading), 'Type' (Gyro Compass), 'Driver' (Ping DSP 3DSS-DX-450 Heading), 'Port number' (23848), 'IP address' (127.0.0.1), and 'Maximum update rate' (0.000 s). It also includes a 'Notes' section and a 'More info...' link. The right window, titled 'Gyro Observation Parameters', contains the following fields: 'Name' (Ping Heading), 'Location' (Baracuda), 'Type' (Bearing (True)), 'Unit' (Degrees), 'Apriori SD' (1.00000), 'Fixed C-O' (0.0000000), 'Variable C-O' (0.00000), 'Scale factor' (1.000000000), and 'Slot Identifiers' (Slot number 1 and 2). Both windows have 'Back', 'Next', 'Finish', 'Cancel', and 'Help' buttons at the bottom.

No slot ID is needed for this data. Same as the POSITION driver, this driver does not need to be used if a manufacturer supplied INS driver is used. If the INS manufacturer's driver is used in QINSy, you must ensure that the 3DSS-DX sonar is running on UTC time.

MOTION DATA – SENT THROUGH 3DSS TOP SIDE

There is an internal motion sensor inside the 3DSS-DX-450, and an input port on the top side unit to supply an external motion source. Both of these can be interfaced with QINSy; the slot ID chooses which source is used:

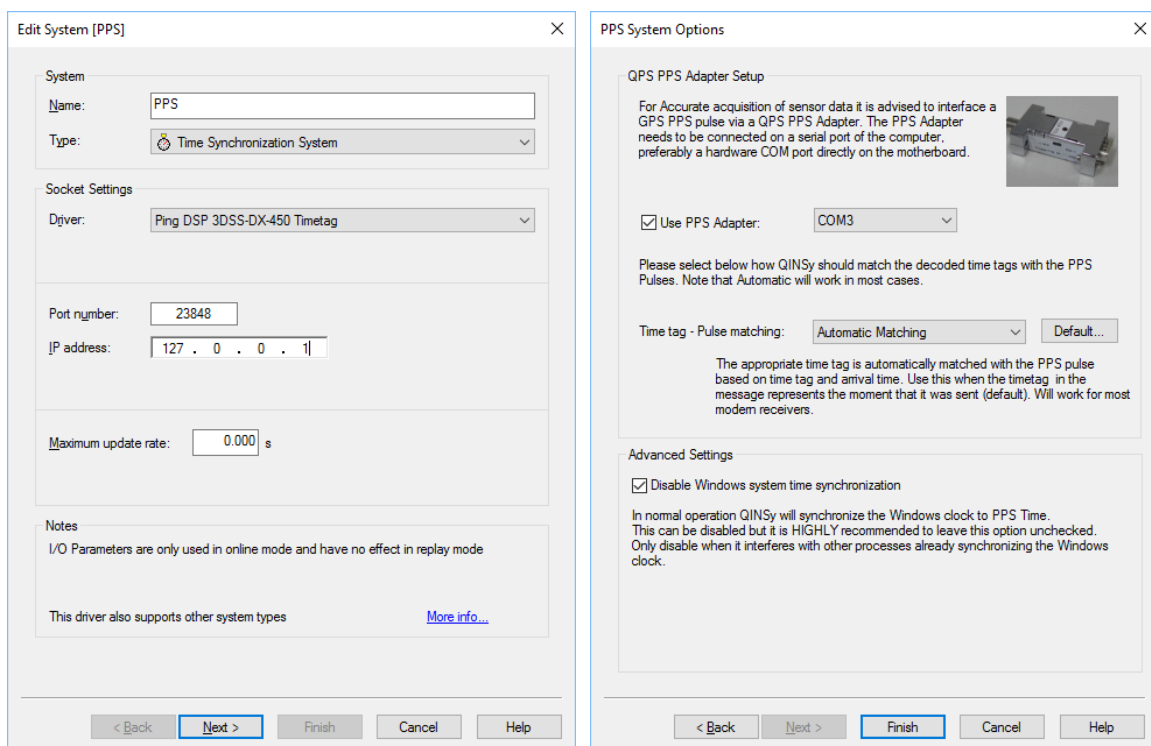
- TSS1 is the external motion source, and
- VNYCM is the internal motion sensor.



The 3DSS-DX-450 (R-P-H) driver should be added as a PITCH ROLL HEAVE SOURCE. Same as the other drivers, the IP address and port must be set to the same values. And similarly, this driver does not need to be used if a manufacturer supplied INS driver is used.

PPS

When there is no other time synchronisation data interfaced with QINSy, the time data from the 3DSS-DX must be interfaced. When the PPS pulse is also interfaced with QINSy, then on page 2 the PPS adapter must be ticked and a serial port must be selected on which the QINSy PPS ADAPTER is connected.

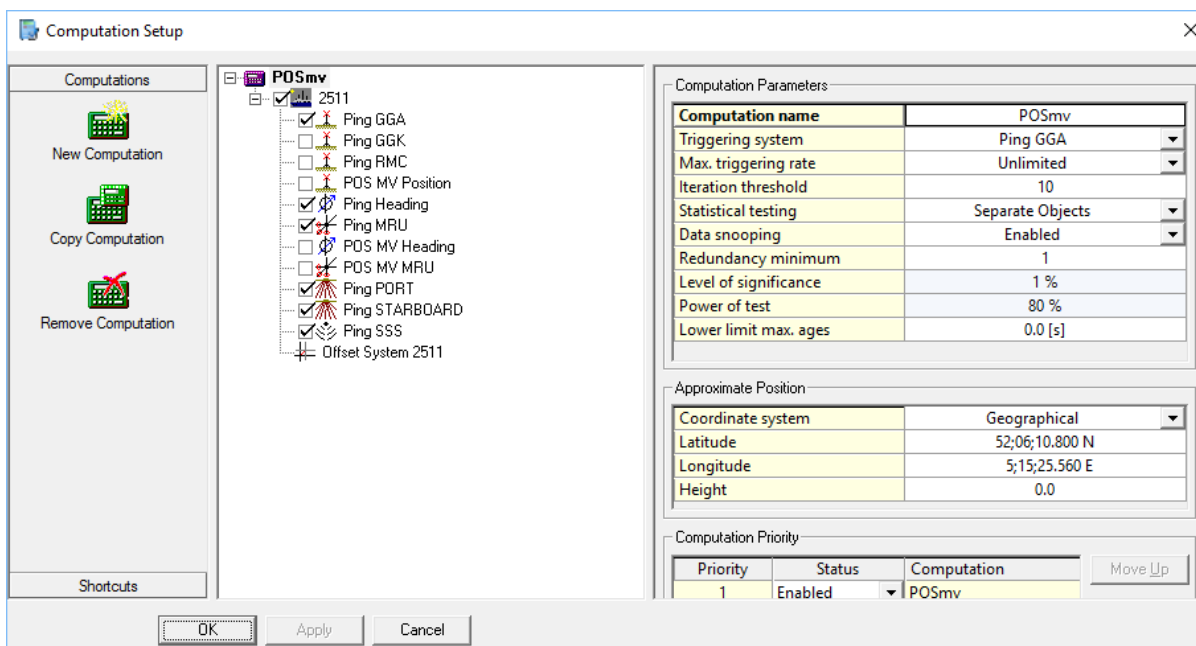


ONLINE CONTROLLER

The data produced by the 3DSS-DX can be stored in the QINSy raw database format (DB), in the QINSY processing format (QPD), and also in the XTF format (XTF). These formats are activated under SESSION -> STORAGE.

In the computation setup you can select which data will be calculated:

- Multibeam (3DSS bathymetry), and/or
- 3D (3DSS 3D point cloud), and/or
- sidescan (conventional 2D).



Computation Setup

Computations

New Computation

Copy Computation

Remove Computation

Shortcuts

POSmv

- ☒ 2511
 - ☒ Ping GGA
 - ☐ Ping GGK
 - ☐ Ping RMC
 - ☐ POS MV Position
 - ☒ Ping Heading
 - ☒ Ping MRU
 - ☐ POS MV Heading
 - ☐ POS MV MRU
 - ☒ Ping PORT
 - ☒ Ping STARBOARD
 - ☒ Ping SSS
 - ☐ Offset System 2511

Computation Parameters

Computation name	POSmv
Triggering system	Ping GGA
Max. triggering rate	Unlimited
Iteration threshold	10
Statistical testing	Separate Objects
Data snooping	Enabled
Redundancy minimum	1
Level of significance	1 %
Power of test	80 %
Lower limit max. ages	0.0 [s]

Approximate Position

Coordinate system	Geographical
Latitude	52;06;10.800 N
Longitude	5;15;25.560 E
Height	0.0

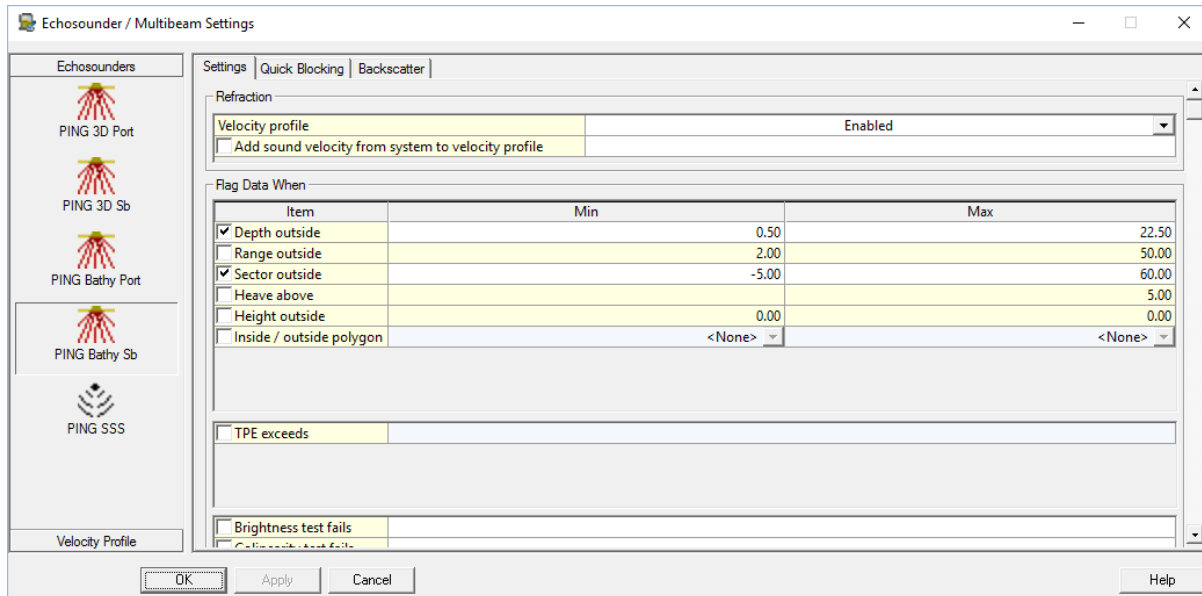
Computation Priority

Priority	Status	Computation
1	Enabled	POSmv

Move Up

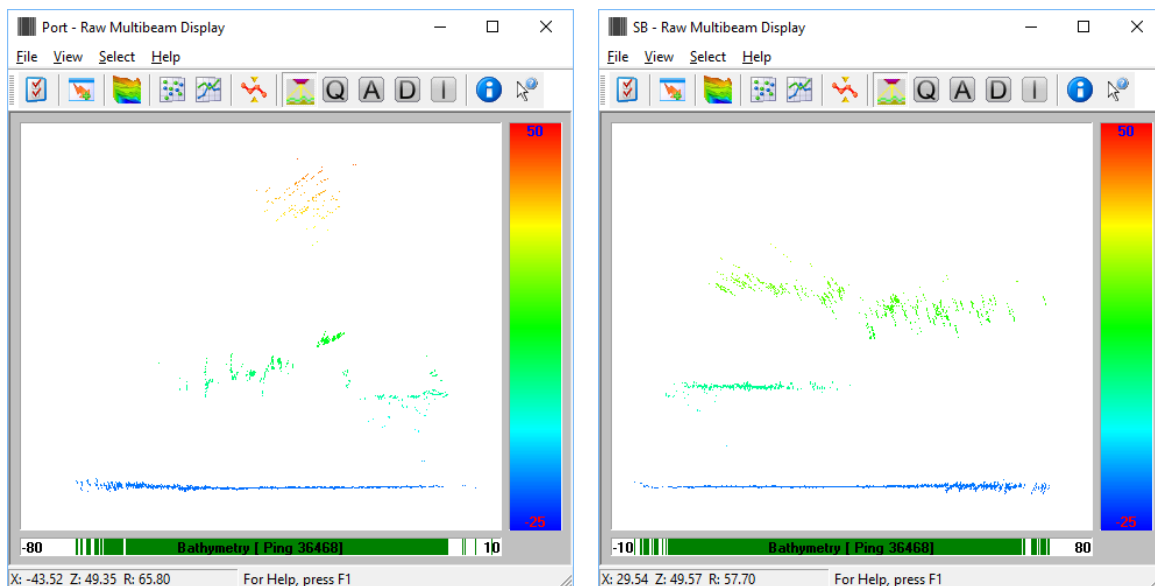
OK Apply Cancel

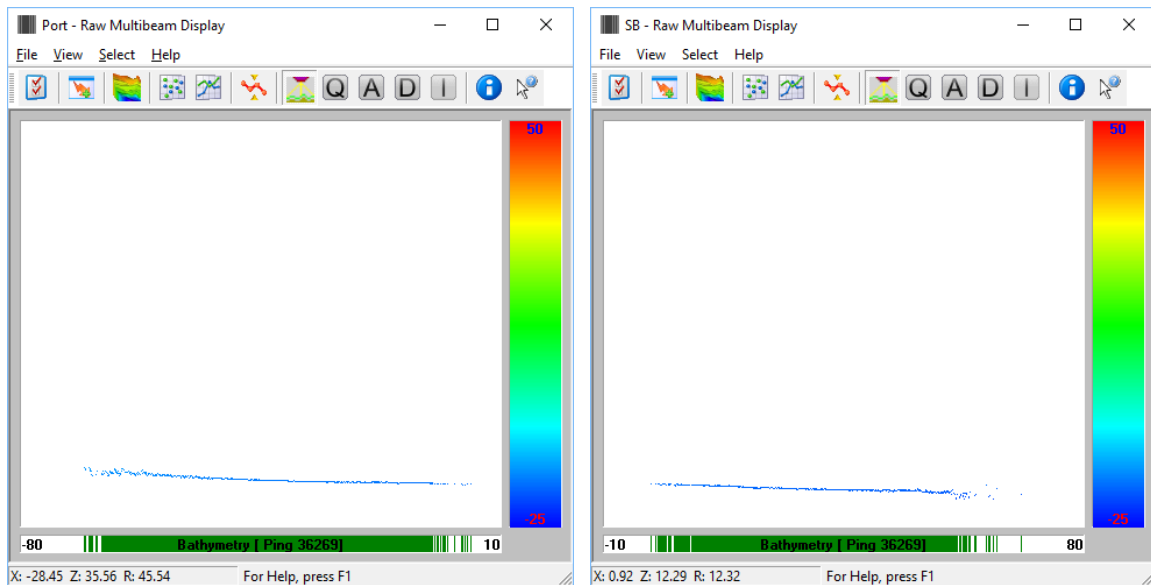
With the multibeam setting, the amount of noise in the QPD files can be reduced. These settings work in the same way as for a normal multibeam.



RAW MULTIBEAM DISPLAY

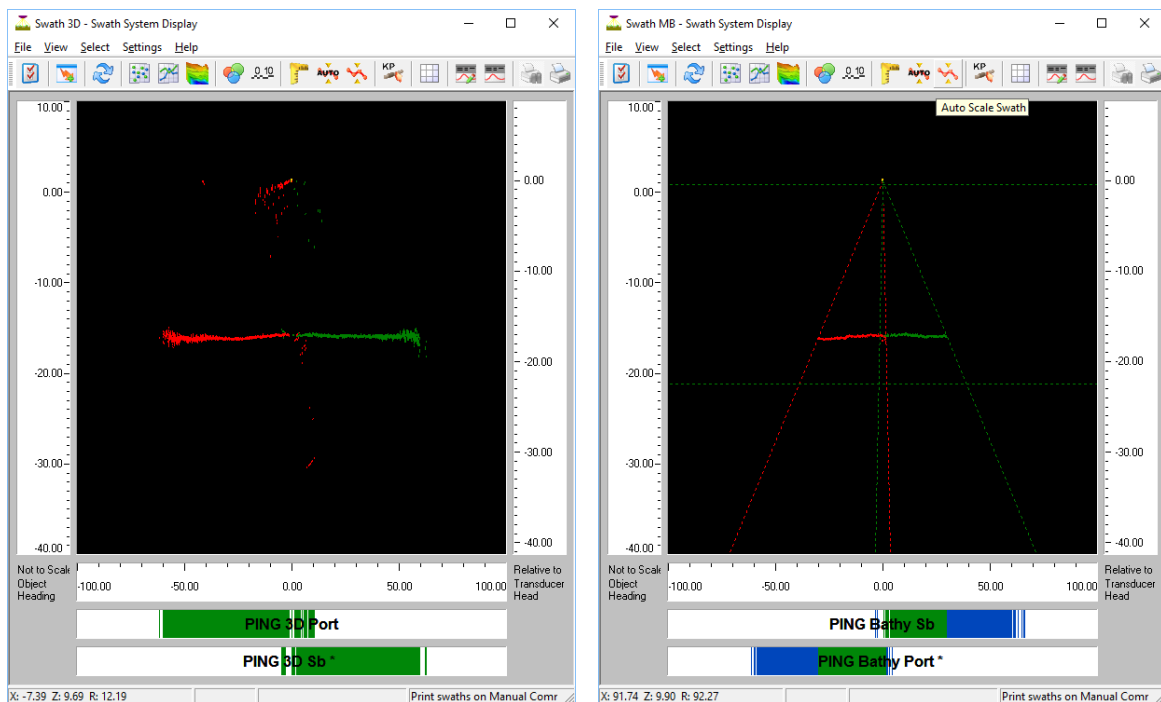
The raw multibeam display shows the data that comes directly from the 3DSS-DX sonar without any filtering or depth or sector blocking. Both the binned bathymetry and the 3D point cloud data can be displayed in the raw multibeam display. The only disadvantage is that only 1 system can be displayed in a display (i.e. only port or starboard, bathymetry or 3D points):





SWATH DISPLAY

In the swath display, the data is corrected for depth and the selected blocking filters are applied. In a swath display multiple systems can be selected and displayed at once. This means that the result of the two transducers can be shown in a single display window:



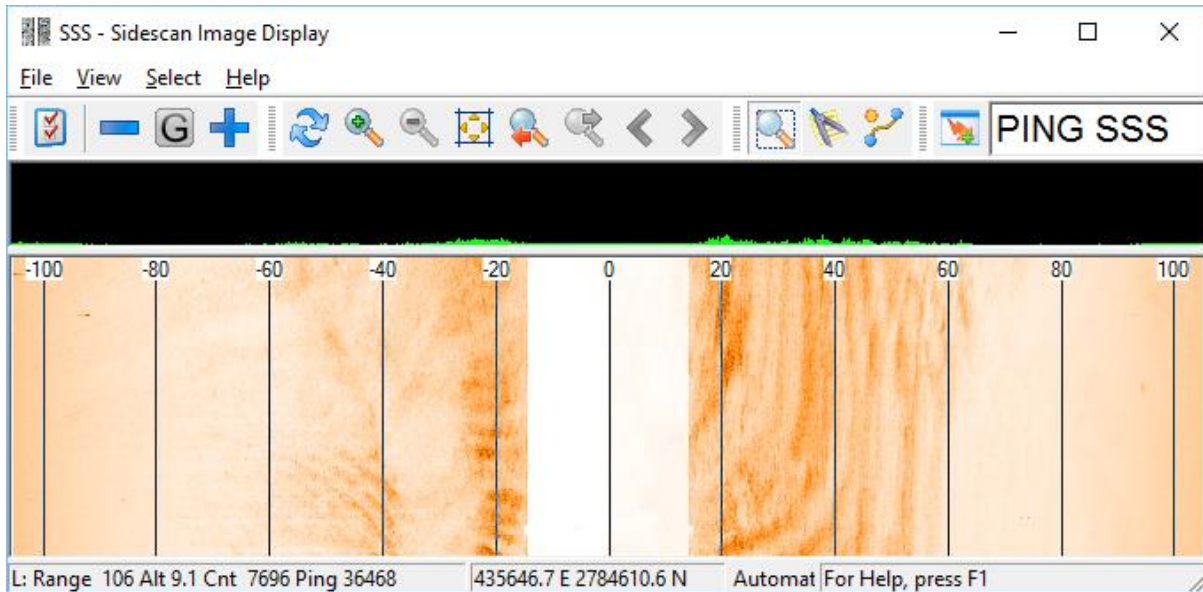
3DSS-DX 3D POINTS

3DSS-DX BATHYMETRY



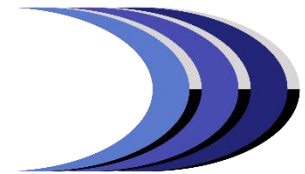
SIDE SCAN SONAR DISPLAY

The side scan sonar display is the classic view / display for side scan data:



ACKNOWLEDGMENT

This section was provided curtesy of Mr. Hans Vos from Van Oord. We thank him for this great work.



CHAPTER 13: INTERFACING WITH PDS 2000



PDS2000 SOFTWARE SETUP

NOTE: The PDS 2000 section is in a preliminary state at the moment.

Drivers for the 3DSS-DX were introduced in PDS 2000 version 4.1.7.0. This version of the drivers supports

- the bathymetry data output,
- side scan sonar data output, and
- the internal MRU.

At present there is no support for the 3D point cloud data output.

INTERFACING BATHYMETRIC DATA

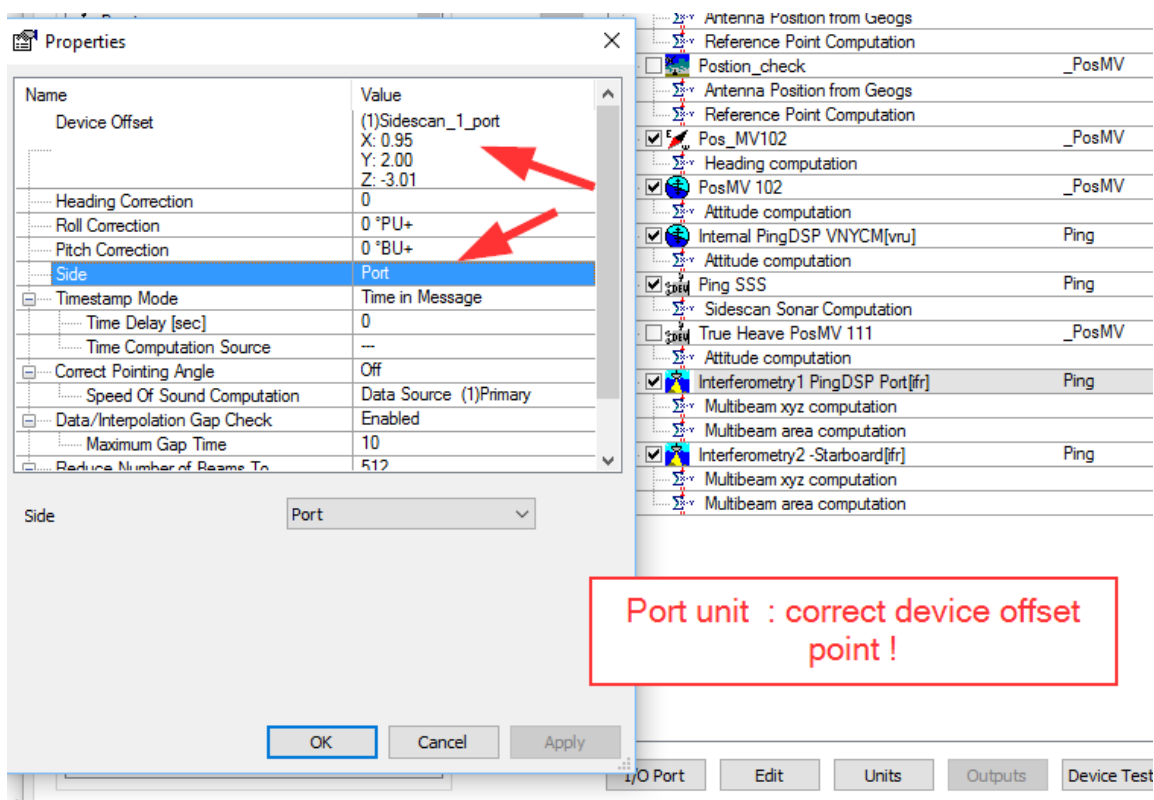
The PDS driver for the 3DSS-DX only supports single side operation, hence two instances must be used; one for port and one for starboard. The 3DSS-DX driver can be found under the INTERFEROMETRY group, and an example configuration is shown below:

Vessel - Ping example Barracuda[Multibeam Survey] X

Device	Port
☒ Pos_MV_sensor1_position_102	_PosMV
☒ Antenna Position from Geogs	
☒ Reference Point Computation	
☒ Position_check	_PosMV
☒ Antenna Position from Geogs	
☒ Reference Point Computation	
☒ Pos_MV102	_PosMV
☒ Heading computation	
☒ PosMV 102	_PosMV
☒ Attitude computation	
☒ Internal PingDSP VNYCM[vru]	Ping
☒ Attitude computation	
☒ Ping SSS	Ping
☒ Sidescan Sonar Computation	
☒ True Heave PosMV 111	_PosMV
☒ Attitude computation	
☒ Interferometry1 PingDSP Port[lfr]	Ping
☒ Multibeam xyz computation	
☒ Multibeam area computation	
☒ Interferometry2 -Starboard[lfr]	Ping
☒ Multibeam xyz computation	
☒ Multibeam area computation	

Two Interferometer devices

The correct device offsets must be specified for each transducer. The Appendix of this manual includes a drawing showing the acoustic center of each transducer of the 3DSS-DX relative to its center and the pole mount adapter. The computed offsets must be specified under the PROPERTIES of each of the transducers:



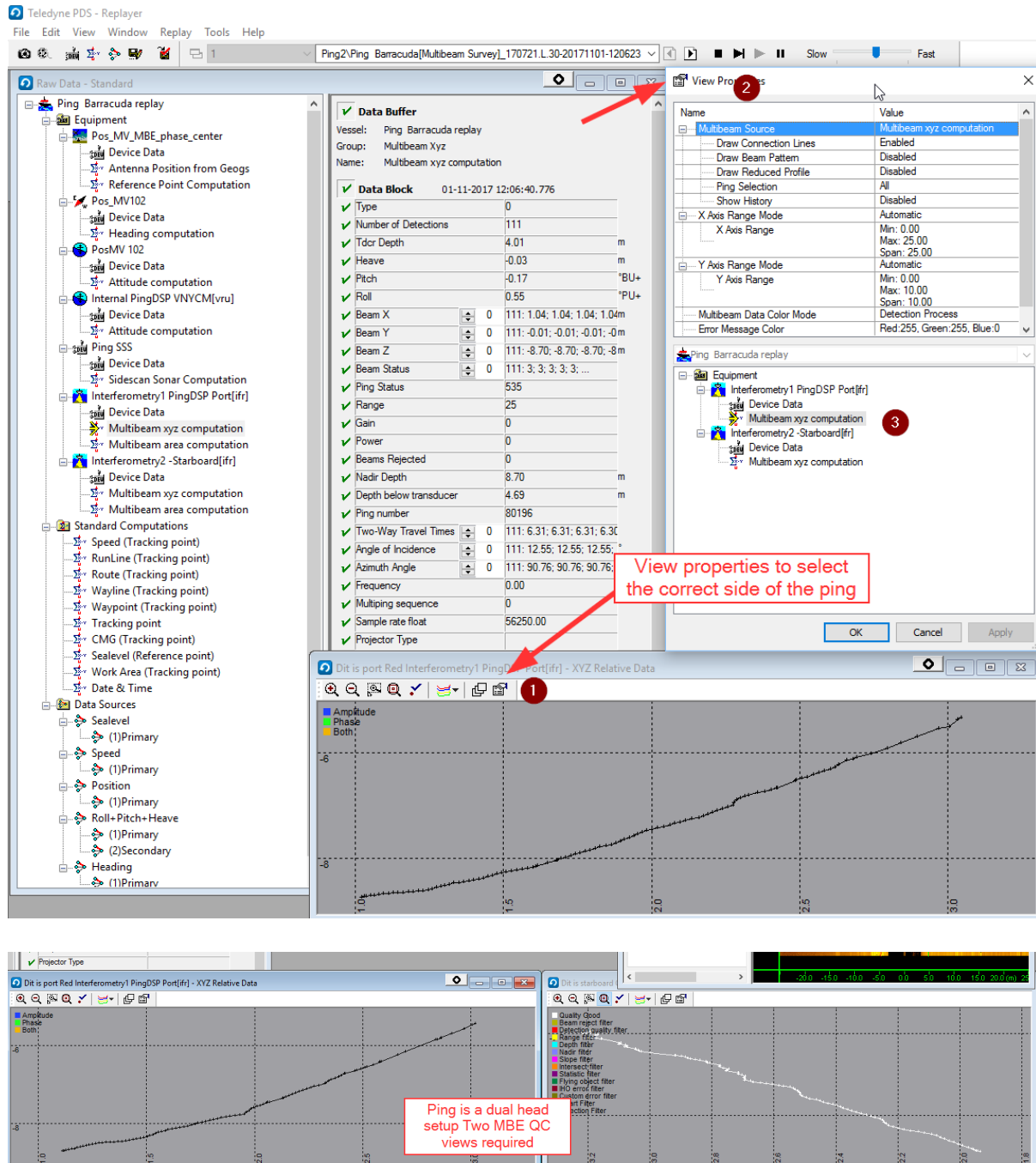
The ROLL correction value for both transducers can be left at 0° until a patch test is conducted to determine a more appropriate value.

NOTE: Ping DSP is in the process of verifying with Teledyne RESON that the $\pm 70^\circ$ roll offsets are not required.

The configurations of the MRU and side scan sonar drivers follow the typical procedures for PDS 2000.

ONLINE DISPLAY

Two display windows need to be created in PDS 2000 to display both the port and the starboard data sets:



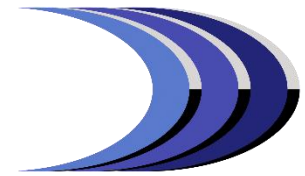
The screenshot shows the Teledyne PDS - Replayer interface. The main window displays a tree view of the data structure. The 'Data Buffer' section shows the selected data block for 'Multibeam xyz computation'. The 'View Properties' dialog box is open, showing the 'Multibeam Source' and 'Multibeam Data Color Mode' settings. A red arrow points to the 'View Properties' dialog box, with a text box stating: "View properties to select the correct side of the ping". Another red arrow points to the 'Data Buffer' section, with a text box stating: "Ping is a dual head setup Two MBE QC views required".

View Properties Dialog Box:

Name	Value
Multibeam Source	Multibeam xyz computation
Draw Connection Lines	Enabled
Draw Beam Pattern	Disabled
Draw Reduced Profile	Disabled
Ping Selection	All
Show History	Disabled
X Axis Range Mode	Automatic
X Axis Range	Min: 0.00 Max: 25.00 Span: 25.00
Y Axis Range Mode	Automatic
Y Axis Range	Min: 0.00 Max: 10.00 Span: 10.00
Multibeam Data Color Mode	Detection Process
Error Message Color	Red:255, Green:255, Blue:0

Data Buffer Section:

Property	Value
Vessel	Ping Barracuda replay
Group	Multibeam Xyz
Name	Multibeam xyz computation
Data Block	01-11-2017 12:06:40.776
Type	0
Number of Detections	111
Tdcr Depth	4.01 m
Heave	-0.03 m
Pitch	-0.17 °BU+
Roll	0.55 °PU+
Beam X	0 111: 1.04; 1.04; 1.04; 1.04m
Beam Y	0 111: -0.01; -0.01; -0.01; -0.01; -0m
Beam Z	0 111: -8.70; -8.70; -8.70; -8.70; -8m
Beam Status	0 111: 3; 3; 3; 3; 3; ...
Ping Status	535
Range	25
Gain	0
Power	0
Beams Rejected	0
Nadir Depth	8.70 m
Depth below transducer	4.69 m
Ping number	80196
Two-Way Travel Times	0 111: 6.31; 6.31; 6.31; 6.31; 6.31; ...
Angle of Incidence	0 111: 12.55; 12.55; 12.55; 12.55; 12.55; ...
Azimuth Angle	0 111: 90.76; 90.76; 90.76; 90.76; 90.76; ...
Frequency	0.00
Multiping sequence	0
Sample rate float	56250.00
Projector Type	



APPENDICES



TROUBLESHOOTING THE 3DSS-DX-450

This troubleshooting guide identifies some symptoms that could occur during use of the 3DSS system, and presents one or more possible causes, along with the recommended corrective action for each. Probable causes and associated corrective actions are listed in the order of likeliness as well as difficulty of the solution. For greatest efficiency, is recommended to try corrective actions in the order that they appear.

<i>Symptom</i>	<i>Potential Causes</i>	<i>Corrective Action</i>
Power Indicator on the Topside Box is off.	The power switch is not turned on.	Verify the power switch is in the on position.
	No 24VDC power is applied.	Verify the Topside Box is connected to a suitable power source. If so, check power source.
	The Fuse in the Topside Box is blown.	Check the fuse for continuity, replace if necessary.

<i>Symptom</i>	<i>Potential Causes</i>	<i>Corrective Action</i>
Cannot Connect to the Sonar	The sonar is not being supplied power, or the sonar is not connected to the Topside Box.	Verify the Topside Box is connected to a suitable power source.
		Verify the switch on the Topside Box is in the ON position and that the power indication LED is ON.
		Verify the connection between the Topside Box and the 3DSS sonar and make sure both connectors are free of debris and water.
Network is not setup correctly.		Verify the Host computer is connected to the Topside Box with a cat5e Ethernet cable.
		Verify Ethernet link is active by viewing a green blinking LED near the Ethernet connection to the PC, or by navigating to "Network Connections" and noting the status of the associated network adapter.
		Verify the Ethernet adapter on the Host computer is enabled. The IPv4 IP address should be set to 192.168.228.XXX (where XXX is not 001), and the subnet mask set to 255.255.255.0.



3DSS-DX and 3DSS-iDX Sonar Manual

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APRIL 18TH, 2024

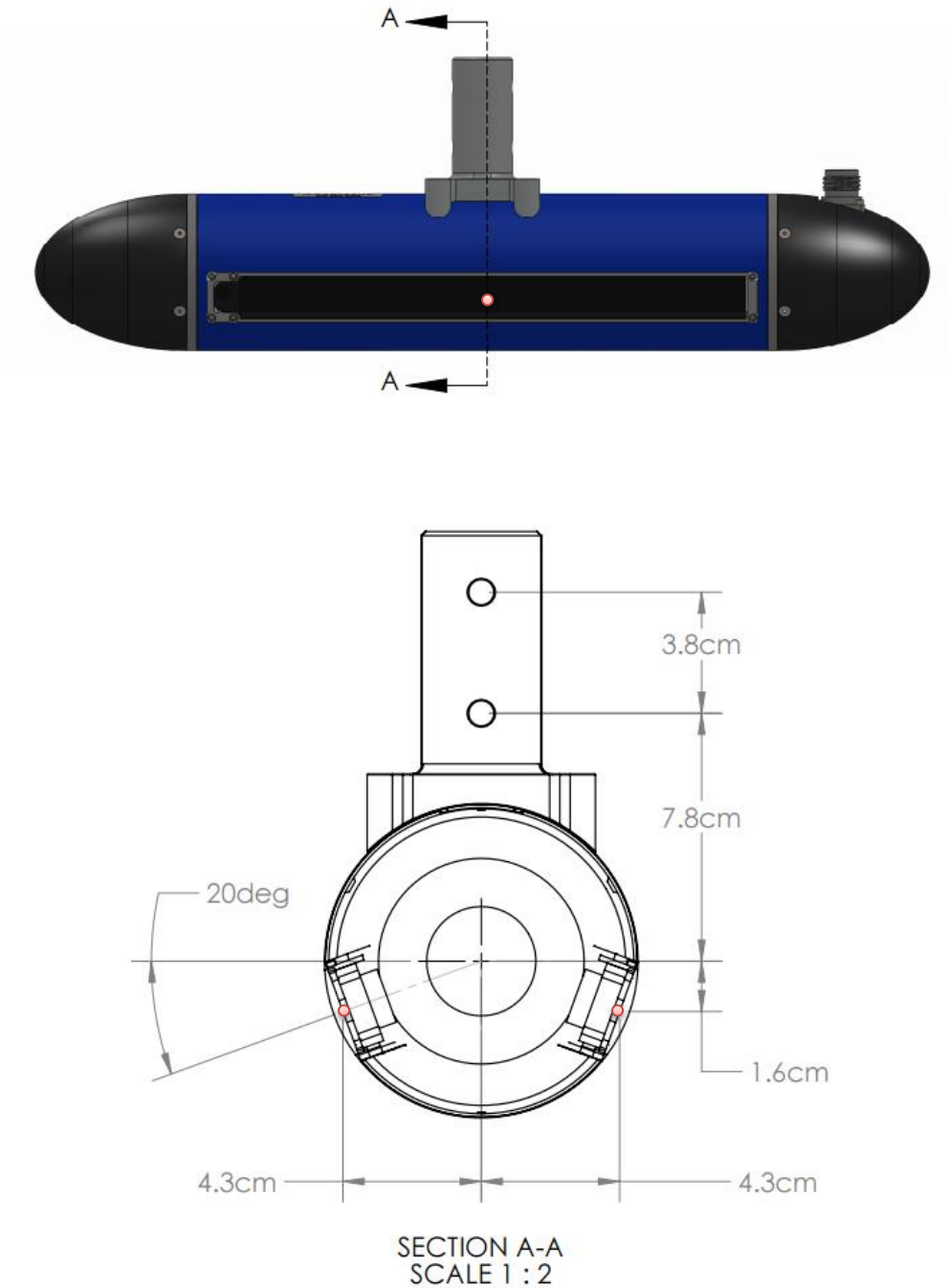
Internal software or Firmware crash.	Turn off power to the sonar and restart the 3DSS Control Software. Next, power the sonar, wait 30 seconds and then try to connect through the 3DSS Control software.
Firewall Blocking the Sonar	Disable the Windows Firewall and/or other firewalls on the PC associated with the LAN adapter used by the sonar. If this fixes the issue either leave the firewall(s) disabled, or add an exception to the firewall for the 3DSS-DX Control Application.

Symptom	Potential Causes	Corrective Action
GPS, PPS, or MRU data is not displayed in the Control Software Status.	Sonar is not connected and/or not pinging.	Connect to the 3DSS sonar through the Control Application. Then Commit -> Update Time -> Play to start pinging. (NEMA settings come in through ping data).
	Device is not connected properly, or not powered.	Ensure the external MRU, GPS, or PPS device is powered and connected to the Topside Box via an RS232 DP9 connector. If it is, the signal could be coming in on pin 2 of the DB9 connector instead of pin 3; try using a suitable RS232 crossover cable. Check connection between topside box and sonar and make sure both connectors are free of debris and water.
	Baud Rates Incorrect	(Applies to MRU and GPS only) The GPS device should be set to a baud rate of 38400 and the MRU devices should be set to 115200 baud. Contact technical support if this is not possible.



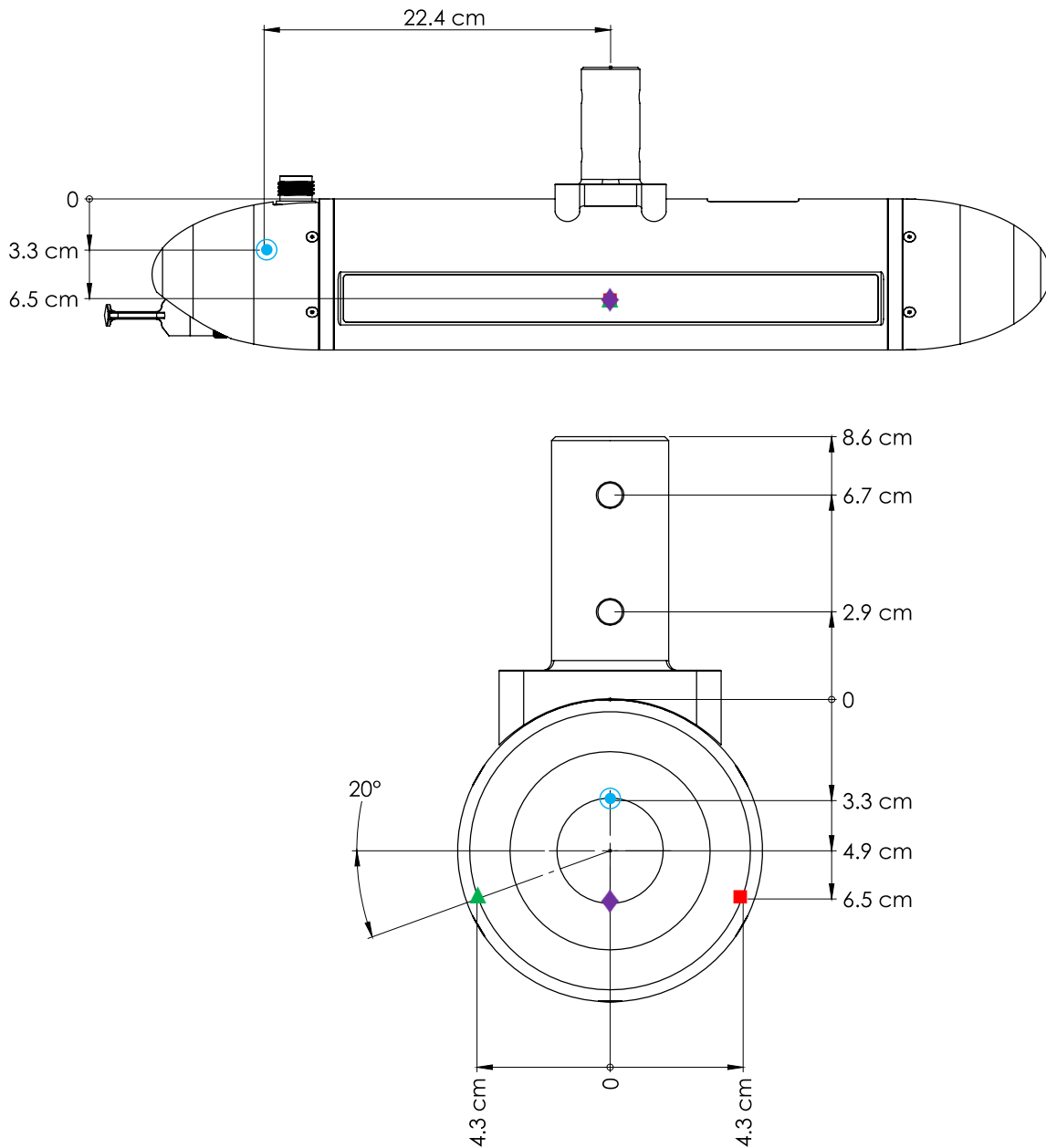
<i>Symptom</i>	<i>Potential Causes</i>	<i>Corrective Action</i>
Distortion or Noise in the side-scan images.	Acoustic noise or interference from the ships wake or engines.	Try reducing engine throttle or shut off engines (if possible) and see if the noise was reduced. If so, position the sonar further from the engines / source of noise.
	Acoustic interference from other sonar systems on board (echo sounder, multi beam, side-scan systems).	Turn off other sonar systems and see if noise is improved.
	Electrically noisy power supply.	If possible, try powering the sonar directly off a 12-24V battery (with nothing else attached to the battery) and see if noise improves.

ACOUSTIC CENTER OF THE 3DSS-DX-450 SONAR



where  marks the acoustic center of each transducer

ACOUSTIC CENTER AND ELLIPSE-2 & EKINOX POSITION FOR 3DSS-IDX



- The position of the center of measurement for the Ellipse-2 and Ekinox.
- The acoustic center of the port transducer.
- ▲ The acoustic center of the starboard transducer.
- ◆ The reference for INS EKF processed positions output by the Ellipse-2 and Ekinox.